Math 140H — Honors Calculus I: Lecture Notes

Jan Reimann

2026-10-26

Table of contents

Welcome! 3

I Real Numbers 4

1 The Rational Numbers and the Number Line 5
- 1.1 From Counting to Measuring . . . 5
- 1.2 The Number Line . . . 6
- 1.3 Distances and Intervals . . . 7
- 1.4 Incommensurable Lengths . . . 8
- 1.5 Summary . . . 9

2 The Real Numbers via Nested Intervals 10
- 2.1 Nested Intervals . . . 10
- 2.2 The Nested Interval Axiom . . . 11
- 2.3 Decimal Expansions . . . 12
- 2.4 Other Bases . . . 13
- 2.5 Summary . . . 14

3 Jupyter Activity: p-ary Expansions 15
- 3.1 Before we start: how to use a Jupyter notebook . . . 15
- 3.2 Importing Python modules . . . 16
- 3.3 From a rational number to its p-ary expansion . . . 17
- 3.4 Checking the examples . . . 20
- 3.5 A few more instructive examples . . . 20
- 3.6 Try it yourself . . . 21
- 3.7 Food for thought . . . 22

II Functions 23

4 The Concept of Function 24
- 4.1 Functions, Domain, and Range . . . 24
- 4.2 Injective Functions . . . 25
- 4.3 Monotone Functions . . . 25
- 4.4 Even and Odd Functions . . . 26

4.5 Building New Functions from Old . . . 26
4.6 Summary . . . 28

5 The Elementary Functions 29
5.1 Polynomials . . . 29
5.2 Rational Functions . . . 30
5.3 Algebraic and Power Functions . . . 30
5.4 Exponential Functions . . . 31
5.5 Logarithms . . . 31
5.6 Trigonometric Functions . . . 32
5.7 Summary . . . 32

6 Continuity 34
6.1 The ε–δ Definition . . . 35
6.2 Types of Discontinuities . . . 36
6.3 Summary . . . 37

7 Activity: Properties of Functions 38
7.1 Background . . . 38
7.2 Step 1: Draw the graphs . . . 38
7.3 Step 2: Make an educated guess . . . 38
7.4 Verification: injectivity . . . 39
7.5 Verification: even / odd . . . 40
7.6 Verification: monotonicity . . . 40
7.7 Continuity: the ε–δ game . . . 40
7.8 Discontinuities . . . 40

III Limits 41

8 Sequences and Convergence 42
8.1 Sequences . . . 42
8.2 The Limit of a Sequence . . . 42
8.3 Convergence to a General Limit . . . 43
8.4 Limit Laws . . . 45
8.5 Comparison and the Squeeze Theorem . . . 45
8.6 Monotone and Bounded Sequences . . . 46
8.7 Applications . . . 47
8.8 Summary . . . 49

9 Convergence of Sequences: Reading Data 50
9.1 A Simple Query Function . . . 53
9.2 Task 1: Explore the Data . . . 53
9.3 Task 2: The ε Challenge . . . 55

9.4 Dataset 2: Damped Oscillator . . . 56
9.5 Task 3: Explore the Sequence . . . 57
9.6 Task 4: The ε Challenge, Again . . . 59
9.7 Task 5: Convergence of $\sqrt[n]{p}$ . . . 60

10 Limits of Functions **62**
10.1 From Sequences to Functions . . . 62
10.2 The ε–δ Version . . . 63
10.3 Examples . . . 64
10.4 One-Sided Limits . . . 65
10.5 Limit Laws for Functions . . . 66
10.6 Continuity of Combinations and Compositions . . . 67
10.7 The Intermediate Value Theorem . . . 68
10.8 Summary . . . 68

IV The Riemann Integral **70**

11 Riemann Sums **71**
11.1 Area of the Circle . . . 71
11.2 Approximation of Areas by Rectangles . . . 72
11.3 Equipartitions and the Definition of the Integral . . . 73
11.4 Why the Choice Does Not Matter (optional) . . . 74
11.5 Verifying the Riemann Integral . . . 77
11.6 "New" Functions: x^2 . . . 78
11.7 Summary . . . 78

12 Lab: Riemann Sums **80**
12.1 Recap: the setup from lecture . . . 85
12.2 Part 1: Start . . . 86
12.3 Part 2: Left, right, and midpoint sums . . . 88
12.4 Part 3: Refining the partition . . . 91
12.5 Part 4: Back to the circle . . . 93
12.6 Part 5: Experiment on your own . . . 95
12.7 Part 6: when Riemann sums fail . . . 96

13 Properties of the Integral **98**
13.1 The Integral as Signed Area . . . 98
13.2 Additivity . . . 99
13.3 Integrals with $a \geq b$ . . . 100
13.4 Sums and Constant Multiples . . . 101
13.5 Basic Rules for Integrals . . . 102
13.6 Estimating Integrals . . . 103

13.7 The Integral as a Mean Value . . . 104
13.8 Summary . . . 107

14 The Indefinite Integral **108**
14.1 Definition and Basic Formula . . . 108
14.2 Properties of Φ as a Function . . . 109
14.3 Examples: Volumes and Dipsticks . . . 111

V The Derivative **115**

15 The Derivative **116**
15.1 The Tangent Problem . . . 116
15.2 Secants . . . 117
15.3 From Angles to Slopes . . . 119
15.4 The Derivative . . . 121
15.5 Examples . . . 122
15.6 Summary . . . 123

16 Working with Derivatives **126**
16.1 Interpretations of the Derivative . . . 126
16.2 Computing Derivatives . . . 128
16.3 Sum and Constant Multiple Rules . . . 132
16.4 Differentiability vs. Continuity . . . 133
16.5 Infinite Derivatives . . . 135
16.6 What the Derivative Tells Us About a Function . . . 137
16.7 Summary . . . 138

17 Activity: Reading Derivatives from Graphs **140**
17.1 Background . . . 140
17.2 Warm-up: a dictionary . . . 140
17.3 Part I: From f to f' . . . 141
17.4 Part II: From f' back to f . . . 144
17.5 Going further . . . 150

18 The Mean Value Theorem **154**
18.1 Local and Global Information . . . 154
18.2 The Mean Value Theorem . . . 154
18.3 Rolle's Theorem . . . 155
18.4 Proof of the Mean Value Theorem . . . 157
18.5 Application: Lipschitz Continuity . . . 158
18.6 Summary . . . 159

19 The Derivative as Linear Approximation **160**
19.1 The Tangent Line as an Approximation 160
19.2 Estimating the Error 162
19.3 Summary 165

20 Activity: Linear Approximation and Euler's Method **166**
20.1 Background: Differential Equations 166
20.2 Step 1: A worked example 166
20.3 Step 2: Exponential growth 168
20.4 Step 3: Logistic growth 168
20.5 Step 4: Pollution in a lake 169
20.6 Over or under? 169

21 Lab: The Mean Value Theorem and Linear Approximation **170**
21.1 Finding ξ 178
21.2 The tangent line as an approximation 182
21.3 Bounding the error with the second derivative 186

VI The Fundamental Theorem **189**

22 The Fundamental Theorem of Calculus **190**
22.1 The Derivative of the Indefinite Integral 190
22.2 Antiderivatives 192
22.3 The Fundamental Theorem 194
22.4 Evaluating Definite Integrals 195
22.5 Summary 196

Welcome!

These notes accompany Math 140H, Honors Calculus I, Fall 2026. Each lecture here corresponds to one class session and covers more or less the material presented in class, in the same order, so that it can serve as a compact study reference. HTML "hardcopies" of the Jupyter notebook activities from class are also included. Lecture notes will generally be available immediately **after** the corresponding class session.

State College, Fall 2026

Part I

Real Numbers

1 The Rational Numbers and the Number Line

> Companion reading
>
> This lecture corresponds to C&J §1.1a, together with the neighborhood definition from §1.1d — covered a bit early here since we will use it constantly for the rest of the semester.

Calculus is built on the real numbers, so before we can talk about limits, derivatives, or integrals we need to understand exactly what a real number *is* — and, just as importantly, why the rational numbers alone are not enough. This lecture makes that precise: we recall how the natural numbers extend to the rationals, set up the number line, and then prove that a length as ordinary as the hypotenuse of a simple right triangle cannot be a rational number at all.

1.1 From Counting to Measuring

There are two major uses for numbers: **counting** and **measuring**. The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ are the tool for counting — abstract symbols for how many elements are in a collection, stripped of any reference to what is actually being counted. They come equipped with addition and multiplication satisfying a short list of familiar rules.

Remark 1.1. For all natural numbers a, b, c:

- $a + b = b + a$ and $ab = ba$ (commutative laws)
- $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ (associative laws)
- $a(b + c) = ab + ac$ (distributive law)
- $a + c = b + c \;\Rightarrow\; a = b$ (cancellation law)

These are the *rules* of arithmetic, not its *content* — everything else in this course is built on top of them, and we will not dwell on them further.

Natural numbers can be added and multiplied, but not always subtracted or divided. There is no natural number a with

$$5 + a = 2,$$

so subtraction is not always possible in $\mathbb{N}$. Extending $\mathbb{N}$ to the **integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ fixes this: now $5 + (-3) = 2$.

In $\mathbb{Z}$, we still cannot always divide: there is no integer $b \in \mathbb{Z}$ with

$$5 \cdot b = 2.$$

Extending $\mathbb{Z}$ to the **rational numbers** fixes this in turn.

Definition 1.1.
A **rational number** is a number that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$. Every rational number has a unique representation p/q with $q > 0$ and $\gcd(p, q) = 1$ (no common factor larger than 1); we call this representation **in lowest terms**.

Within $\mathbb{Q}$, all four rational operations — addition, subtraction, multiplication, and division (except by 0) — can be carried out freely, and they obey exactly the same laws as in Remark 1.1. In this precise sense, $\mathbb{Q}$ is the smallest extension of the natural numbers in which counting-type arithmetic becomes measuring-type arithmetic.

1.2 The Number Line

To turn numbers into geometry, fix a line L, a point O on it called the **origin**, and a second point I to the right of O; the segment OI serves as our unit scale for measuring — a "prototype meter." The direction from O to I is called **positive**.

Every point on L is now uniquely determined by two pieces of information: its **distance** from O, and its **direction** (left or right) from O. For example, the point at distance $\frac{1}{2}$ in the positive direction represents the rational number $\frac{1}{2}$; the point at distance 2 in the positive direction represents 2.

Definition 1.2.
The distance of a number x from the origin is written $|x|$ and called the **absolute value** of x:

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Note $|x| \geq 0$ always, with equality only when $x = 0$.

Every rational point can in fact be *constructed*, not just described: to locate p/q on L with ruler and compass, construct a subdivision of the segment OI into q equal parts, then extend the part closest to O, p times. We will identify each rational number with its representation on L.

1.3 Distances and Intervals

For $x, y \in L$, the number $|x - y|$ is the **distance between x and y**. Observe: if $x, y \in \mathbb{Q}$, then $|x-y| \in \mathbb{Q}$ as well — distances between rational points are themselves rational.

Definition 1.3.
For $a < b$, the **interval** between a and b comes in four flavors:

- open: $(a, b) := \{x : a < x < b\}$ (both endpoints excluded)
- closed: $[a, b] := \{x : a \leq x \leq b\}$ (both endpoints included)
- half-open: $[a, b) := \{x : a \leq x < b\}$ and $(a, b] := \{x : a < x \leq b\}$

Definition 1.4.
Given a number a and $\varepsilon > 0$, the **ε-neighborhood of a** is the open interval

$$\{y : |y - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon),$$

i.e. the set of points less than ε away from a.

We will meet ε-neighborhoods constantly once we get to limits — keep this definition close by.

1.3.1 Density

Proposition 1.1.
Between any two distinct rational numbers $a < b$ there is another rational number c with $a < c < b$. Consequently, between a and b there are infinitely many rational numbers.

Proof. Let $c = \dfrac{a+b}{2}$. Since $a, b \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under addition and division by 2, $c \in \mathbb{Q}$; and since $a < b$, averaging gives $a < c < b$.

Now repeat the argument on the pair a, c to get a rational number strictly between a and c, hence strictly between a and b. Doing this again and again produces infinitely many rational numbers between a and b. □

We say the rationals are **dense** on the number line: no matter how closely you look, you will always find another one.

1.4 Incommensurable Lengths

Given how constructible and how dense $\mathbb{Q}$ is, a natural question arises: is there anything "left in between" the rational numbers? Are there any *gaps*? Clearly, such a gap could not itself be an interval — density rules that out, since any interval of positive length contains infinitely many rational points. If a gap exists at all, it can only be a single missing point.

Remark 1.2. The discovery that not every length is a rational multiple of a chosen unit is due to the school of Pythagoras, in the fifth or sixth century BCE. Consider a right triangle with both legs of length 1; by the Pythagorean theorem, its hypotenuse a satisfies

$$1^2 + 1^2 = a^2 = 2,$$

and a can certainly be constructed with ruler and compass. The theorem below shows this length cannot be written as p/q for integers p, q — the hypotenuse and the legs of this triangle are, in the language of the ancients, **incommensurable**: there is no common unit of which both are integer multiples.

Theorem 1.1.
There is no rational number x with $x^2 = 2$. Equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, toward a contradiction, that $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and $\gcd(p, q) = 1$ (i.e. p/q is in lowest terms).

1. Square both sides and clear the denominator: $p^2 = 2q^2$.
2. Conclude that p^2 is even — if p were odd, p^2 would be odd, but $p^2 = 2q^2$ is even. Write $p = 2r$ for some integer r.
3. Substitute $p = 2r$ into the equation from Step 1 and simplify to get an equation for q^2 in terms of r: $(2r)^2 = 2q^2 \;\Rightarrow\; 4r^2 = 2q^2 \;\Rightarrow\; q^2 = 2r^2$.
4. Conclude that q is also even, by the same reasoning as in Step 2.
5. Steps 2 and 4 show that 2 divides both p and q. This contradicts $\gcd(p, q) = 1$, which we assumed when we wrote p/q in lowest terms.
6. Conclude: $\sqrt{2}$ is irrational — no integers p, q satisfy $\sqrt{2} = p/q$, so $\sqrt{2} \notin \mathbb{Q}$.

□

Exercise 1.1. Adapt the proof of Theorem 1.1 to show that $\sqrt{3}$ is irrational. Then try to see where the argument would break if you attempted the same proof for $\sqrt{4}$ — since $\sqrt{4} = 2$ is certainly rational, something in the argument *must* fail. Where, exactly?

1.5 Summary

- The rationals $\mathbb{Q}$ satisfy all the usual laws of arithmetic (Remark 1.1) and are **dense**: between any two rationals lies another (Proposition 1.1).
- Distances $|x-y|$, intervals, and ε-neighborhoods (Definition 1.3, Definition 1.4) give us the basic vocabulary for talking about "closeness" on the number line — vocabulary we will lean on heavily once limits arrive.
- Density notwithstanding, $\mathbb{Q}$ does not exhaust the number line: Theorem 1.1 exhibits an ordinary, constructible length that is not a rational number.
- The rationals therefore leave *gaps* in the line. Lecture 2 closes these gaps rigorously, using **nested intervals** to construct the real numbers.

2 The Real Numbers via Nested Intervals

> Companion reading
>
> This lecture corresponds to C&J §1.1b–c.

Last time we found a gap: the length $\sqrt{2}$ is constructible with ruler and compass, yet Theorem 1.1 shows it is not a rational number. So what *is* $\sqrt{2}$, as a number? We are used to describing such quantities through infinite decimal expansions, e.g.

$$\pi = 3.1415926\ldots,$$

but so far this is just a string of digits — it is not clear what it *means* mathematically. The key idea of this lecture is that an infinite decimal (or any other description of an irrational number) is really a recipe for **approximating** a point by rational numbers, to any desired degree of accuracy. We make this precise using shrinking intervals with rational endpoints.

2.1 Nested Intervals

Suppose x is a point on the number line that we want to pin down using only rational numbers. Start with any closed interval $I_1 = [a_1, b_1]$ with $a_1, b_1 \in \mathbb{Q}$ and $a_1 \le x \le b_1$. Now choose a subinterval $I_2 = [a_2, b_2] \subseteq I_1$ that still contains x:

$$a_1 \le a_2 \le x \le b_2 \le b_1,$$

for instance by splitting I_1 in half and keeping whichever half contains x. Repeating this forever produces a chain

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots, \qquad x \in I_n \text{ for every } n.$$

For this to genuinely *pin down* x — rather than just hover near it — we need one more ingredient: the intervals must shrink all the way down to a point.

Definition 2.1.
A **nested sequence of closed intervals with rational endpoints** is a sequence $I_n = [a_n, b_n]$, $n = 1, 2, 3, \ldots$, with $a_n, b_n \in \mathbb{Q}$, such that

1. $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, and
2. the lengths $|I_n| := b_n - a_n$ tend to 0 as $n \to \infty$: for every $\varepsilon > 0$ there is an N such that $|I_n| < \varepsilon$ for all $n \geq N$.

Proposition 2.1.
If a nested sequence of closed intervals (I_n) contains a point in every I_n, that point is unique: there is at most one x with $x \in I_n$ for all n.

Proof. Suppose x and y both lie in every I_n, and toward a contradiction suppose $x \neq y$. Let $\varepsilon_0 = |x - y| > 0$. Since $|I_n| \to 0$, there is some n with $|I_n| < \varepsilon_0$. But $x, y \in I_n$ means $|x - y| \leq |I_n| < \varepsilon_0 = |x - y|$, which is absurd. Hence $x = y$. □

So a nested sequence of intervals can "address" at most one point — and, geometrically, every point x on the line does arise this way:

Theorem 2.1.
Every real number x is the unique point contained in every interval of some nested sequence of closed intervals with rational endpoints.

Proof. Uniqueness is Proposition 2.1. For existence, pick any rational bracket $I_1 = [a_1, b_1]$ with $a_1 \leq x \leq b_1$, and repeatedly bisect: given $I_n = [a_n, b_n]$, let $m_n = \frac{a_n + b_n}{2}$ (rational, since $a_n, b_n \in \mathbb{Q}$), and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } x \leq m_n \\ [m_n, b_n] & \text{if } x > m_n. \end{cases}$$

Then $x \in I_{n+1} \subseteq I_n$ and $|I_{n+1}| = |I_n|/2$, so $|I_n| = |I_1|/2^{n-1} \to 0$. This is a nested sequence of closed intervals with rational endpoints containing x. □

In this sense, every real number can be described — completely and unambiguously — by an infinite sequence of rational numbers (the endpoints a_n, b_n).

2.2 The Nested Interval Axiom

The construction above starts from a real number x and produces a nested sequence around it. What about the converse: if we are simply *handed* a nested sequence of closed intervals with rational endpoints, must there be a point in $\bigcap_n I_n$ at all?

For **open** intervals, the answer is no: $I_n = (0, \frac{1}{n})$ satisfies $I_1 \supseteq I_2 \supseteq \cdots$ and $|I_n| = \frac{1}{n} \to 0$, yet no point lies in every I_n — any candidate $x > 0$ fails once $n > 1/x$, and $x = 0$ is excluded from every I_n by openness.

For **closed** intervals, this is far less obvious, and cannot be deduced from the algebraic properties of $\mathbb{Q}$ alone (after all, $\mathbb{Q}$ itself has a gap at $\sqrt{2}$). We take it as a basic postulate about the number line.

Axiom (Nested Interval Property)

If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a nested sequence of closed intervals with rational endpoints, then

$$\bigcap_{n\in\mathbb{N}} I_n \neq \emptyset,$$

i.e. there is a point x contained in every I_n.

This is a genuine **axiom**: a basic postulate we accept without proof, rather than a consequence of things we already know. Together with Proposition 2.1, it guarantees that no gaps exist on the number line — every nested sequence of rational brackets zeroes in on exactly one real number, no more and no less. This completeness of the number line is fundamental for everything that follows in this course; we will meet it again, in other guises, once we get to limits.

2.3 Decimal Expansions

Decimal expansions are simply a very concrete, systematic way of producing a nested sequence of intervals — one that subdivides by tens at every step.

Definition 2.2.
Let x be a point on the number line. Set

$$c_0 := \lfloor x \rfloor,$$

the greatest integer with $c_0 \leq x \leq c_0 + 1$, and $I_0 = [c_0, c_0 + 1]$, so $|I_0| = 1$.

Divide I_0 into ten equal parts using the points $c_0 + \frac{1}{10}, c_0 + \frac{2}{10}, \ldots, c_0 + \frac{9}{10}$, and let $c_1 \in \{0, 1, \ldots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} \;\leq\; x \;\leq\; c_0 + \frac{c_1}{10} + \frac{1}{10},$$

and set I_1 to be this interval, so $|I_1| = \frac{1}{10}$.

Iterating, having chosen digits $c_1, \ldots, c_{n-1} \in \{0, \ldots, 9\}$, subdivide I_{n-1} into ten equal parts and let $c_n \in \{0, \ldots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} + \cdots + \frac{c_n}{10^n} \;\leq\; x \;\leq\; c_0 + \frac{c_1}{10} + \cdots + \frac{c_n}{10^n} + \frac{1}{10^n},$$

and let I_n be this interval, so $|I_n| = 10^{-n}$.

By construction $(I_n)_{n\geq 0}$ is exactly a nested sequence of closed intervals with rational endpoints in the sense of Definition 2.1: $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$, each contains x, and $|I_n| = 10^{-n} \to 0$. We abbreviate this whole sequence by writing

$$x = c_0 + 0.c_1c_2c_3\ldots,$$

the **decimal expansion** of x — a notational shortcut for the nested intervals, not a new kind of number. (Careful when $c_0 < 0$: the digits after the point are still *added* to c_0. For instance $x = -2.75$ has $c_0 = \lfloor -2.75 \rfloor = -3$ and $x = -3 + 0.25$, not $-3 - 0.25$.)

Combining Definition 2.2 with Proposition 2.1 and the Nested Interval Axiom:

- Every real number has an (infinite) decimal expansion.
- Conversely, every such expansion — i.e. every choice of $c_0 \in \mathbb{Z}$ and digits $c_1, c_2, \cdots \in \{0, \ldots, 9\}$ — is a nested sequence of closed intervals with rational endpoints, so by the axiom it defines a real number.

Decimal expansions are not always unique

Some real numbers have *two* decimal expansions: for instance

$$1.000\ldots = 0.999\ldots.$$

Exercise 2.1. Why does $1.000\ldots = 0.999\ldots$? (Hint: what would $1 - 0.999\ldots$ have to be, if it were positive?) More generally, can you characterize *exactly* which real numbers have two decimal expansions?

Solution. If $1 - 0.999\ldots = \delta > 0$, then δ would have to be smaller than 10^{-n} for every n (since $0.\underbrace{9\cdots9}_{n} = 1 - 10^{-n}$ already exceeds $1 - \delta$), forcing $\delta = 0$ — contradiction. So the two expansions name the same point.

In general, a real number has two decimal expansions exactly when it is a **terminating** decimal, i.e. of the form $m/10^n$ for some $m \in \mathbb{Z}$, $n \geq 0$: one expansion ends in all 0's, the other in all 9's. Every other real number has a unique decimal expansion.

2.4 Other Bases

Nothing in Definition 2.2 is special about the number 10 — we could subdivide each interval into any number $p > 1$ of equal parts instead of ten, producing digits $c_i \in \{0, 1, \ldots, p-1\}$ and a **base-p (or p-ary) expansion** of x.

The case $p = 2$, the **binary expansion**, is exactly the bisection procedure from the proof of Theorem 2.1: at each step we only need to decide whether x lies in the lower or upper half of the current interval.

Example 2.1.
Let's find the first three binary digits of $x = 0.76$. Start with $I_0 = [0, 1]$, so $c_0 = 0$.

- Midpoint 0.5: since $0.76 \geq 0.5$, digit $c_1 = 1$ and $I_1 = [0.5, 1]$.
- Midpoint 0.75: since $0.76 \geq 0.75$, digit $c_2 = 1$ and $I_2 = [0.75, 1]$.
- Midpoint 0.875: since $0.76 < 0.875$, digit $c_3 = 0$ and $I_3 = [0.75, 0.875]$.

So $0.76 = 0.110\ldots$ in base 2.

Exercise 2.2. Find the first three fractional digits of the base-2 expansion of $x = 0.3$, and the first three fractional digits of the base-3 expansion of $x = 0.76$.

2.5 Summary

- A **nested sequence of closed intervals with rational endpoints** (Definition 2.1) can contain at most one point (Proposition 2.1), and every real number arises as the unique point of some such sequence (Theorem 2.1).
- The converse — that every nested sequence of closed intervals with rational endpoints *does* contain a point — is not provable from what came before; it is the **Nested Interval Axiom**, the basic postulate that guarantees there are no gaps in the number line.
- **Decimal expansions** (Definition 2.2) are the special case of base-10 subdivision; every real number has one, though numbers of the form $m/10^n$ have exactly two (Exercise 2.1).
- The same construction works in any base $p > 1$, giving p-ary expansions; base 2 is exactly the bisection algorithm behind Theorem 2.1.

3 Jupyter Activity: p-ary Expansions

Math 140H, Honors Calculus I

3.1 Before we start: how to use a Jupyter notebook

This is the first notebook of the semester, so we start a quick orientation:

- A notebook is a sequence of **cells**. This text lives in a *markdown cell* (formatted text); cells with `code` in the toolbar are *code cells* (Python).
- Click a cell and press **Shift+Enter** to run it and move to the next one. Code cells must be run in order the first time through, top to bottom — a later cell may use a function or variable that an earlier cell defined.
- If something breaks, it is often because a cell was skipped or run twice out of order. Use the menu command *Run All* (or *Restart Kernel and Run All*) to run the whole notebook cleanly from the top.
- You do **not** need to know Python already. Every function below is commented line by line — read the comments (the lines starting with `#`) as a description of the math, and the code as one particular way of writing it down.

3.1.1 Code cells

You can think of a code cell very much like a fancy calculator. Move to the next cell an run it.

```
16+(53*8.7777-128)
```

```
353.2181
```

Careful: In Python, exponentiation is `**`, not `^`.

```
2**4
```

```
16
```

```
2^4
```

```
6
```

`^` in Python is the operation called `bitwise XOR`; that explains the strange result above.

We will learn more about Jupyter notebooks (and a little bit more about Python) in the course of this semester. Today, we will simply experiment a bit with pre-defined Python function that converts rational numbers to p-ary expansions. Go ahead and run the cells below one after the other (do not forget to read the text inbetween.)

3.2 Importing Python modules

A lot functionality in Python is provided by *modules*. The following cell loads the modules we need for this notebook.

```
# We import three tools before doing anything else:
#
#   math     -- the standard library module with math.floor(y), the greatest integer <=
#               (We need floor of *exact fractions*, not of floating-point approximation
#               see the note below.)
#   Fraction -- from the 'fractions' module. A Fraction stores a rational number as an e
#               numerator/denominator pair, e.g. Fraction(1, 3) is *exactly* one third,
#               This matters a lot here: a computer's ordinary decimal numbers (called '
#               are themselves binary approximations, so 1/3 as a float is already sligh
#               and the tiny error would snowball after enough steps of the algorithm be
#               Using Fraction means every digit we compute is provably correct, not jus
#   numpy    -- imported as 'np' by convention. We use it only to store the list of digi
#               compute as a numpy array, which is a convenient container for a sequence
import math
from fractions import Fraction
import numpy as np
```

The cells below show the advantage of the `Fraction` function:

```
1/2+1/3   # will use decimal expansions
```

```
0.8333333333333333
```

```
Fraction(1,2)+Fraction(1,3)   # will preserve the precise fraction representation
```

```
Fraction(5, 6)
```

3.3 From a rational number to its *p*-ary expansion

We generalize the nested interval construction to base p: given x, set

$$c_0 := \lfloor x \rfloor,$$

and then repeatedly zoom in. Having peeled off $c_0, c_1, \ldots, c_{n-1}$, look at what is left over,

$$r_{n-1} \;=\; x - c_0 - \frac{1}{p}c_1 - \cdots - \frac{1}{p^{n-1}}c_{n-1} \;\in [0, p^{-(n-1)}),$$

multiply it by p^{n-1} so it lands back in $[0,1)$, and read off the next digit as the integer part of p times that:

$$c_n = \lfloor p \cdot (p^{n-1} r_{n-1}) \rfloor.$$

(Multiplying by p here means zooming in by a factor of p.)

The function below does exactly this, but it tracks the *rescaled* remainder (called `remainder` in the code, always kept in $[0,1)$) rather than r_{n-1} itself, since that is what you actually multiply by p at each step:

1. `remainder = x - floor(x)`, the fractional part of x, in $[0,1)$.
2. Repeat: multiply `remainder` by `p`. Its integer part is the next digit; subtract that integer part off, leaving a new `remainder` back in $[0,1)$, and go again.

```
def rational_to_p_ary(x, p, max_digits=50):
    """
    Compute the base-p expansion of a rational number x.

    Parameters
    ----------
    x : Fraction, int, or a string like "7/12"
        The rational number to expand. (Do NOT pass a plain float such as 0.76 -- floats
        already binary approximations, so we would be expanding the wrong number. Use
        Fraction(76, 100) instead. See the discussion above.)
    p : int
        The base, an integer > 1 (p = 2 for binary, p = 10 for the usual decimal expansi
    max_digits : int, optional (default = 50)
        how many fractional digits to compute

    Returns
    -------
    integer_part : int
        c_0, the integer part of x (i.e. floor(x)).
    digits : numpy array of int
        The fractional digits c_1, c_2, c_3, ... that were computed, in order.
    repeat_start : int or None
        If the expansion is eventually periodic, the index (0-based, into `digits`) wher
        repeating block begins. If the expansion terminates exactly (all later digits ar
        this is None.
    """
    if not isinstance(p, int) or p <= 1:
        raise ValueError("the base p must be an integer greater than 1")

    # Convert x to an exact fraction, no matter what form it was given in.
    x = Fraction(x)

    # Step 1: peel off the integer part c_0 = floor(x). What remains, x - c_0, is the
    # 'fractional part' of x and always lies in [0, 1).
    integer_part = math.floor(x)
    remainder = x - integer_part

    digits = []             # the digits c_1, c_2, ... found so far, in order
    seen_remainders = {}    # remembers, for each remainder value we've seen, which step
                             # first showed up at -- this is how we notice a repeat
    repeat_start = None
```

```
    for step in range(max_digits):
        # Zooming into the next sub-interval of
        # length 1/p means multiplying the remainder by p; the integer part of the resul
        # exactly which of the p equal pieces x fell into, i.e. the next digit.
        remainder = remainder * p
        next_digit = math.floor(remainder)
        digits.append(next_digit)

        # Subtract off the digit we just read, leaving a fresh remainder back in [0, 1),
        # ready for the next iteration.
        remainder = remainder - next_digit

    return integer_part, np.array(digits, dtype=int), repeat_start
```

The cell below defines a small helper function to print an expansion in a readable form. We deliberately write it as `c_0 + 0.c_1c_2c_3...` with an explicit `+`, matching the lecture notes' warning: if c_0 is negative, the fractional digits are still *added* to it, not subtracted, so `-3.25` written the ordinary way would be dangerously ambiguous. Writing `-3 + 0.25` instead leaves no room for that mistake.

```
def format_expansion(integer_part, digits, p, repeat_start=None):
    """
    Turn the output of rational_to_p_ary into a readable string, e.g.
        format_expansion(0, np.array([1, 1, 0]), 2, None) -> '0 + 0.110  (base 2)'
        format_expansion(0, np.array([3]), 10, 0)          -> '0 + 0.(3)  (base 10)'   #
    """
    # Turn each digit into a one-character string. In any base p <= 10 these are ordinar
    # for p > 10 we would need letters (like hexadecimal's a-f), which we don't handle h
    digit_chars = [str(int(d)) for d in digits]

    if repeat_start is None:
        # No repeating block: just list every digit we have.
        fractional_part = "".join(digit_chars) if len(digit_chars) > 0 else "0"
    else:
        # Split into the digits before the repeat (head) and the repeating block itself
        # and wrap the tail in parentheses to mark it as repeating forever.
        head = "".join(digit_chars[:repeat_start])
        tail = "".join(digit_chars[repeat_start:])
        fractional_part = f"{head}({tail})" if head else f"({tail})"

    return f"{integer_part} + 0.{fractional_part}  (base {p})"
```

3.4 Checking the examples

In the lecture notes, we found the first three *binary* digits of $x = 0.76$ by hand: $c_1 = 1,\ c_2 = 1,\ c_3 = 0$, i.e. $0.76 = 0.110\ldots$ in base 2. Let's confirm that.

```
integer_part, digits, repeat_start = rational_to_p_ary(Fraction(76, 100), 2, max_digits=
print(format_expansion(integer_part, digits, 2, repeat_start))
print("first three fractional digits:", digits[:3], " -- matches c1, c2, c3 = 1, 1, 0 fr
```

```
0 + 0.11000010100011111010111000010l0  (base 2)
first three fractional digits: [1 1 0]  -- matches c1, c2, c3 = 1, 1, 0 from Example 2.6
```

Now find the first three fractional digits of the base-2 expansion of $x = 0.3$, and of the base-3 expansion of $x = 0.76$. Work those out on paper first – then run the cell below to check yourself.

```
ip, d, r = rational_to_p_ary(Fraction(3, 10), 2, max_digits=20)
print("0.3 in base 2:", format_expansion(ip, d, 2, r))
print("first three digits:", d[:3])
print()
ip, d, r = rational_to_p_ary(Fraction(76, 100), 3, max_digits=20)
print("0.76 in base 3:", format_expansion(ip, d, 3, r))
print("first three digits:", d[:3])
```

```
0.3 in base 2: 0 + 0.01001100110011001100  (base 2)
first three digits: [0 1 0]

0.76 in base 3: 0 + 0.20211200100201102212  (base 3)
first three digits: [2 0 2]
```

3.5 A few more instructive examples

- $1/3$ in base 10 has *no* terminating part at all – the repeating block starts immediately.
- $1/6$ in base 10 has one non-repeating digit followed by a repeating block (matches $1/6 = 0.1\overline{6}$).
- $1/10$ terminates in base 10 (as expected, $1/10 = 0.1$ exactly) but is *purely repeating* in base 2 – this is the mathematical reason `0.1` cannot be stored exactly as an ordinary floating-point number, and is why we insisted on `Fraction` above instead of Python floats.

- Negative x: $x = -11/4 = -2.75$. Its integer part is $c_0 = \lfloor -2.75 \rfloor = -3$, and the digits describe 0.25, which gets *added* to -3 (giving back -2.75), not subtracted.

```
for label, value, base, n in [
    ("1/3", Fraction(1, 3), 10, 10),
    ("1/6", Fraction(1, 6), 10, 10),
    ("1/10", Fraction(1, 10), 10, 10),
    ("1/10", Fraction(1, 10), 2, 30),
    ("-11/4", Fraction(-11, 4), 10, 10),
]:
    ip, d, r = rational_to_p_ary(value, base, max_digits=n)
    print(f"{label:>6} = {format_expansion(ip, d, base, r)}")
```

```
   1/3 = 0 + 0.3333333333  (base 10)
   1/6 = 0 + 0.1666666666  (base 10)
  1/10 = 0 + 0.1000000000  (base 10)
  1/10 = 0 + 0.000110011001100110011001100110  (base 2)
 -11/4 = -3 + 0.2500000000  (base 10)
```

3.6 Try it yourself

Pick your own rational number x and a base $p > 1$. Use `rational_to_p_ary` to get its p-ary expansion.

```
# Replace the values below with your own choice of x (as a Fraction), p.
x = Fraction(1, 2)
p = 10


ip, d, r = rational_to_p_ary(x, p, max_digits=30)
print(f"x in base {p}:", format_expansion(ip, d, p, r))
```

```
x in base 10: 0 + 0.500000000000000000000000000000  (base 10)
```

3.7 Food for thought

Question 1: For a fixed base p, which fractions will have a p-ary expansion that will eventually be identically 0? (such as 1/2 in base 10) Try to formulate a hypothesis and test it using the cell above.

Question 2: Is it true that all p-ary expansions of rational numbers eventually become periodic? If yes, why is this the case?

Part II

Functions

4 The Concept of Function

With the number line in hand, we can start doing calculus — but calculus is not really about individual numbers. It is about *dependence*: how one quantity changes as another one does. The area of a circle depends on its radius, the height of a falling stone depends on the elapsed time, the pressure of a gas depends on its volume. The mathematical object that captures such a dependence is a **function**, and getting comfortable with the vocabulary now will pay off for the rest of the course.

4.1 Functions, Domain, and Range

Definition 4.1.
A **function** f is a rule that assigns to every element x of a set D — the **domain** of f — one single, unambiguously determined object $f(x)$, its **value** at x, lying in a set E — the **codomain**. We write

$$f\colon D \to E, \qquad x \mapsto f(x).$$

In this course the domain is almost always a subset of the number line, $D \subseteq \mathbb{R}$, and the codomain is $E = \mathbb{R}$; we then call f a **real-valued function of a real variable** and write simply $f\colon D \to \mathbb{R}$.

The word *unambiguously* is the heart of the definition: for each input $x \in D$ there must be **exactly one** output $f(x)$ — no more (that would not be a rule) and no fewer (every point of the domain must get a value). The domain is part of the data of a function: $x \mapsto x^2$ on $\mathbb{R}$ and $x \mapsto x^2$ on $[0, \infty)$ are, strictly speaking, different functions.

Remark 4.1. When a function is specified by a formula with no domain stated, we adopt the convention that its domain is the **natural domain**: the largest set of real numbers for which the formula makes sense. For $x \mapsto 1/(x-1)$ this is $\mathbb{R} \setminus \{1\}$; for $x \mapsto \sqrt{x}$ it is $[0, \infty)$. We are always free to *restrict* to a smaller domain when we want to.

Definition 4.2.
The **range** (or **image**) of $f\colon D \to \mathbb{R}$ is the set of values it actually attains:

$$\operatorname{ran}(f) \;=\; \{\, y \in \mathbb{R} : y = f(x) \text{ for some } x \in D \,\}.$$

Remark 4.2. The range need not fill up the codomain. For $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, the codomain is $\mathbb{R}$ but

$$\operatorname{ran}(f) = [0, \infty),$$

since squares are never negative and every $y \geq 0$ is $(\sqrt{y})^2$. Saying "f maps into $\mathbb{R}$" is a statement about where the values are *allowed* to live; the range records where they *actually* land.

4.2 Injective Functions

Definition 4.3.
A function $f\colon D \to \mathbb{R}$ is **injective** (or **one-to-one**) if it sends distinct inputs to distinct outputs:

$$x \neq y \quad \Rightarrow \quad f(x) \neq f(y),$$

or equivalently, taking the contrapositive, $f(x) = f(y) \quad \Rightarrow \quad x = y$.

Example 4.1.

- $f(x) = x^3$ is injective on $\mathbb{R}$: if $x^3 = y^3$ then $x = y$ (taking cube roots).
- $g(x) = x^2$ is **not** injective on $\mathbb{R}$: $g(-1) = 1 = g(1)$, yet $-1 \neq 1$. Restricted to $[0, \infty)$, however, g *is* injective.

Injectivity is exactly the property that lets us run a function backwards; we return to this when we discuss inverse functions below.

4.3 Monotone Functions

Definition 4.4.
Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$. For all $x, w \in D$, we say f is

- **increasing** if $x \leq w \quad \Rightarrow \quad f(x) \leq f(w)$;
- **strictly increasing** if $x < w \quad \Rightarrow \quad f(x) < f(w)$;
- **decreasing** if $x \leq w \quad \Rightarrow \quad f(x) \geq f(w)$;
- **strictly decreasing** if $x < w \quad \Rightarrow \quad f(x) > f(w)$.

A function of any of these four types is called **monotone**.

Example 4.2.

- $f(x) = x^3$ is strictly increasing on all of $\mathbb{R}$.

- $g\colon [0,\pi] \to \mathbb{R}$, $g(x) = \cos x$, is strictly decreasing: as the angle runs from 0 to π, the x-coordinate on the unit circle drops steadily from 1 to -1.

Exercise 4.1. Show that a strictly increasing function is injective (and likewise for strictly decreasing). Is the converse true — must an injective function be monotone?

4.4 Even and Odd Functions

Definition 4.5.
Suppose the domain D is symmetric about 0, i.e. $x \in D \quad \Rightarrow \quad -x \in D$. A function $f\colon D \to \mathbb{R}$ is

- **even** if $f(-x) = f(x)$ for all $x \in D$ — its graph is symmetric about the y-axis;
- **odd** if $f(-x) = -f(x)$ for all $x \in D$ — its graph is symmetric about the origin.

Example 4.3.
x^2, $|x|$, and $\cos x$ are even; x, x^3, and $\sin x$ are odd; $x + x^2$ is neither. Most functions are neither even nor odd.

Exercise 4.2. Show that every function f on a symmetric domain can be written *uniquely* as a sum $f = f_{\mathrm{e}} + f_{\mathrm{o}}$ of an even function f_{e} and an odd function f_{o}. (Hint: what must $\frac{1}{2}(f(x) + f(-x))$ and $\frac{1}{2}(f(x) - f(-x))$ be?)

4.5 Building New Functions from Old

Most functions we work with are assembled from simpler ones by a few standard operations.

4.5.1 Arithmetic combinations

Definition 4.6.
Let $f, g\colon D \to \mathbb{R}$ be two functions on the same domain, and $c \in \mathbb{R}$. Then

$$f + g, \qquad c\,f, \qquad f \cdot g$$

are again functions on D, defined **pointwise**: $(f+g)(x) = f(x) + g(x)$, $(c\,f)(x) = c\,f(x)$, $(f \cdot g)(x) = f(x)\,g(x)$. The **quotient** f/g, given by $(f/g)(x) = f(x)/g(x)$, is a function on the smaller domain

$$D \setminus \{\, x \in D : g(x) = 0 \,\}.$$

4.5.2 Composition

Definition 4.7.
Let $g\colon D \to \mathbb{R}$ and $f\colon D' \to \mathbb{R}$ with $\operatorname{ran}(g) \subseteq D'$. The **composition** $f \circ g$ ("first g, then f") is

$$(f \circ g)(x) \;:=\; f(g(x)), \qquad x \in D,$$

a function with domain $\operatorname{dom}(f \circ g) = \operatorname{dom}(g)$.

Example 4.4.
With $f(x) = \sqrt{x}$ and $g(x) = 1 + x^2$:

$$(f \circ g)(x) = \sqrt{1 + x^2} \quad (\text{all } x \in \mathbb{R}), \qquad (g \circ f)(x) = 1 + x \quad (x \geq 0).$$

The order matters: $f \circ g$ and $g \circ f$ are different functions.

4.5.3 Inverse functions

Definition 4.8.
If $f\colon D \to \mathbb{R}$ is injective, then for each $y \in \operatorname{ran}(f)$ there is **exactly one** $x \in D$ with $f(x) = y$. Sending each such y to that x defines the **inverse function**

$$f^{-1}\colon \operatorname{ran}(f) \to D, \qquad f^{-1}(f(x)) = x \;\text{ for all } x \in D, \qquad f(f^{-1}(y)) = y \;\text{ for all } y \in \operatorname{ran}(f).$$

Its domain is $\operatorname{ran}(f)$ and its range is D.

> Notation trap
>
> f^{-1} denotes the **inverse function**, not the reciprocal $1/f$. For $f(x) = x^3$ we have $f^{-1}(y) = \sqrt[3]{y}$, whereas $\dfrac{1}{f(x)} = x^{-3}$.

Remark 4.3. The point (a, b) lies on the graph of f exactly when (b, a) lies on the graph of f^{-1}. So the graph of f^{-1} is the mirror image of the graph of f across the line $y = x$.

Example 4.5.
$f(x) = x^3$ is injective on $\mathbb{R}$ with $\operatorname{ran}(f) = \mathbb{R}$, and $f^{-1}(y) = \sqrt[3]{y}$. By contrast $g(x) = x^2$ on $\mathbb{R}$ has no inverse (it is not injective); restricted to $[0, \infty)$ it does, namely $g^{-1}(y) = \sqrt{y}$.

4.6 Summary

- A **function** $f\colon D \to \mathbb{R}$ (Definition 4.1) assigns to each point of its **domain** D exactly one real value; the set of values attained is its **range** (Definition 4.2), which may be a proper subset of the codomain.
- f is **injective** (Definition 4.3) if distinct inputs give distinct outputs, and **monotone**

  (1) if it consistently preserves or reverses order; strict monotonicity implies injectivity (Exercise 4.1).

- On a symmetric domain, f may be **even** or **odd** (Definition 4.5), i.e. symmetric about the y-axis or the origin.
- New functions are built from old ones by pointwise **arithmetic** (Definition 4.6), **composition** (Definition 4.7), and — when f is injective — by forming the **inverse** f^{-1} (Definition 4.8), whose graph is the reflection of that of f across $y = x$.

5 The Elementary Functions

> Companion reading
>
> This lecture corresponds to C&J §1.3.

Almost every function met in a first calculus course is built out of a short catalogue of standard functions: polynomials, rational functions, powers, exponentials, logarithms, and the trigonometric functions. These are the **elementary functions**. This lecture is a tour of the list, noting for each type its natural domain, its range, and its basic shape — information we will need constantly once we start differentiating and integrating.

5.1 Polynomials

Definition 5.1.
A **polynomial** is a function of the form

$$p(x) \;=\; a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

with real coefficients $a_0, a_1, \ldots, a_n$. If $a_n \neq 0$, the integer n is the **degree** of p.

The lowest degrees have familiar names:

degree	form	name
0	$y = a_0$	constant
1	$y = a_0 + a_1 x$	linear
2	$y = a_0 + a_1 x + a_2 x^2$	quadratic
3	$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3$	cubic

A polynomial is a finite sum of constant multiples of the **power functions** $1, x, x^2, x^3, \ldots$, each of which is defined for every real number. Hence every polynomial has natural domain

$$\mathrm{dom}(p) = \mathbb{R}$$

(unless we deliberately restrict it).

Remark 5.1. The pure powers x^n share a few features worth remembering. Every graph passes through $(0,0)$ and $(1,1)$. On the interval $[0,1]$ a higher power lies *below* a lower one ($x^3 \le x^2 \le x$), while on $[1,\infty)$ the order reverses. The power x^n is even when n is even and odd when n is odd; for odd n it is strictly increasing on $\mathbb{R}$ (hence injective), while for even n it is not injective on $\mathbb{R}$.

5.2 Rational Functions

Definition 5.2.
A **rational function** is a quotient of two polynomials,

$$r(x) \;=\; \frac{p(x)}{q(x)}, \qquad q \not\equiv 0.$$

Its natural domain is $\mathbb{R} \setminus Z$, where

$$Z = \{\, x \in \mathbb{R} : q(x) = 0 \,\}$$

is the (finite) set of zeros of the denominator.

Example 5.1.

- $\dfrac{x}{x^2+1}$ is a rational function. Since $x^2 + 1 > 0$ for every real x, the set Z is empty and the domain is all of $\mathbb{R}$.
- $\dfrac{\sqrt{x}}{x^3 - x}$ is **not** a rational function: $\sqrt{x}$ is not a polynomial. (It is, however, still an elementary function, as it is the arithmetic combination of two elementary functions - see next section)

5.3 Algebraic and Power Functions

Definition 5.3.
A **power function** is $f(x) = x^\alpha$ with a fixed real exponent $\alpha \in \mathbb{R}$. For $x > 0$ this is defined for every α (via $x^\alpha = \exp(\alpha \ln x)$, once we have exp and ln). For an **integer** exponent the domain extends to

$$\operatorname{dom}(x^\alpha) = \begin{cases} \mathbb{R} & \alpha \ge 0, \\ \mathbb{R} \setminus \{0\} & \alpha < 0. \end{cases}$$

Remark 5.2. For a non-integer exponent one usually keeps the domain $[0,\infty)$ (or $(0,\infty)$ if $\alpha<0$). The roots $x^{1/2}=\sqrt{x}$, $x^{1/3}=\sqrt[3]{x}$, ... are examples; odd roots such as $\sqrt[3]{x}$ can in fact be extended to all of $\mathbb{R}$. Power functions with rational exponent, and more generally functions obtained from polynomials by the four arithmetic operations together with root extraction, are called **algebraic functions**. The exponential and logarithm below are *not* algebraic — they are **transcendental**.

5.4 Exponential Functions

Definition 5.4.
Fix a **base** $a>0$. The **exponential function** with base a,

$$\exp_a:\mathbb{R}\to\mathbb{R},\qquad \exp_a(x)=a^x,$$

is defined for every real x. For $a\neq 1$ it takes only positive values, and every positive value is attained:

$$\operatorname{dom}(\exp_a)=\mathbb{R},\qquad \operatorname{ran}(\exp_a)=(0,\infty)=\mathbb{R}^{>0}.$$

For $a>1$ the function a^x is strictly increasing; for $0<a<1$ it is strictly decreasing; for $a=1$ it is the constant 1. In all cases $a^0=1$ and $a^{x+y}=a^xa^y$.

5.5 Logarithms

Definition 5.5.
Fix a base $a>0$ with $a\neq 1$. Then $\exp_a$ is injective from $\mathbb{R}$ onto $(0,\infty)$, and its inverse function (in the sense of Definition 4.8) is the **logarithm to base** a,

$$\log_a:(0,\infty)\to\mathbb{R},\qquad \log_a(a^x)=x,\qquad a^{\log_a y}=y.$$

Thus

$$\operatorname{dom}(\log_a)=(0,\infty),\qquad \operatorname{ran}(\log_a)=\mathbb{R}.$$

Because it is the inverse of a^x, the graph of $\log_a$ is the reflection of the graph of a^x across the line $y=x$ (Remark 4.3). It converts products to sums: $\log_a(xy)=\log_a x+\log_a y$.

5.6 Trigonometric Functions

Let x be a real number, thought of as an angle measured in **radians**. Rotating the point $(1, 0)$ counter-clockwise through the angle x about the origin lands on a point of the unit circle whose coordinates are, by definition, $(\cos x, \sin x)$.

Both sin and cos are defined for all $x \in \mathbb{R}$, take values in $[-1, 1]$, and repeat after a full turn.

Definition 5.6.
A function f is **periodic with period** $P > 0$ if $f(x + P) = f(x)$ for every x in its domain.

Thus sin and cos are periodic with period 2π. The **tangent**

$$\tan x = \frac{\sin x}{\cos x}$$

is defined wherever $\cos x \neq 0$, i.e. for $x \neq \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$, and is periodic with period π.

Remark 5.3. A general **sinusoid** has the form

$$f(x) = A \sin(bx) + C,$$

with **amplitude** $|A|$ (half the vertical spread of the graph), **period** $\dfrac{2\pi}{b}$, and **vertical offset** C (the horizontal line the graph oscillates around).

Exercise 5.1. The function $f(x) = 3\sin(2x) + 1$ is a sinusoid. Find its amplitude, period, and range, and the smallest $x > 0$ at which it attains its maximum.

5.7 Summary

- **Polynomials** (Definition 5.1) are finite combinations of the powers $1, x, x^2, \ldots$; they are defined on all of $\mathbb{R}$.
- **Rational functions** (Definition 5.2) are quotients of polynomials, defined off the finite zero set of the denominator.
- **Power functions** x^α (Definition 5.3) have domain $\mathbb{R}$ or $\mathbb{R} \setminus \{0\}$ for integer exponents, and are typically taken on $[0, \infty)$ otherwise; together with roots they form the **algebraic** functions.
- **Exponentials** a^x (Definition 5.4) have domain $\mathbb{R}$ and range $(0, \infty)$; their inverses, the **logarithms** $\log_a$ (Definition 5.5), have domain $(0, \infty)$ and range $\mathbb{R}$.

- The **trigonometric functions** $\sin, \cos$ are 2π-periodic (Definition 5.6) with range $[-1, 1]$; $\tan = \sin / \cos$ is π-periodic and undefined where cos vanishes.

6 Continuity

> Companion reading
>
> This lecture corresponds to C&J §1.2.d.

Intuitively, a function is continuous if a small change in the independent variable produces only a small change in the dependent variable — the graph has no jumps, gaps, or blow-ups. Before making this precise, it helps to see why the intuitive idea is not yet a *definition*: it uses the word "small" twice without saying how small, or how the two smallnesses are related.

Example 6.1.
Let

$$f(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

At $x_0 = 0$, an arbitrarily small change in x — moving from $x = 0$ to some $x < 0$ — causes f to jump from 1 to 0, a change of a full unit. No matter how small a step we take to the left of 0, the output change stays large. We call this a **jump discontinuity** at $x_0 = 0$.

Q: How do we make "small change forces small change" precise? The idea is to formalize "small" using neighborhoods.

Definition 6.1.
Given $a \in \mathbb{R}$ and $\varepsilon > 0$, the **ε-neighborhood** of a is the interval

$$(a - \varepsilon,\, a + \varepsilon).$$

Think of ε as a *tolerance*: saying "x is within tolerance of a" means $x \in (a-\varepsilon, a+\varepsilon)$, i.e. $|x - a| < \varepsilon$. "Small change" becomes "ε small."

Fix a point x_0. We want: the distance between $f(x)$ and $f(x_0)$ stays within tolerance ε, i.e.

$$|f(x) - f(x_0)| < \varepsilon,$$

whenever x is *sufficiently close* to x_0 — say, within some tolerance δ of x_0. The key point is that this has to work for **any** $\varepsilon > 0$, no matter how small; but we are allowed to choose δ *in response to* ε — a smaller tolerance on the output may well force a smaller tolerance on the input.

6.1 The ε–δ Definition

Definition 6.2.
Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $x_0 \in D$. We say f is **continuous at** x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in D \text{ and } |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

Example 6.2.
Take f as in Example 6.1 and $x_0 = 0$. Pick $\varepsilon = \frac{1}{2}$. No matter how small we choose $\delta > 0$, there is always some $x < 0$ with $|x - x_0| < \delta$ (i.e. x is δ-close to $x_0 = 0$), and for any such x,

$$|f(x) - f(x_0)| = |0 - 1| = 1 > \tfrac{1}{2}.$$

So no δ works for this ε, and f is **not** continuous at $x_0 = 0$.

Remark 6.1. Continuity is a property of f **at a point**. The function f above is continuous at every $x_0 \neq 0$ (Exercise 6.1) and discontinuous only at $x_0 = 0$ — a function can be continuous at some points of its domain and discontinuous at others.

Exercise 6.1. Show that f from Example 6.1 is continuous at every $x_0 \neq 0$.

6.1.1 The ε–δ game

Remark 6.2. It helps to think of the definition as a game between an opponent and you. The opponent challenges you with some $\varepsilon > 0$; you must exhibit a matching $\delta > 0$ so that

$$|x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

f is continuous at x_0 exactly when you can win this game **no matter which** ε **the opponent picks**.

Example 6.3.
$g(x) = 3x + 2$ is continuous at every $x_0 \in \mathbb{R}$.

Proof. Let $\varepsilon > 0$ and $x_0 \in \mathbb{R}$ be arbitrary. We want $|g(x) - g(x_0)| < \varepsilon$. Compute

$$|g(x) - g(x_0)| = |(3x + 2) - (3x_0 + 2)| = |3x - 3x_0| = 3|x - x_0|.$$

So if we let $\delta = \varepsilon/3$, then

$$|x - x_0| < \delta \quad \Rightarrow \quad |g(x) - g(x_0)| = 3|x - x_0| < 3\delta = \varepsilon.$$

□

Exercise 6.2. Show, directly from Definition 6.2, that $h(x) = \frac{1}{2}x - 1$ is continuous at every $x_0 \in \mathbb{R}$. (What choice of δ in terms of ε makes the argument work?)

A Jupyter demonstration accompanying this lecture lets you play the ε–δ game interactively: pick x_0 and a δ, and watch the resulting range of f-values shrink (or fail to shrink) as $\delta \to 0$.

6.2 Types of Discontinuities

If f fails to be continuous at x_0, the failure typically has one of three flavors.

Remark 6.3. **Jump discontinuities.** The values of f approach one number from the left of x_0 and a different number from the right, as in Example 6.1. We will make this precise next week with *one-sided limits*: a jump discontinuity is exactly a point where the one-sided limits exist but disagree.

Remark 6.4. **Unbounded (infinite) discontinuities.** Consider

$$f(x) = \begin{cases} 1/x^2 & x \neq 0 \\ 0 & x = 0. \end{cases}$$

As $x \to 0$, $f(x) \to \infty$, so $|f(x) - f(0)|$ is **unbounded** on every neighborhood of 0 — no ε-tolerance, however large, can be met near $x_0 = 0$ by any δ.

Remark 6.5. **Oscillation discontinuities.** Consider

$$g(x) = \begin{cases} \sin(1/x) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Near $x_0 = 0$, g oscillates wildly: arbitrarily close to 0 we can always find points where g equals 1 and points where g equals -1. Indeed, for any $n \in \mathbb{N}$, taking

$$x_1 = \frac{1}{2\pi n + \pi/2} \qquad \text{gives} \qquad \sin(1/x_1) = 1,$$

and similarly $x_{-1} = \dfrac{1}{2\pi n + 3\pi/2}$ gives $\sin(1/x_{-1}) = -1$; both $x_1, x_{-1} \to 0$ as $n \to \infty$. So on every neighborhood of 0, no matter how small, g takes values ε apart for ε close to 2 — g cannot be made continuous at 0 by any choice of $g(0)$.

Example 6.4.
Contrast this with

$$x\sin(1/x) \quad (x \neq 0), \qquad \text{value } 0 \text{ at } x = 0,$$

which **is** continuous at $x_0 = 0$: although $\sin(1/x)$ still oscillates without settling down, it is squeezed between -1 and 1, so $|x\sin(1/x) - 0| \leq |x| \to 0$ as $x \to 0$. Oscillation alone does not cause a discontinuity — it must fail to be *damped out* as $x \to x_0$.

6.3 Summary

- The intuitive idea "small change in, small change out" is formalized using **ε-neighborhoods** (Definition 6.1): a tolerance interval $(a-\varepsilon, a+\varepsilon)$ around a point.
- f is **continuous at** x_0 (Definition 6.2) if for every tolerance $\varepsilon > 0$ on the output, there is a matching tolerance $\delta > 0$ on the input, so that x within δ of x_0 forces $f(x)$ within ε of $f(x_0)$ — think of it as a game (Remark 6.2) you must be able to win against any ε.
- Continuity is a property of f **at a point** (Remark 6.1); a function can be continuous at some points and not others.
- Failures of continuity come in three standard flavors: **jump** discontinuities (Remark 6.3), where one-sided behavior disagrees; **unbounded** discontinuities (Remark 6.4), where f blows up; and **oscillation** discontinuities (Remark 6.5), where f never settles down near x_0 — unless the oscillation is damped, as in $x\sin(1/x)$ (Example 6.4), which remains continuous.

7 Activity: Properties of Functions

About this activity

An in-class, pencil-and-paper worksheet. It revisits the properties of functions built up in the previous chapters — *injective*, *monotone*, *even/odd*, and, above all, *continuous* — by working them out in full for two specific functions. Companion reading: C&J §1.2.d.

7.1 Background

We have met several properties a function can have: **injective**, **monotone**, **even or odd**, and — most important — **continuous**. In this activity we investigate all of them for the two functions

$$f(x) = \begin{cases} \dfrac{x}{|x|} & x \neq 0, \\ 0 & x = 0, \end{cases} \qquad g(x) = \begin{cases} \dfrac{1}{x^2} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

7.2 Step 1: Draw the graphs

Sketch the graph of each function on its own set of axes. Plot a few sample points first (e.g. $x = \pm\frac{1}{2},\ \pm 1,\ \pm 2,\ \pm 3$), then connect them.

7.3 Step 2: Make an educated guess

From your graphs, try to answer the following.

1. Is the function injective?
2. Is it monotone? If so, is it *strictly* monotone? If not, can you find intervals — as large as possible — on which it is (strictly) monotone?
3. Is it even, odd, or neither?

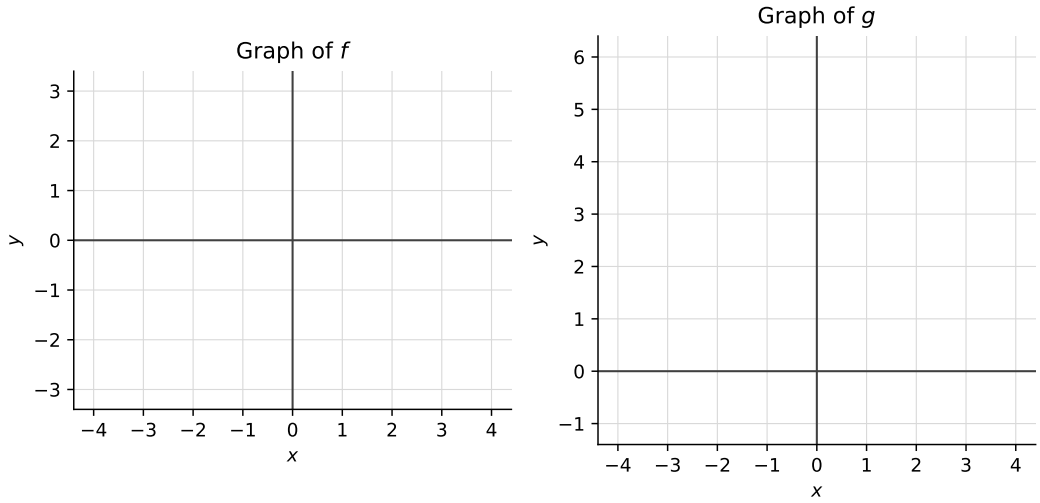


Figure 7.1: Blank axes for sketching f (left) and g (right).

4. Is it continuous everywhere? If not, at which x is it discontinuous?

Record short answers in the table; you will verify each one in the steps that follow.

Property	f	g
Injective? (yes / no)		
Monotone? Strictly? On which interval(s)?		
Even, odd, or neither?		
Continuous everywhere? If not, at which x?		

7.4 Verification: injectivity

Neither f nor g is injective. To prove this, it is enough to exhibit two inputs $x \neq y$ with the same output.

Exercise 7.1. Find explicit values:

- for f: $x =$ ____ and $y =$ ____ with $f(x) = f(y)$;
- for g: $x =$ ____ and $y =$ ____ with $g(x) = g(y)$.

7.5 Verification: even / odd

Exercise 7.2. For whichever symmetry you guessed, verify it directly from the definition of the function: compute $f(-x)$ and check that it equals $-f(x)$ (odd) or $f(x)$ (even). Then do the same for g.

7.6 Verification: monotonicity

Exercise 7.3. Verify your monotonicity guess on a suitable interval, directly from the definition. For instance, to show a function is increasing on an interval, assume $x \leq y$ in that interval and deduce $f(x) \leq f(y)$ (use strict inequalities for *strictly* increasing); argue similarly for decreasing.

7.7 Continuity: the ε–δ game

Both f and g are continuous at $x_0 = 1$. Play the ε–δ game from Definition 6.2.

Exercise 7.4.

1. Take $\varepsilon = 1$. Find a number $\delta > 0$ such that

$$|x - 1| < \delta \quad \Rightarrow \quad |f(x) - f(1)| < 1$$

   (recall $f(1) = 1$). Then find such a δ for g.
2. Repeat with a smaller tolerance, $\varepsilon = \frac{1}{10}$. Does the argument really change?
3. Can you produce a δ that works for an **arbitrary** $\varepsilon > 0$?

7.8 Discontinuities

Both functions are discontinuous at $x_0 = 0$. Here it is *you* who loses the game.

Exercise 7.5. Find an $\varepsilon > 0$ for which **no** δ works — that is, however close x is taken to 0, there is always some such x with

$$|f(x) - f(0)| \geq \varepsilon.$$

Give such an ε for f, and one for g.

Part III

Limits

8 Sequences and Convergence

8.1 Sequences

A **sequence** is an unending list of real numbers

$$a_1,\ a_2,\ a_3,\ \ldots$$

We write $(a_n)_{n\in\mathbb{N}}$, or just (a_n), and call a_n the **n-th term**. Formally, a sequence is nothing but a function

$$a\colon \mathbb{N} \to \mathbb{R},$$

where we write a_n instead of $a(n)$.

Example 8.1.

$$\begin{aligned}
&a_n = \tfrac{1}{n}: && 1,\ \tfrac{1}{2},\ \tfrac{1}{3},\ \tfrac{1}{4},\ \ldots\\
&b_n = \tfrac{(-1)^{n+1}}{n}: && 1,\ -\tfrac{1}{2},\ \tfrac{1}{3},\ -\tfrac{1}{4},\ \ldots\\
&c_n = \tfrac{n}{n+1}: && \tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{3}{4},\ \tfrac{4}{5},\ \ldots\\
&d_n = \sqrt[n]{p}, \quad p>0 \text{ fixed}\\
&e_n = \alpha^n, \quad \alpha > 1.
\end{aligned}$$

8.2 The Limit of a Sequence

Look at $a_n = \frac{1}{n}$. What happens to the terms as n gets larger and larger? The values get smaller and smaller, coming **arbitrarily close to** 0 — though no term is ever equal to 0. We want to make "arbitrarily close" precise.

The idea is to use ε-neighborhoods (Definition 6.1): given a tolerance $\varepsilon > 0$, we ask that *from some point on*, every term of the sequence lies within ε of 0.

Definition 8.1.
We say (a_n) **converges to** 0, written

$$\lim_{n\to\infty} a_n = 0 \qquad \text{or} \qquad a_n \to 0,$$

if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n| = |a_n - 0| < \varepsilon \qquad \text{for all } n \geq N.$$

The number N is allowed to depend on ε — a smaller tolerance will in general require going further out in the sequence. To emphasize this dependence one sometimes writes $N(\varepsilon)$.

Example 8.2.
$\lim_{n\to\infty} \frac{1}{n} = 0$.

Proof. Let $\varepsilon > 0$. We must find N so that $\frac{1}{n} < \varepsilon$ for all $n \geq N$. Now

$$\tfrac{1}{n} < \varepsilon \quad \Leftrightarrow \quad \tfrac{1}{\varepsilon} < n,$$

so any $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$ works: if $n \geq N > \frac{1}{\varepsilon}$, then $\frac{1}{n} < \varepsilon$.

Concretely, for $\varepsilon = 0.05 = \frac{5}{100}$ we need $n > \frac{100}{5} = 20$, so $N = 21$ will do. □

Exercise 8.1. Show that $\lim_{n\to\infty} \dfrac{(-1)^{n+1}}{n} = 0$. (Note $|b_n| = \frac{1}{n}$.)

8.3 Convergence to a General Limit

Now consider $c_n = \frac{n}{n+1}$. The terms get closer and closer to 1, which suggests $\lim_{n\to\infty} c_n = 1$. The definition is the same as before, with 0 replaced by the candidate limit.

Definition 8.2.
Let $a \in \mathbb{R}$. We say (c_n) **converges to** a, written $\lim_{n\to\infty} c_n = a$ or $c_n \to a$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|c_n - a| < \varepsilon \qquad \text{for all } n \geq N.$$

Equivalently: from the N-th term on, all terms lie in the ε-neighborhood $(a-\varepsilon,\, a+\varepsilon)$. A sequence is **convergent** if it converges to some a, and **divergent** otherwise.

Example 8.3.
$\lim_{n\to\infty} \dfrac{n}{n+1} = 1$. Here

$$|c_n - 1| = \left|\frac{n}{n+1} - 1\right| = 1 - \frac{n}{n+1} = \frac{1}{n+1}.$$

Let us check two tolerances. For $\varepsilon = 1$ we need $\frac{1}{n+1} < 1$, which holds for every $n \geq 1$; so $N = 1$ works. For $\varepsilon = \frac{1}{4}$ we need

$$\frac{1}{n+1} < \frac{1}{4} \quad \Leftrightarrow \quad 3 < n,$$

so $N = 4$ works.

In general, $|c_n - 1| = \frac{1}{n+1} < \varepsilon \quad \Leftrightarrow \quad n > \frac{1}{\varepsilon} - 1$, so any $N > \frac{1}{\varepsilon} - 1$ works, and $c_n \to 1$.

Remark 8.1. When a limit exists it is **unique**: if $c_n \to a$ and $c_n \to a'$, then for every $\varepsilon > 0$ we have $|a - a'| \leq |a - c_n| + |c_n - a'| < 2\varepsilon$ for large n, so $|a - a'| = 0$. This is what justifies speaking of *the* limit and the notation $\lim_{n\to\infty} c_n$.

8.3.1 Divergence

To show a sequence **diverges** we must show it converges to no limit at all. Negating Definition 8.2: (c_n) does not converge to a if there is some $\varepsilon > 0$ such that, no matter how far out we go, some later term is at least ε away from a.

Example 8.4.
The sequence $((-1)^n) = -1, 1, -1, 1, \ldots$ diverges.

Proof. Suppose $(-1)^n \to a$ for some a. Take $\varepsilon = 1$. Then $|(-1)^n - a| < 1$ for all large n; applying this to an even and an odd n gives $|1 - a| < 1$ and $|-1 - a| < 1$. But then

$$2 = |1 - (-1)| \leq |1 - a| + |a - (-1)| < 1 + 1 = 2,$$

a contradiction. □

Exercise 8.2. Investigate the remaining sequences from Example 8.1: $d_n = \sqrt[n]{p}$ (with $p > 0$ fixed) and $e_n = \alpha^n$ (with $\alpha > 1$). One converges; the other diverges. Which is which, and what is the limit?

Remark 8.2. For reference: $\sqrt[n]{p} \to 1$ for every $p > 0$, while α^n diverges when $\alpha > 1$ (its terms grow without bound). Both facts are proved in C&J §1.6.

8.4 Limit Laws

Computing $N(\varepsilon)$ by hand for every sequence quickly becomes tedious. The following rules let us build limits of complicated sequences out of simpler ones.

Proposition 8.1 (Arithmetic of limits)**.** *Suppose $a_n \to a$ and $b_n \to b$. Then*

$$a_n + b_n \to a + b, \quad a_n - b_n \to a - b,$$
$$a_n \cdot b_n \to a \cdot b, \qquad \frac{a_n}{b_n} \to \frac{a}{b} \quad (\text{provided } b \neq 0).$$

Example 8.5.
Find $\lim_{n\to\infty} \dfrac{2n^2 - 1}{n^2 + n + 1}$.

Divide numerator and denominator by n^2:

$$\frac{2n^2 - 1}{n^2 + n + 1} = \frac{2 - \frac{1}{n^2}}{1 + \frac{1}{n} + \frac{1}{n^2}}.$$

Now $\frac{1}{n} \to 0$ and $\frac{1}{n^2} \to 0$, so by the limit laws the numerator tends to $2 - 0 = 2$ and the denominator to $1 + 0 + 0 = 1$. Since the denominator limit is nonzero,

$$\lim_{n\to\infty} \frac{2n^2 - 1}{n^2 + n + 1} = \frac{2}{1} = 2.$$

8.5 Comparison and the Squeeze Theorem

Exercise 8.3. Warm-up. Show that $\lim_{n\to\infty} \dfrac{1}{2^n} = 0$.

Rather than hunting for $N(\varepsilon)$ from scratch every time, it is often easier to compare a sequence to one whose limit we already know.

Proposition 8.2 (Comparison test)**.** *Suppose $0 \le b_n \le a_n$ for all n, and $a_n \to 0$. Then $b_n \to 0$.*

Proof. Let $\varepsilon > 0$. Since $a_n \to 0$ there is $N \in \mathbb{N}$ with $a_n < \varepsilon$ for all $n \ge N$. Then for $n \ge N$,

$$0 \le b_n \le a_n < \varepsilon,$$

so $|b_n - 0| = b_n < \varepsilon$. Hence $b_n \to 0$. □

We can generalize this by sandwiching a sequence between two others that converge to the *same* limit works for any target value.

Proposition 8.3 (Squeeze theorem)**.** *Suppose $a_n \le b_n \le c_n$ for all n, and $a_n \to L$, $c_n \to L$ for the same $L \in \mathbb{R}$. Then $b_n \to L$.*

Proof. Let $\varepsilon > 0$. Choose N_1 with $|a_n - L| < \varepsilon$ for $n \ge N_1$, and N_2 with $|c_n - L| < \varepsilon$ for $n \ge N_2$. Let $N = \max(N_1, N_2)$. For $n \ge N$,

$$L - \varepsilon < a_n \le b_n \le c_n < L + \varepsilon,$$

so $|b_n - L| < \varepsilon$. Hence $b_n \to L$. □

Remark 8.3. Proposition 8.2 is the special case of the squeeze theorem where the lower bounding sequence is constantly 0 and $L = 0$.

8.6 Monotone and Bounded Sequences

Definition 8.3.
A sequence (a_n) is **increasing** if $a_n \le a_{n+1}$ for all n, and **decreasing** if $a_n \ge a_{n+1}$ for all n. It is **monotone** if it is increasing or decreasing.

Definition 8.4.
A sequence (a_n) is **bounded** if there is a constant C such that $|a_n| \le C$ for all n.

Theorem 8.1 (Monotone Convergence Theorem)**.** *Every bounded monotone sequence converges.*

Remark 8.4. This theorem expresses the same completeness of $\mathbb{R}$ as the Nested Interval Axiom from The Nested Interval Axiom — each can be derived from the other. We will not reprove the general statement here; instead we isolate the paired form that we actually use below, which follows directly from the axiom together with the comparison test.

Corollary 8.1. *Suppose (a_n) is increasing, (b_n) is decreasing, $a_n \le b_n$ for all n, and $b_n - a_n \to 0$. Then (a_n) and (b_n) converge to a common limit c, the unique point in $\bigcap_n [a_n, b_n]$.*

Proof. Since (a_n) is increasing and (b_n) is decreasing with $a_n \le b_n$, the intervals $[a_n, b_n]$ are nested: $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ for all n. By the Nested Interval Axiom there is a point c with $a_n \le c \le b_n$ for every n. Then

$$0 \le c - a_n \le b_n - a_n \to 0 \qquad \text{and} \qquad 0 \le b_n - c \le b_n - a_n \to 0,$$

so the comparison test (Proposition 8.2) gives $a_n \to c$ and $b_n \to c$. □

8.7 Applications

Lemma 8.1 (Bernoulli's inequality)**.** *For $c \geq -1$ and $n \in \mathbb{N}$,*

$$(1+c)^n \geq 1 + nc.$$

(For $c \geq 0$ one can prove this by considering just the first two terms of the binomial expansion of $(1+c)^n$. The remaining terms are all non-negative, so by leaving them out, we obtain something smaller.)

8.7.1 Roots: $\sqrt[n]{p} \to 1$

Theorem 8.2.
For every $p > 0$, $\lim_{n\to\infty} \sqrt[n]{p} = 1$.

Proof. For $p = 1$ the sequence is constant, so there is nothing to show.

Suppose $p > 1$. Write $a_n := \sqrt[n]{p} = 1 + \delta_n$; since $p > 1$ we have $a_n > 1$, so $\delta_n > 0$. Raising to the n-th power and applying Bernoulli's inequality (with $c = \delta_n \geq 0$),

$$p = (1+\delta_n)^n \geq 1 + n\delta_n,$$

so $0 < \delta_n \leq \dfrac{p-1}{n}$. Since $\frac{p-1}{n} \to 0$, the comparison test gives $\delta_n \to 0$, and hence $a_n = 1 + \delta_n \to 1$.

Suppose $0 < p < 1$. Then $q := 1/p > 1$, and by the case just proved $\sqrt[n]{q} \to 1$. Since $\sqrt[n]{q} \neq 0$ for all n and its limit is nonzero, the quotient rule in Proposition 8.1 gives

$$\sqrt[n]{p} = \frac{1}{\sqrt[n]{q}} \longrightarrow \frac{1}{1} = 1.$$

□

This proves the claim recorded in Remark 8.2 for $d_n = \sqrt[n]{p}$.

8.7.2 The Number e

Proposition 8.4.
The sequence $e_n := \left(1+\frac{1}{n}\right)^n$ *is increasing.*

Proof. We must show

$$\left(1+\frac{1}{n}\right)^n \le \left(1+\frac{1}{n+1}\right)^{n+1}. \tag{$*$}$$

Writing both sides with a common fractional base and splitting off one factor of $\frac{n+2}{n+1}$ on the right,

$$(*) \iff \left(\frac{n+1}{n}\right)^n \le \left(\frac{n+2}{n+1}\right)^n \cdot \frac{n+2}{n+1} \iff \frac{n+1}{n+2} \le \left(\frac{n(n+2)}{(n+1)^2}\right)^n.$$

Since $(n+1)^2 = n^2+2n+1$, we have $n(n+2) = n^2+2n = (n+1)^2-1$, so

$$\left(\frac{n(n+2)}{(n+1)^2}\right)^n = \left(1-\frac{1}{(n+1)^2}\right)^n.$$

By Bernoulli's inequality (with $c = -\frac{1}{(n+1)^2} \ge -1$),

$$\left(1-\frac{1}{(n+1)^2}\right)^n \ge 1-\frac{n}{(n+1)^2}.$$

So it suffices to show $\dfrac{n+1}{n+2} \le 1-\dfrac{n}{(n+1)^2}$, i.e.

$$\frac{1}{n+2} \ge \frac{n}{(n+1)^2} \iff (n+1)^2 \ge n(n+2) \iff n^2+2n+1 \ge n^2+2n,$$

which is always true. Chaining the inequalities back together proves $(*)$. □

Exercise 8.4. Show that $f_n := \left(1+\frac{1}{n}\right)^{n+1}$ is decreasing. (Adapt the computation in the proof of Proposition 8.4.)

Theorem 8.3.
The limit

$$e := \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

exists; numerically $e \approx 2.71828\ldots$.

Proof. Let $e_n = \left(1+\frac{1}{n}\right)^n$ and $f_n = \left(1+\frac{1}{n}\right)^{n+1} = e_n\left(1+\frac{1}{n}\right)$, so $e_n \le f_n$ for all n. By Proposition 8.4, (e_n) is increasing; by Exercise 8.4, (f_n) is decreasing. In particular $e_n \le f_n \le f_1 = 4$ for all n, so (e_n) is bounded.

Moreover

$$0 \le f_n - e_n = e_n \cdot \frac{1}{n} \le \frac{f_1}{n} = \frac{4}{n} \longrightarrow 0,$$

so $f_n - e_n \to 0$ by the comparison test. By Corollary 8.1, (e_n) and (f_n) converge to a common limit — which we call e. Numerically, $e_{10} \approx 2.594$, $e_{100} \approx 2.705$, $e_{1000} \approx 2.7169$, consistent with $e \approx 2.71828\ldots$. □

8.8 Summary

- A **sequence** is a function $a\colon \mathbb{N} \to \mathbb{R}$, written (a_n).
- (a_n) **converges to** a (Definition 8.2) if for every tolerance $\varepsilon > 0$ there is an index N such that $|a_n - a| < \varepsilon$ for all $n \ge N$ — from some point on, all terms lie in the ε-neighborhood of a. The special case $a = 0$ is Definition 8.1.
- The limit, when it exists, is **unique** (Remark 8.1). A sequence with no limit, such as $((-1)^n)$ (Example 8.4), is **divergent**.
- The **limit laws** (Proposition 8.1) compute limits of sums, differences, products, and quotients termwise; dividing through by the highest power of n turns a ratio of polynomials into a form where they apply (Example 8.5).
- The **comparison test** and **squeeze theorem** (Proposition 8.2, Proposition 8.3) let us find a limit by trapping a sequence between two others whose limits we already know.
- A **bounded monotone sequence converges** (2), a fact equivalent to the Nested Interval Axiom; the paired form (Corollary 8.1) — an increasing and a decreasing sequence squeezing together — is the version we use in practice.
- **Bernoulli's inequality** $(1+c)^n \ge 1+nc$ (Lemma 8.1) drives two applications: every $\sqrt[n]{p} \to 1$ (Theorem 8.2), and $\left(1+\frac{1}{n}\right)^n$ increases while $\left(1+\frac{1}{n}\right)^{n+1}$ decreases to the same limit, defining the number $e \approx 2.71828\ldots$ (Theorem 8.3).

9 Convergence of Sequences: Reading Data

```
#
# !!! RUN THIS FIRST !!!
#
# (loads the data and defines all necessary functions)
#


import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option('display.max_rows', 20)  # long tables show head/tail with '...' automatic

df = pd.read_csv('data/rc_circuit_data.csv')

def a(n):
    """Return the n-th term of the sequence, a_n, as a plain float."""
    row = df.loc[df['n'] == n]
    if row.empty:
        raise ValueError(f"n = {n} is out of range (data covers n = 0 .. {df['n'].max()}
    return float(row['measured_value'].values[0])

def find_smallest_N(data, L_guess, epsilon, value_col='measured_value'):
    """Smallest N (present in the data) such that all later points are within epsilon of
    vals = data[value_col].values
    n_vals = data['n'].values
    for i in range(len(vals)):
        if np.all(np.abs(vals[i:] - L_guess) < epsilon):
            return int(n_vals[i])
    return None
```

```
def check_convergence(data, L_guess, N_guess, epsilon, value_col='measured_value', revea
    vals = data[value_col].values
    n_vals = data['n'].values
    mask = n_vals >= N_guess

    if not mask.any():
        print(f"N_guess = {N_guess} is beyond the range of available data (max n = {n_va
        return False

    tail_n = n_vals[mask]
    tail_vals = vals[mask]
    bad = np.abs(tail_vals - L_guess) >= epsilon

    if bad.any():
        first_bad_n = tail_n[bad][0]
        print(f"  Not valid: at n = {first_bad_n}, |a_n - {L_guess}| >= {epsilon}.")
        print("  The sequence leaves the epsilon-band after your N_guess. Try a larger N
              "or reconsider your L_guess.")
        return False

    print(f"  Valid: for all n >= {N_guess}, |a_n - {L_guess}| < {epsilon}.")

    # Don't just hand over the smallest N -- only reveal it once the guess is close.
    best_N = find_smallest_N(data, L_guess, epsilon, value_col)
    if best_N is not None:
        gap = N_guess - best_N
        if gap == 0:
            print("  This is also the smallest N that works for this L_guess.")
        elif gap <= reveal_threshold:
            print(f"  Close! The smallest N that works is N = {best_N} (you were only {g
        else:
            print("  This N works, but it's probably not the smallest one -- "
                  "try a smaller N and see if it still holds.")
    return True


def plot_challenge(data, L_guess, N_guess=None, epsilon=None, value_col='measured_value'
                   zoom=True, buffer=50):
    """
    Plot the sequence with the guessed L (and epsilon band, and N_guess) marked.

    By default (zoom=True), the plot starts `buffer` samples before N_guess and
```

```
    runs to the end of the data -- enough to see whether the sequence was
    outside the band right before N_guess, and to see it stay inside the band
    all the way to the end. Set zoom=False to see the whole sequence instead,
    or pass a bigger `buffer` to look further back.
    """
    plot_data = data
    if zoom and N_guess is not None:
        start = max(int(data['n'].min()), N_guess - buffer)
        mask = data['n'] >= start
        plot_data = data[mask]

    plt.figure(figsize=(8, 4))
    plt.plot(plot_data['n'], plot_data[value_col], '.', alpha=0.6)
    plt.axhline(L_guess, color='green', linestyle='--', label=f'L_guess = {L_guess}')
    if epsilon is not None:
        plt.fill_between(plot_data['n'], L_guess - epsilon, L_guess + epsilon,
                         color='green', alpha=0.15, label=f'epsilon band (+/- {epsilon}
    if N_guess is not None:
        plt.axvline(N_guess, color='red', linestyle=':', label=f'N_guess = {N_guess}')
    plt.xlabel('n')
    plt.ylabel(value_col)
    plt.legend()
    plt.show()
```

You've been handed a log file, `data/rc_circuit_data.csv`, from a lab sensor. It records a voltage measurement `measured_value` at each sample index `n` (and the corresponding time `t_seconds`).

You do not know what circuit or process produced this data. Your job is to:

1. Explore the data and form a hypothesis about its limiting behavior.
2. Given a challenge value ε, **guess** the limit L and the index N after which the data stays within ε of L — then verify your guess rigorously, not just by eye.

Recall the definition:

$$\lim_{n\to\infty} a_n = L \iff \text{for all } \varepsilon > 0, \text{ there is } N \text{ s.t. } n \geq N \implies |a_n - L| < \varepsilon.$$

9.1 A Simple Query Function

Sometimes you just want a single term of the sequence, a_n, without pulling up the whole table. `a(n)` does that.

```
# try it:

a(0), a(5000), a(9999)
```

```
(0.0, 5.0165, 4.9733)
```

9.2 Task 1: Explore the Data

The full log has 10,000 samples. Set `n_start` and `n_end` below to pick a window and re-run the cell to see the table and plot over just that range — e.g. zoom into the early samples to see the initial behavior, or into the tail to see whether it's still moving. Change the numbers and re-run as many times as you like.

```
# EDIT HERE

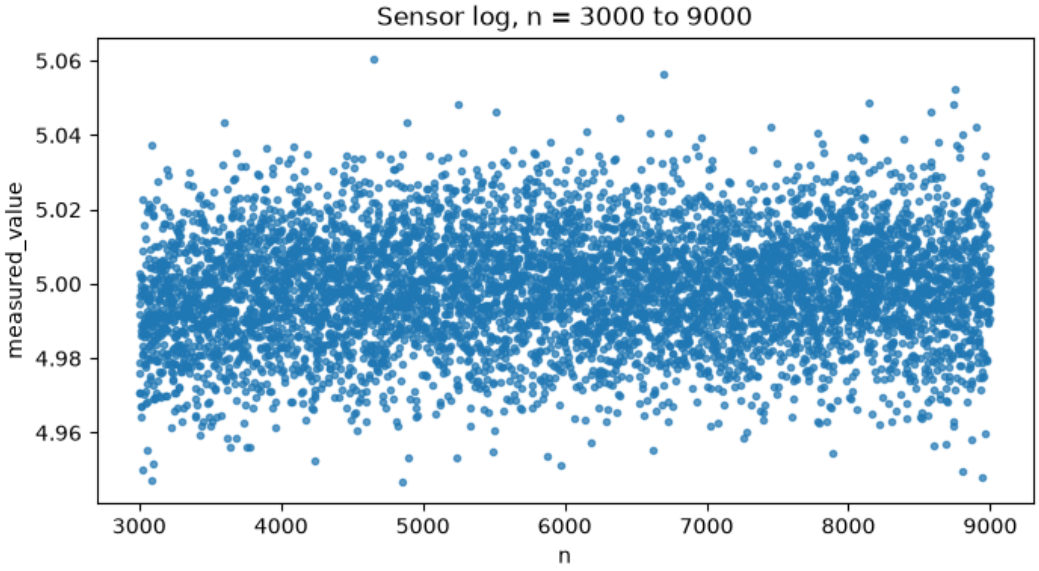
n_start = 3000
n_end = 9000


#
# No need to change anything down here
#
view = df[(df['n'] >= n_start) & (df['n'] <= n_end)]
display(view)

plt.figure(figsize=(8, 4))
plt.plot(view['n'], view['measured_value'], '.', alpha=0.7)
plt.xlabel('n')
plt.ylabel('measured_value')
plt.title(f'Sensor log, n = {n_start} to {n_end}')
plt.show()
```

	n	t_seconds	measured_value
3000	3000	0.3000	4.9917

	n	t_seconds	measured_value
3001	3001	0.3001	4.9966
3002	3002	0.3002	4.9946
3003	3003	0.3003	4.9709
3004	3004	0.3004	5.0027
...	...	...	...
8996	8996	0.8996	5.0257
8997	8997	0.8997	4.9995
8998	8998	0.8998	4.9907
8999	8999	0.8999	4.9952
9000	9000	0.9000	5.0096



9.2.1 Questions to answer before moving on

- What value does the sequence appear to be approaching? Write down your best guess for L.
- Roughly how many samples does it take before the data "settles down" near that value?
- Does the data ever exceed your guessed L, or only approach it from below?
- Try `a(n)` for a few large `n` (e.g. 7000, 8500, 9999) — does it match what the plot suggested?

9.3 Task 2: The ε Challenge

Your instructor assigns a value of ε. Using **only the data** (no formulas, no ground truth), you must:

1. State your guessed limit `L_guess`.
2. State the smallest index `N_guess` you believe works.
3. Check your answer with `check_convergence` below, which verifies directly against the definition: it checks whether **every** remaining data point from `N_guess` onward is within `epsilon` of your `L_guess`.

Note this checks self-consistency with your own guessed L — it does not tell you the "true" limit, because the whole point is that you don't get to see that. Two different reasonable guesses for L can each have a valid N.

The checker also won't just hand you the smallest possible N: if your guess is close (within 50), it'll tell you the exact smallest N; if you're further off, it'll just tell you to try smaller.

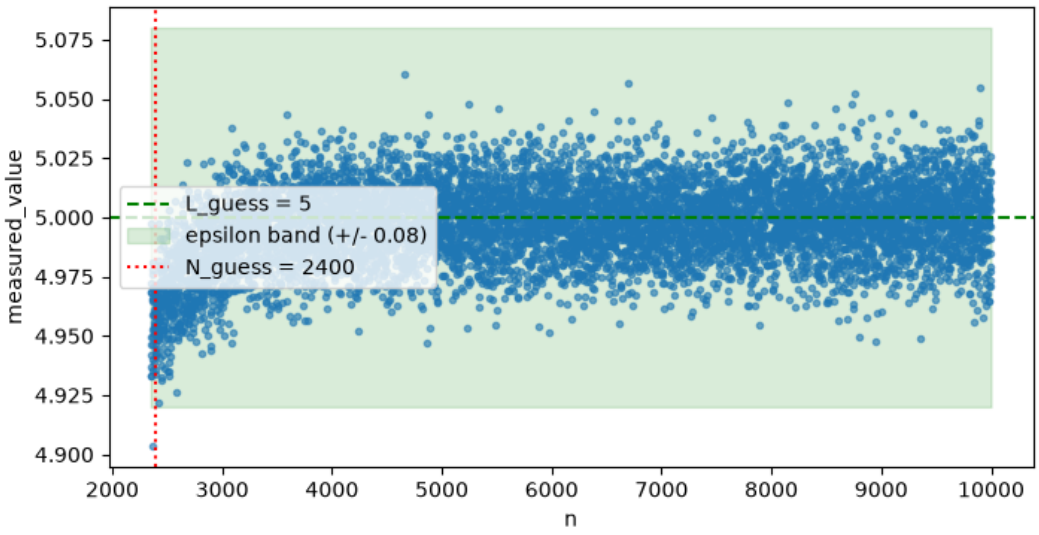
`plot_challenge` zooms in by default: it starts a bit before your `N_guess` and runs to the end of the data, so you can see both whether the sequence was outside the band just before `N_guess` and that it stays inside all the way to the end. Pass `zoom=False` if you want to see the whole sequence instead.

```
# --- Your answer ---
EPSILON = 0.08          # <- instructor-assigned challenge value

L_guess = 5         # TODO: your guessed limit
N_guess = 2400           # TODO: your guessed N

check_convergence(df, L_guess, N_guess, EPSILON)
plot_challenge(df, L_guess, N_guess, EPSILON)
```

```
Valid: for all n >= 2400, |a_n - 5| < 0.08.
 Close! The smallest N that works is N = 2361 (you were only 39 away).
```



9.4 Dataset 2: Damped Oscillator

This time there's no CSV and no noise. Instead, you're given the sequence directly as a **function** you can call for any index `n` — exactly the way a sequence (b_n) is usually presented in a calculus course. It models the position of an underdamped mechanical system (e.g. a suspension, a sensor needle) responding to a step input:

$$b_n = x_{ss}\left(1 - e^{-\zeta\omega_n t_n}\left(\cos(\omega_d t_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin(\omega_d t_n)\right)\right), \qquad t_n = n \cdot dt$$

with $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Unlike Dataset 1, this sequence **oscillates** around its limit before settling — so "getting close once" is not enough; you have to check it *stays* close.

Since it's an exact formula (not a fixed table), you can evaluate it at *any* `n` you like, no matter how large — there's no upper limit like there was with `a(n)`.

```
x_ss = 2.0      # steady-state value
zeta = 0.1      # damping ratio
wn = 10.0       # natural frequency (rad/s)
wd = wn * np.sqrt(1 - zeta**2)
dt2 = 0.001     # seconds per sample
```

```
def b(n):
    """Return the n-th term of the damped step-response sequence, b_n, as a
    plain float (or array of floats if n is array-like). No noise: this is
    the exact closed-form value."""
    scalar_input = np.ndim(n) == 0
    t = np.atleast_1d(np.asarray(n, dtype=float)) * dt2
    vals = x_ss * (1 - np.exp(-zeta * wn * t) *
                   (np.cos(wd * t) + (zeta / np.sqrt(1 - zeta**2)) * np.sin(wd * t)))
    return float(vals[0]) if scalar_input else vals

# try it -- note n can be far larger than anything in Dataset 1's table:
b(0), b(3000), b(1_000_000)
```

```
(0.0, 2.0095601007352615, 2.0)
```

9.5 Task 3: Explore the Sequence

Same idea as Task 1: set `n_start` and `n_end` and re-run to see a table and plot of b_n over that window. Try zooming into the first couple thousand terms to see the oscillation, then look further out to see it settle.

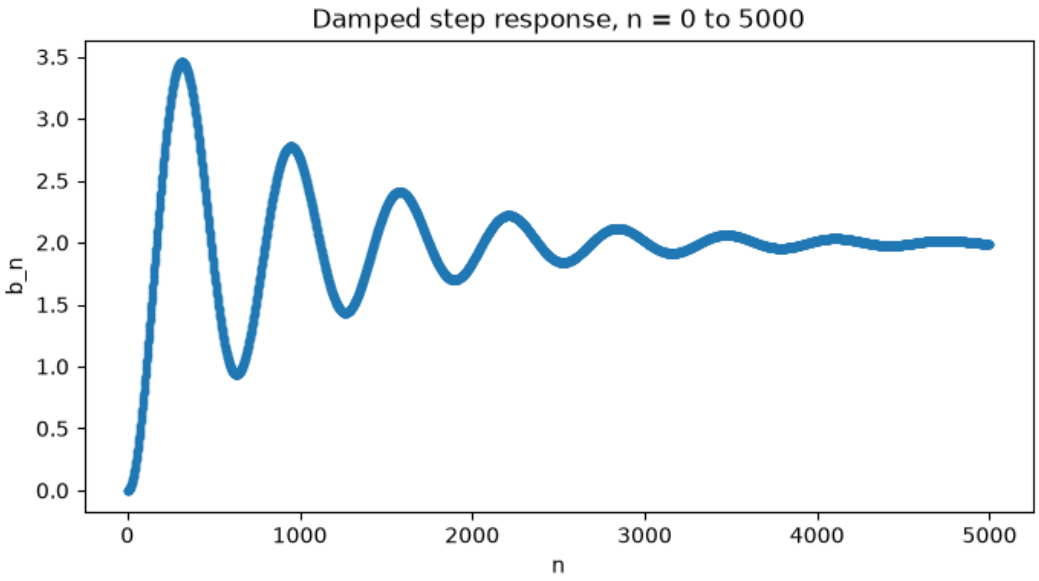
```
n_start = 0
n_end = 5000

n_vals = np.arange(n_start, n_end + 1)
view2 = pd.DataFrame({'n': n_vals, 'b_n': b(n_vals)})
display(view2)

plt.figure(figsize=(8, 4))
plt.plot(view2['n'], view2['b_n'], '.', alpha=0.7)
plt.xlabel('n')
plt.ylabel('b_n')
plt.title(f'Damped step response, n = {n_start} to {n_end}')
plt.show()
```

	n	b_n
0	0	0.000000
1	1	0.000100
2	2	0.000399

	n	b_n
3	3	0.000898
4	4	0.001596
...	...	...
4996	4996	1.989225
4997	4997	1.989154
4998	4998	1.989084
4999	4999	1.989015
5000	5000	1.988948



9.5.1 Questions to answer before moving on

- What value does b_n appear to be converging to? Compare to your guess for Dataset 1 — how did you decide this time, given there's no noise to average out?
- Around how many terms does it take before the oscillation dies down to something you'd call "close" to the limit?
- Pick an `n` where the plot looks close to the limit, then check a later `n` — does it ever swing back out? Why might that matter for choosing `N`?

9.6 Task 4: The ε Challenge, Again

Same process as Task 2 — guess `L_guess` and `N_guess` for a given `epsilon`, and check them. The `check_convergence`, `find_smallest_N`, and `plot_challenge` functions from Task 2 work unchanged here; we just need a table of b_n values to check against. Since b_n is exact (no noise), we can build that table out to a very large `n` — far past where anything interesting happens — so the check is effectively verifying "forever."

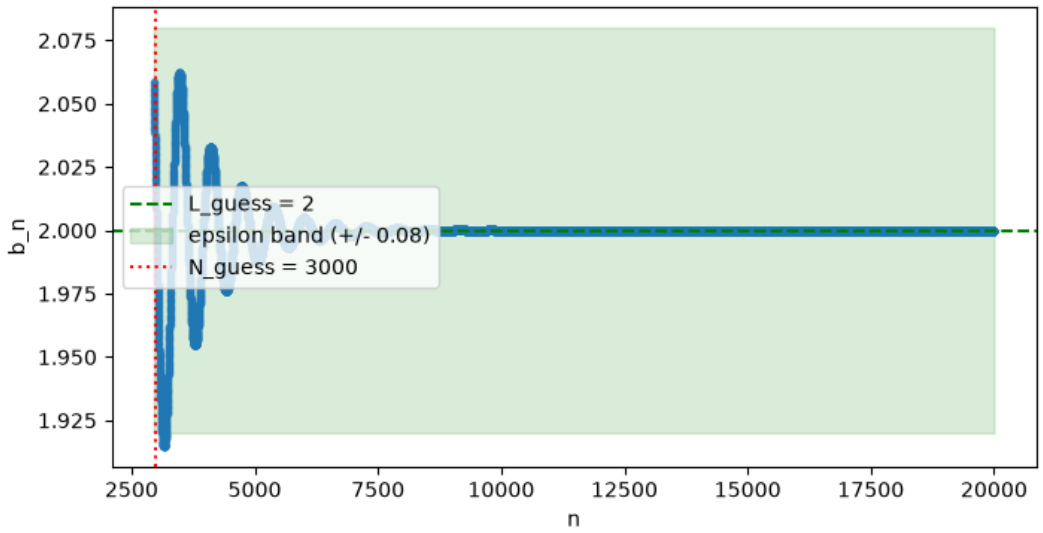
```
n_check = np.arange(0, 20000)      # generous horizon -- well past where b_n settles
osc_df = pd.DataFrame({'n': n_check, 'b_n': b(n_check)})

# --- Your answer ---
EPSILON2 = 0.08          # <- instructor-assigned challenge value

L_guess2 = 2           # TODO: your guessed limit
N_guess2 = 3000          # TODO: your guessed N

check_convergence(osc_df, L_guess2, N_guess2, EPSILON2, value_col='b_n')
plot_challenge(osc_df, L_guess2, N_guess2, EPSILON2, value_col='b_n')
```

```
Not valid: at n = 3124, |a_n - 2| >= 0.08.
 The sequence leaves the epsilon-band after your N_guess. Try a larger N, or reconsider
```



9.7 Task 5: Convergence of $\sqrt[n]{p}$

Finally, we want to study the convergence behavior of $d_n = \sqrt[n]{p}$ for various p. Set the parameters in the cell below.

```
# Set p
p = 2    # Experiment with different p

# Set the range of n you want to consider
d_start = 1
d_end = 100

def d(n):
    return float(p**(1/n))
```

As above, to return just a single term of the sequence, use `d(n)`:

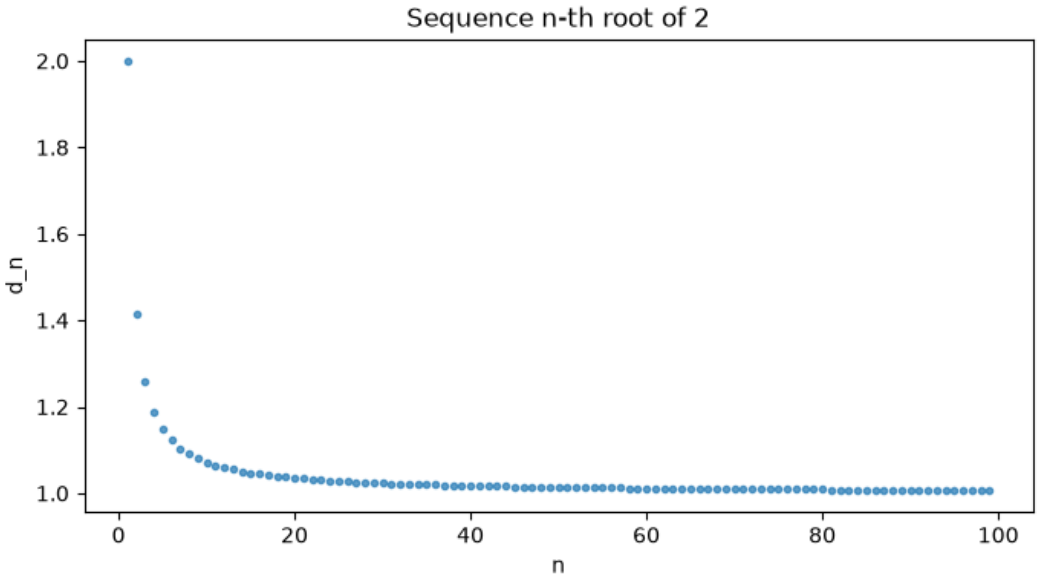
```
d(2)
```

```
1.4142135623730951
```

Now let's plot the sequence:

```
d_array = np.arange(d_start, d_end)
d_vals = p**(1/d_array)
view3 = pd.DataFrame({'n': d_array, 'd_n': d_vals})
# display(view3)

plt.figure(figsize=(8, 4))
plt.plot(view3['n'], view3['d_n'], '.', alpha=0.7)
plt.xlabel('n')
plt.ylabel('d_n')
plt.title(f'Sequence n-th root of {p}')
plt.show()
```



Questions: - Does this sequence converge for all $p > 0$? - If so, does the limit depend on p?

10 Limits of Functions

Companion reading

This lecture corresponds to C&J §1.8 (the concept of limit for a function of a continuous variable), tying together §1.2.d (continuity) with the Intermediate Value Theorem of §1.2.e.

10.1 From Sequences to Functions

We already know what it means for a *sequence* to converge (Definition 8.2). We now transfer this idea to functions: instead of asking what happens to a_n as $n \to \infty$, we ask what happens to $f(x)$ as x approaches some fixed point a. The cheapest way to make this precise is to reduce it to something we already understand — sequences — by feeding *every* sequence approaching a into f and demanding that the outputs all converge to the same value.

Definition 10.1.
Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $a \in \mathbb{R}$. We say

$$\lim_{x \to a} f(x) = L$$

if for **every** sequence (x_n) with $x_n \in D$, $x_n \neq a$, and $x_n \to a$, we have $f(x_n) \to L$.

Remark 10.1. The word "every" is doing all the work: it is not enough to find *some* sequence $x_n \to a$ with $f(x_n) \to L$ — the same L must come out no matter which approaching sequence we feed in. This mirrors uniqueness of sequential limits (Remark 8.1): if two different sequences $x_n \to a$ and $x_n' \to a$ produced different outputs $f(x_n) \to L$ and $f(x_n') \to L' \neq L$, the limit $\lim_{x \to a} f(x)$ would simply not exist.

Example 10.1.
Let $f(x) = x^2$. Then $\lim_{x \to 0} f(x) = 0$.

Proof. Let (x_n) be an arbitrary sequence with $x_n \to 0$. We may assume $|x_n| \le 1$ for all n (true eventually, since $x_n \to 0$). Then

$$0 \le f(x_n) = x_n^2 \le |x_n|,$$

and $|x_n| \to 0$, so by the comparison test (Proposition 8.2), $x_n^2 \to 0$. Since (x_n) was an arbitrary sequence converging to 0, Definition 10.1 gives $\lim_{x\to 0} f(x) = 0$. □

10.2 The ε–δ Version

Just as with continuity (Definition 6.2), the sequential definition of a function limit has an equivalent ε–δ formulation.

Proposition 10.1 (ε–δ characterization of a limit)**.** *Let $f\colon D \to \mathbb{R}$, $a \in \mathbb{R}$, $L \in \mathbb{R}$. Then*

$$\lim_{x\to a} f(x) = L$$

if and only if

for every $\varepsilon > 0$ there exists $\delta > 0$ s.t. $x \in D$, $0 < |x-a| < \delta \quad \Rightarrow \quad |f(x)-L| < \varepsilon$.

The proof is the same idea as the equivalence used implicitly for continuity: if the ε–δ condition holds and $x_n \to a$, choosing N so that $|x_n - a| < \delta$ for $n \ge N$ forces $|f(x_n) - L| < \varepsilon$ for $n \ge N$, i.e. $f(x_n) \to L$. Conversely, if the ε–δ condition *fails* for some $\varepsilon > 0$, then for every $\delta = \frac{1}{n}$ we can pick a point x_n with $0 < |x_n - a| < \frac{1}{n}$ but $|f(x_n) - L| \ge \varepsilon$ — producing a sequence $x_n \to a$ with $f(x_n) \not\to L$, so the sequential definition fails too.

The only difference from Definition 6.2 is the punctured condition $0 < |x - a|$: a limit does not care what happens (or fails to happen) *at* a itself, only nearby. This is what lets us connect the two notions.

Proposition 10.2 (Continuity, restated)**.** *Let $f\colon D \to \mathbb{R}$ and $a \in D$. Then f is continuous at a (Definition 6.2) if and only if*

$$\lim_{x\to a} f(x) \text{ exists and equals } f(a).$$

This is exactly Definition 6.2 with the roles of x and a relabeled: continuity says $|x - a| < \delta \quad \Rightarrow \quad |f(x) - f(a)| < \varepsilon$, which is the ε–δ condition of Proposition 10.1 with $L = f(a)$ (the point $x = a$ itself trivially satisfies $|f(x) - f(a)| = 0 < \varepsilon$, so punctured and unpunctured versions agree once $L = f(a)$).

10.3 Examples

Example 10.2.
$f(x) = x^3$ is continuous at every $a \in \mathbb{R}$.

Proof. Let $x_n \to a$. By the limit laws for sequences (Proposition 8.1),

$$x_n^3 = x_n \cdot x_n \cdot x_n \longrightarrow a \cdot a \cdot a = a^3.$$

Hence $f(x_n) \to f(a)$ for every sequence $x_n \to a$, so $\lim_{x\to a} f(x) = f(a)$, and Proposition 10.2 gives continuity at a. □

Example 10.3.
The **Dirichlet function**

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

is discontinuous at every $a \in \mathbb{R}$.

Take $a = 1$. Consider the two sequences

$$x_n = 1 + \frac{1}{n} \quad \text{(rational)}, \qquad\qquad y_n = 1 + \frac{\sqrt{2}}{n} \quad \text{(irrational)}.$$

Both converge to 1, but

$$f(x_n) \equiv 1 \to 1, \qquad\qquad f(y_n) \equiv 0 \to 0 \neq f(1) = 1.$$

Since two sequences converging to $a = 1$ produce different limits for f, $\lim_{x\to 1} f(x)$ does not exist (Remark 10.1), so f is not continuous at 1.

Exercise 10.1. Adapt the argument of Example 10.3 to an arbitrary $a \in \mathbb{R}$: exhibit a rational sequence and an irrational sequence, both converging to a, on which the Dirichlet function takes different limiting values. Conclude that f is discontinuous at *every* $a \in \mathbb{R}$.

Example 10.4.
Let

$$f(x) = \begin{cases} x^3 + 16 & x < 0 \\ -2x + 3 & x \geq 0. \end{cases}$$

Take $x_n = -\frac{1}{n} \to 0$. Then

$$f(x_n) = \left(-\frac{1}{n}\right)^3 + 16 \longrightarrow 16 \neq f(0) = 3,$$

so f is discontinuous at 0. We will see below exactly what kind of discontinuity this is.

10.4 One-Sided Limits

For Example 10.4, the failure of continuity comes from the two "branches" of f disagreeing at 0 — approaching from the left gives a different answer than approaching from the right. To capture this we restrict which sequences $x_n \to a$ we are allowed to test with.

Definition 10.2.
Let $f\colon D \to \mathbb{R}$ and $a \in \mathbb{R}$.

- $\lim\limits_{x\to a^-} f(x) = L$ (the **left-sided limit**) if $f(x_n) \to L$ for every *increasing* sequence $x_n \nearrow a$ with $x_n \in D$.
- $\lim\limits_{x\to a^+} f(x) = L$ (the **right-sided limit**) if $f(x_n) \to L$ for every *decreasing* sequence $x_n \searrow a$ with $x_n \in D$.

Remark 10.2. The left- and right-sided limits may disagree. For f as in Example 10.4,

$$\lim_{x\to 0^-} f(x) = 16, \qquad \lim_{x\to 0^+} f(x) = 3,$$

since $x^3 + 16 \to 16$ along any $x_n \nearrow 0$, while $-2x + 3 \to 3$ along any $x_n \searrow 0$.

Definition 10.3.
f has a **jump discontinuity** at a if both one-sided limits exist and are finite, but

$$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x).$$

This makes precise the informal description from Remark 6.3, and excludes the unbounded case of Remark 6.4, where no finite one-sided limit exists at all. The function of Example 10.4 has a jump discontinuity at 0.

Theorem 10.1.
Let $f\colon D \to \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if

$$\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x).$$

Like the two-sided limit (Proposition 10.1), each one-sided limit has an equivalent ε–δ form obtained by restricting to one side:

$$\lim_{x\to a^-} f(x) = L \quad \Leftrightarrow \quad \forall \varepsilon > 0\ \exists \delta > 0 :\ a - \delta < x < a \quad \Rightarrow \quad |f(x) - L| < \varepsilon,$$

and symmetrically for $\lim_{x\to a^+} f(x) = L$ with $a < x < a + \delta$.

Proof. If f is continuous at a, then $f(x_n) \to f(a)$ for *every* sequence $x_n \to a$ (Proposition 10.2) — in particular for every increasing sequence $x_n \nearrow a$ and every decreasing sequence $x_n \searrow a$, giving both one-sided limits equal to $f(a)$.

Conversely, suppose both one-sided limits equal $f(a)$, and let $\varepsilon > 0$. By the ε–δ forms above there exist $\delta_1, \delta_2 > 0$ with

$$a-\delta_1 < x < a \quad \Rightarrow \quad |f(x)-f(a)| < \varepsilon, \qquad\qquad a < x < a+\delta_2 \quad \Rightarrow \quad |f(x)-f(a)| < \varepsilon.$$

Let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x-a| < \delta$, then either $a-\delta < x < a$ or $a < x < a+\delta$, so in both cases $|f(x)-f(a)| < \varepsilon$; and $x = a$ trivially gives $|f(x)-f(a)| = 0 < \varepsilon$. So $|x-a| < \delta \quad \Rightarrow \quad |f(x)-f(a)| < \varepsilon$, which is exactly Definition 6.2. □

10.5 Limit Laws for Functions

Because Definition 10.1 reduces everything to sequences, the arithmetic of function limits carries over directly from the arithmetic of sequence limits (Proposition 8.1).

Proposition 10.3 (Arithmetic of function limits)**.** *Suppose* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*. Then*

$$\lim_{x\to a} \big(f(x)+g(x)\big) = L+M, \qquad\qquad \lim_{x\to a} f(x)g(x) = LM,$$

and, if $M \neq 0$*,*

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Corollary 10.1. *If* $a \in \operatorname{dom}(f) \cap \operatorname{dom}(g)$ *and* f, g *are continuous at* a*, then* $f+g$*,* $f \cdot g$*, and (provided* $g(a) \neq 0$*)* f/g *are continuous at* a*.*

Example 10.5.

$$\lim_{x\to 0} \frac{x^3-3x+4}{x^2+\cos(x)} = \frac{\lim_{x\to 0} x^3 - \lim_{x\to 0} 3x + \lim_{x\to 0} 4}{\lim_{x\to 0} x^2 + \lim_{x\to 0} \cos(x)} = \frac{0-0+4}{0+1} = 4.$$

Example 10.6.
Careful: the quotient rule requires the *denominator's* limit to be nonzero — it says nothing when both numerator and denominator vanish. For

$$\lim_{x\to a} \frac{x^2-a^2}{x-a},$$

we have $\lim_{x\to a}(x-a) = 0$, so Proposition 10.3 simply does not apply. Instead, factor first:

$$\frac{x^2-a^2}{x-a} = \frac{(x-a)(x+a)}{x-a} = x+a \qquad (x \neq a),$$

and only *then* take the limit: $\lim_{x\to a}(x+a) = 2a$.

10.6 Continuity of Combinations and Compositions

Proposition 10.2 lets us move freely between "f is continuous at a" and "$\lim_{x\to a} f(x) = f(a)$". The next fact is the one we have already been using informally whenever we wrote something like $\lim_{x\to a} \cos(h(x)) = \cos(\lim_{x\to a} h(x))$: a limit may be passed from the *outside* of a continuous function to its *inside*.

Proposition 10.4 (Limits pass through continuous functions)**.** *Suppose $\lim_{x\to a} f(x) = L$, and g is continuous at L. Then*

$$\lim_{x\to a} g(f(x)) = g(L) = g\Big(\lim_{x\to a} f(x)\Big).$$

Proof. Let (x_n) be an arbitrary sequence with $x_n \in \mathrm{dom}(f)$, $x_n \neq a$, $x_n \to a$. By hypothesis (Definition 10.1), $f(x_n) \to L$. Since g is continuous at L, feeding the sequence $(f(x_n)) \to L$ into Definition 6.2 gives $g(f(x_n)) \to g(L)$. As (x_n) was an arbitrary sequence approaching a, Definition 10.1 gives $\lim_{x\to a} g(f(x)) = g(L)$. □

Taking $L = f(a)$ recovers continuity of a composition:

Corollary 10.2 (Continuity of compositions)**.** *If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a.*

Proof. Since f is continuous at a, Proposition 10.2 gives $\lim_{x\to a} f(x) = f(a)$. Applying Proposition 10.4 with $L = f(a)$,

$$\lim_{x\to a} (g \circ f)(x) = \lim_{x\to a} g(f(x)) = g(f(a)) = (g \circ f)(a),$$

so $g \circ f$ is continuous at a by Proposition 10.2. □

Together with Corollary 10.1 — continuity of $f + g$, $f \cdot g$, and f/g (where $g \neq 0$) from continuity of f and g — Corollary 10.2 means that *any* function built out of continuous building blocks by finitely many arithmetic operations and compositions is again continuous wherever it is defined. This is exactly the justification behind phrases like "combination of continuous functions" or "composition of continuous functions" used to evaluate a limit by plugging in, as in Example 10.5 above.

Remark 10.3. The elementary functions — polynomials, rational functions, power and algebraic functions, exponentials, logarithms, and the trigonometric functions — are continuous at every point of their domain. Each basic type can be checked directly from Definition 6.2, the way we checked x^3 in Example 10.2; Corollary 10.1 and Corollary 10.2 then extend continuity to every function assembled from these by arithmetic and composition, such as $\sqrt{2 + \sqrt{2x}}$ or $\cos(\pi \sin(x)/x)$. Once we

develop differentiation, this will also follow a second way: every elementary function is differentiable wherever it is defined, and differentiability will turn out to imply continuity.

10.7 The Intermediate Value Theorem

Theorem 10.2 (Intermediate Value Theorem)**.** *Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous, and suppose $a < b$ with $f(a) < 0$ and $f(b) > 0$. Then there exists $c \in (a, b)$ with $f(c) = 0$.*

Proof. We construct c by **bisection**. Set $a_1 = a$, $b_1 = b$, so $f(a_1) \le 0 \le f(b_1)$. Given a_n, b_n with $f(a_n) \le 0$ and $f(b_n) \ge 0$, let $m_n = \frac{a_n + b_n}{2}$.

- If $f(m_n) = 0$, we are done — take $c = m_n$.
- If $f(m_n) < 0$, set $a_{n+1} = m_n$, $b_{n+1} = b_n$.
- If $f(m_n) > 0$, set $a_{n+1} = a_n$, $b_{n+1} = m_n$.

In either of the last two cases $f(a_{n+1}) \le 0 \le f(b_{n+1})$, so the invariant is preserved, and we continue. If the process never terminates, this produces sequences with (a_n) increasing, (b_n) decreasing, $a_n \le b_n$ for all n, and

$$b_n - a_n = \frac{b - a}{2^{n-1}} \longrightarrow 0$$

(the interval length halves at each step; compare Exercise 8.3). By Corollary 8.1, (a_n) and (b_n) converge to a common limit $c \in [a, b]$.

Since f is continuous, Proposition 10.2 gives $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$. Now $f(a_n) \le 0$ for every n; if we had $f(c) > 0$, taking $\varepsilon = f(c)/2$ in the definition of convergence would force $f(a_n) > f(c)/2 > 0$ for large n, a contradiction. So $f(c) \le 0$. The same argument with $f(b_n) \ge 0$ gives $f(c) \ge 0$. Hence $f(c) = 0$, and $a < c < b$ since $f(a) < 0$ and $f(b) > 0$ rule out $c = a$ or $c = b$. □

Remark 10.4. The same bisection idea proves the more general statement: if f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$ — apply Theorem 10.2 to $f(x) - y$.

10.8 Summary

- $\lim\limits_{x \to a} f(x) = L$ (Definition 10.1) means $f(x_n) \to L$ for **every** sequence $x_n \to a$ in $\mathrm{dom}(f)$ — this reduces function limits to sequence limits, and is equivalent to the ε–δ condition of Proposition 10.1.

- **Continuity restated:** f is continuous at a exactly when $\lim_{x\to a} f(x)$ exists and equals $f(a)$ (Proposition 10.2).
- **One-sided limits** (Definition 10.2) test only monotone sequences approaching from one side. A **jump discontinuity** (Definition 10.3) is exactly a point where both one-sided limits exist (finitely) but disagree; f is continuous at a iff both one-sided limits equal $f(a)$ (Theorem 10.1).
- The **limit laws** (Proposition 10.3) compute limits of sums, products, and quotients termwise, and give continuity of $f+g$, $f\cdot g$, f/g from continuity of f and g (Corollary 10.1) — but a $\frac{0}{0}$ form must be simplified *before* the quotient rule applies (Example 10.6).
- **Compositions of continuous functions are continuous** (Corollary 10.2), and more generally a limit may be passed through a continuous function from the outside in (Proposition 10.4). Together with the arithmetic rule above, this shows every **elementary function** is continuous where defined (Remark 10.3).
- The **Intermediate Value Theorem** (Theorem 10.2): a continuous function that is negative at a and positive at b must vanish somewhere in between. The proof bisects $[a,b]$ into nested intervals shrinking to a point c, then uses continuity to pass $f(a_n)\le 0\le f(b_n)$ to the limit.

Part IV

The Riemann Integral

11 Riemann Sums

11.1 Area of the Circle

We start with a problem from antiquity: what is the area of a circle of radius 1? The classical idea (going back to Archimedes) is to approximate the circle by figures whose area we *can* compute — regular n-gons inscribed in the circle.

Split the n-th polygon ($n \geq 3$) into n congruent triangles, each with its apex at the center of the circle, two sides of length 1, and apex angle $\frac{2\pi}{n}$. Taking one of the two unit sides as base, the height of the triangle is $h = \sin(\frac{2\pi}{n})$, so

$$\text{area of one triangle} = \frac{1}{2} \cdot 1 \cdot h = \frac{\sin(\frac{2\pi}{n})}{2},$$

$$A_n := \text{area of the } n\text{-gon} = \frac{n}{2} \sin\Big(\frac{2\pi}{n}\Big).$$

As n grows, the polygons hug the circle more and more tightly, and A_n should approximate the area of the circle better and better.

Formally, we can say two things about the sequence (A_n):

- (A_n) is **increasing**.
- (A_n) is **bounded**: every inscribed n-gon is contained in the square of side length 2 circumscribing the circle, so $A_n \leq 4$ for all n.

By the Monotone Convergence Theorem (2), (A_n) converges. Its limit is π.

Remark 11.1. Two points are glossed over here. Monotonicity is not obvious from the picture, since the n-gon is not contained in the $(n+1)$-gon; it follows from the fact that $\sin(x)/x$ is decreasing on $(0, \pi]$, writing $A_n = \pi \cdot \dfrac{\sin(2\pi/n)}{2\pi/n}$. The same rewriting shows that identifying the limit as π amounts to knowing $\lim_{x\to 0} \frac{\sin x}{x} = 1$ — a limit we will return to when we study derivatives of the trigonometric functions.

How can we generalize this idea to other areas? Triangles could work, but it is not clear how to set this up *methodically* for an arbitrary region. For regions under the graph of a function, there is a better choice: **rectangles**, whose area is as easy to compute as it gets.

11.2 Approximation of Areas by Rectangles

Whatever approximation scheme we use, we want it to have the following properties:

- It uses a geometric figure whose area is easily computable. Rectangles are as easy as it gets.
- There is a **general method** to approximate an area by rectangles.
- There is a **general method** to **refine** any approximation.

11.2.1 Areas under Curves

Let $f\colon [a,b] \to \mathbb{R}$ be continuous with $f(x) \geq 0$ for all $x \in [a,b]$. We want to compute

$A :=$ area between the graph of f and the x-axis, between $x = a$ and $x = b$.

The idea is as follows.

- **Partition** $[a,b]$ into finitely many segments

$$a = x_0 < x_1 < \cdots < x_n = b.$$

  The segments $[x_{i-1}, x_i]$ will form the *bases* of rectangles.
- For each segment $[x_{i-1}, x_i]$, pick a point $x_i^* \in [x_{i-1}, x_i]$. The *height* of the i-th rectangle is $f(x_i^*)$.

If R_i denotes the area of the i-th rectangle, then

$$R_i = |x_i - x_{i-1}| \cdot f(x_i^*) \qquad (i = 1, \ldots, n),$$

and $R_1 + \cdots + R_n$ is an approximation to A.

Definition 11.1.
A finite set of points

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b$$

is called a **partition** of $[a,b]$. Points $S = \{x_1^*, \ldots, x_n^*\}$ with $x_i^* \in [x_{i-1}, x_i]$ are called **sample points**, and

$$\Delta_i = |x_i - x_{i-1}| = x_i - x_{i-1} \qquad (i = 1, \ldots, n)$$

is the length of the i-th segment. The **Riemann sum** for f on $[a,b]$, given P and S, is

$$RS(P,S) = f(x_1^*)\,\Delta_1 + f(x_2^*)\,\Delta_2 + \cdots + f(x_n^*)\,\Delta_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta_i.$$

The symbol $\sum$ is **summation** (or Σ-) **notation**: $\sum_{i=1}^{n} c_i$ stands for $c_1 + c_2 + \cdots + c_n$. Note that Definition 11.1 makes sense for *any* function f on $[a, b]$ — neither continuity nor $f \geq 0$ is needed to *form* the sum. (If f takes negative values, the corresponding terms simply count as negative area.)

Refinement. How do we make the approximation better? We *refine the partition* — add more division points so that the segments get shorter — and choose new sample points. The goal is to find a way to do this that is both *methodological* and *practical*. The natural requirement is that the length of **every** segment tends to 0. It is convenient to measure this by the length of the longest segment,

$$\|P\| := \max_{1 \leq i \leq n} \Delta_i,$$

called the **mesh** of P. (Other authors call it the *span*.)

11.3 Equipartitions and the Definition of the Integral

It turns out the approximation process is remarkably robust — several approaches are possible, and they all give the same answer. The most practical one restricts to a special kind of partition.

Equipartitions. All segments have the *same* length, so only the number of segments matters. With n segments,

$$\Delta_i = \frac{b-a}{n} =: \Delta \quad \text{for all } i, \qquad\qquad x_i = a + i\Delta.$$

There are three standard ways to choose sample points:

- **left endpoints:** $x_i^* = x_{i-1}$,
- **right endpoints:** $x_i^* = x_i$,
- **midpoints:** $x_i^* = \bar{x}_i := \dfrac{x_{i-1} + x_i}{2}$.

This gives three sequences of Riemann sums:

$$LHS_n = \sum_{i=1}^{n} f(x_{i-1})\,\Delta, \qquad RHS_n = \sum_{i=1}^{n} f(x_i)\,\Delta, \qquad MS_n = \sum_{i=1}^{n} f(\bar{x}_i)\,\Delta,$$

the **left-hand sum**, the **right-hand sum**, and the **midpoint sum**.

Theorem 11.1.
If f is continuous on $[a, b]$, then all three limits

$$\lim_{n\to\infty} LHS_n, \qquad \lim_{n\to\infty} RHS_n, \qquad \lim_{n\to\infty} MS_n$$

exist and are equal.

Definition 11.2.
Let f be continuous on $[a, b]$. The **(Riemann) integral** of f from a to b is the common value L of the limits in Theorem 11.1:

$$\int_a^b f(x)\,dx = L = \lim_{n\to\infty} LHS_n = \lim_{n\to\infty} RHS_n = \lim_{n\to\infty} MS_n.$$

The notation is Leibniz's: the integral sign $\int$ is an elongated S for "sum", and dx recalls the lengths Δ. It is only notation — dx on its own has no meaning here. Likewise, the name of the variable is irrelevant: $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$.

11.4 Why the Choice Does Not Matter (optional)

Theorem 11.1 is a special case of a much stronger statement: for a continuous function it does not matter at all *how* we partition or *where* we sample, as long as the segments get uniformly short.

Theorem 11.2 (Existence of the integral)**.** *Let f be continuous on $[a, b]$. There is a unique number I with the following property: for **every** sequence of partitions $P_1, P_2, \ldots$ of $[a, b]$ with $\|P_n\| \to 0$, and **every** choice of sample points S_n for P_n,*

$$\lim_{n\to\infty} RS(P_n, S_n) = I.$$

Since an equipartition into n segments has mesh $\frac{b-a}{n} \to 0$, Theorem 11.1 follows at once, with $\int_a^b f(x)\,dx = I$. We now sketch *why* Theorem 11.2 is true. The argument is informal in two places, flagged in Remark 11.2 below.

11.4.1 Step 1: Trapping Every Riemann Sum

A continuous function on a closed interval attains a maximum and a minimum there. (This is the *Extreme Value Theorem*, which we have not proved yet.) So for a partition P we can put

$$m_i = \min\{f(x) : x_{i-1} \le x \le x_i\}, \qquad M_i = \max\{f(x) : x_{i-1} \le x \le x_i\},$$

and form the **lower** and **upper (Darboux) sums**

$$L(P) = \sum_{i=1}^{n} m_i\,\Delta_i, \qquad\qquad U(P) = \sum_{i=1}^{n} M_i\,\Delta_i.$$

Geometrically (for $f \geq 0$), $L(P)$ is the area of the largest rectangles on the segments that fit *under* the graph, and $U(P)$ is the area of the smallest rectangles that reach *over* it.

Whatever sample points x_i^* we choose, $m_i \leq f(x_i^*) \leq M_i$, so

$$L(P) \;\leq\; RS(P,S) \;\leq\; U(P) \qquad \text{for every choice of sample points } S.$$

In other words, $L(P)$ and $U(P)$ are the *worst cases* over all choices of sample points. If we can show that they are close to each other, the choice of sample points becomes irrelevant.

11.4.2 Step 2: Refining Moves the Sums Toward Each Other

Suppose P' is **finer** than P, i.e. P' contains all points of P and possibly more. Adding a single point z inside a segment $[x_{i-1}, x_i]$ splits it into $[x_{i-1}, z]$ and $[z, x_i]$. The minimum of f over each smaller piece is *at least* the minimum m_i over the whole segment, and the maximum over each piece is *at most* M_i. Adding points one at a time, we get

$$L(P) \leq L(P') \qquad \text{and} \qquad U(P') \leq U(P).$$

So refining can only push the lower sum up and the upper sum down.

This has a useful consequence. Given *any* two partitions P and Q, the partition $P \cup Q$ is finer than both, so

$$L(P) \;\leq\; L(P \cup Q) \;\leq\; U(P \cup Q) \;\leq\; U(Q).$$

That is: **every lower sum is at most every upper sum**, no matter which partitions they come from. All lower sums sit to the left of all upper sums on the number line, and by the completeness of $\mathbb{R}$ (compare the Nested Interval Axiom) there is at least one number I in between:

$$L(P) \leq I \leq U(Q) \qquad \text{for all partitions } P, Q.$$

11.4.3 Step 3: Closing the Gap

The gap between upper and lower sum is

$$U(P) - L(P) = \sum_{i=1}^{n} (M_i - m_i)\,\Delta_i.$$

Here continuity enters. On a very short segment a continuous function cannot vary much. Suppose we know the following: for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - y| < \delta \quad \Rightarrow \quad |f(x) - f(y)| < \varepsilon \qquad \text{for all } x, y \in [a, b]. \tag{11.1}$$

Then, as soon as $\|P\| < \delta$, any two points in the same segment are within δ of each other, so $M_i - m_i < \varepsilon$ for every i, and

$$U(P) - L(P) < \sum_{i=1}^{n} \varepsilon \, \Delta_i = \varepsilon \, (b - a).$$

Since ε was arbitrary, the gap can be made as small as we like by making the mesh small.

11.4.4 Step 4: Conclusion

Now let $\|P_n\| \to 0$ and let S_n be any sample points. By Steps 1 and 2, both $RS(P_n, S_n)$ and I lie in the interval $[L(P_n), U(P_n)]$, so

$$|RS(P_n, S_n) - I| \leq U(P_n) - L(P_n).$$

By Step 3 the right-hand side tends to 0, so $RS(P_n, S_n) \to I$ (by the comparison test, Proposition 8.2). The number I is unique: any other I' trapped between all lower and all upper sums satisfies $|I - I'| \leq U(P_n) - L(P_n) \to 0$ as well. Nothing in this argument depended on *which* partitions or sample points we used — only on the mesh going to 0.

Remark 11.2. The sketch uses three facts we have not yet established:

1. the **Extreme Value Theorem** (continuous functions attain a max and min on closed intervals), used to define m_i and M_i;
2. the existence of a number I between all lower and all upper sums, which is a consequence of the completeness of $\mathbb{R}$;
3. condition Equation 11.1, called **uniform continuity**.

The third point is the subtle one. It looks like Definition 6.2, but the quantifiers are in a different order: in Definition 6.2, δ is chosen *after* the point a, so it may depend on a; in Equation 11.1, one δ must work *simultaneously for all points* of $[a, b]$. For instance, $f(x) = \frac{1}{x}$ is continuous on $(0, 1]$, but near 0 it becomes steeper and steeper, and no single δ works for a given ε. On a **closed, bounded** interval this cannot happen: a continuous $f\colon [a, b] \to \mathbb{R}$ is automatically uniformly continuous. Like the Extreme Value Theorem, this can be proved with a bisection / nested interval argument in the spirit of the proof of Theorem 10.2.

Remark 11.3. Continuity really matters. For the Dirichlet function of Example 10.3 on $[0, 1]$, every segment of every partition contains both rational and irrational points. Choosing all sample points rational gives $RS(P, S) = \sum_i 1 \cdot \Delta_i = 1$; choosing them all irrational gives $RS(P, S) = 0$ — for *every* partition, however fine. So the Riemann sums have no common limit, and the Dirichlet function is not Riemann integrable.

11.5 Verifying the Riemann Integral

Before computing anything new, let us check that Definition 11.2 gives the right answer for areas we already know from geometry. By Theorem 11.1 we may use whichever of the three sums is most convenient. Throughout, $\Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$.

Example 11.1.
Let $f(x) = c$. Every sample point gives height c, so

$$LHS_n = RHS_n = \sum_{i=1}^{n} \frac{b-a}{n} \cdot c = n \cdot \frac{b-a}{n} \cdot c = (b-a)\,c,$$

and hence

$$\int_a^b c\,dx = \lim_{n\to\infty} LHS_n = \lim_{n\to\infty} (b-a)\,c = (b-a)\,c,$$

the area of a rectangle with base $b-a$ and height c.

Example 11.2.
Let $f(x) = x$. Using right endpoints $x_i = a + \frac{b-a}{n}\,i$,

$$\begin{aligned} RHS_n &= \sum_{i=1}^{n} \frac{b-a}{n} \cdot f\Big(a + \frac{b-a}{n}\,i\Big) = \frac{b-a}{n} \sum_{i=1}^{n} \Big(a + \frac{b-a}{n}\,i\Big) \\ &= \frac{b-a}{n}\Big[\Big(a + \frac{b-a}{n}\cdot 1\Big) + \Big(a + \frac{b-a}{n}\cdot 2\Big) + \cdots + \Big(a + \frac{b-a}{n}\cdot n\Big)\Big] \\ &= \frac{b-a}{n}\Big(n\,a + \frac{b-a}{n}\cdot\frac{n(n+1)}{2}\Big) \\ &= (b-a)\,a + \frac{b-a}{n}\cdot\frac{b-a}{n}\cdot\frac{n(n+1)}{2} \\ &= (b-a)\,a + \frac{(b-a)^2}{2}\cdot\frac{n+1}{n} \xrightarrow{n\to\infty} (b-a)\,a + \frac{(b-a)^2}{2}. \end{aligned}$$

Here we used $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ and $\frac{n+1}{n} \to 1$.

For $0 \le a < b$ this is exactly what geometry predicts: the region under $y = x$ between a and b is a rectangle of base $b-a$ and height a (area $(b-a)\,a$), topped by a right triangle with both legs of length $b-a$ (area $\frac{(b-a)^2}{2}$). Simplifying,

$$\int_a^b x\,dx = (b-a)\Big(a + \frac{b-a}{2}\Big) = \frac{(b-a)(b+a)}{2} = \frac{b^2-a^2}{2}.$$

Exercise 11.1. Verify this for an arbitrary linear function $f(x) = mx + c$: compute $\lim_{n\to\infty} RHS_n$ and check that the result agrees with the area of the trapezoid under the graph (when $f \ge 0$ on $[a,b]$).

11.6 "New" Functions: x^2

The examples above only reproduced areas we already knew. The first genuinely new area is the one under a parabola. Let $f(x) = x^2$. Then

$$RHS_n = \frac{b-a}{n} \sum_{i=1}^{n} f\Big(a + \frac{b-a}{n}\, i\Big) = \frac{b-a}{n} \sum_{i=1}^{n} \Big(a + \frac{b-a}{n}\, i\Big)^2.$$

To make life a little easier, take $a = 0$:

$$\begin{aligned} RHS_n &= \frac{b}{n} \sum_{i=1}^{n} \Big(\frac{b}{n}\, i\Big)^2 = \frac{b}{n} \cdot \frac{b^2}{n^2} \left(1^2 + 2^2 + \cdots + n^2\right) \\ &= \frac{b^3}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{b^3}{6}\Big(1 + \frac{1}{n}\Big)\Big(2 + \frac{1}{n}\Big) \xrightarrow{n\to\infty} \frac{b^3 \cdot 2}{6} = \frac{b^3}{3}. \end{aligned}$$

So

$$\int_0^b x^2\, dx = \frac{b^3}{3}.$$

Exercise 11.2. Prove by induction that $1^2 + 2^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Exercise 11.3. Generalize the computation above to show that for arbitrary $a < b$,

$$\int_a^b x^2\, dx = \frac{b^3 - a^3}{3}.$$

(Expand $\left(a + \frac{b-a}{n}\, i\right)^2$ and use the formulas for $\sum i$ and $\sum i^2$.)

11.7 Summary

- The area of the unit circle is the limit of the areas $A_n = \frac{n}{2} \sin(\frac{2\pi}{n})$ of inscribed regular n-gons; the sequence is increasing and bounded, so it converges by

  2.

- For areas under a graph we use **rectangles**: a **partition** P of $[a, b]$ and **sample points** S give the **Riemann sum** $RS(P, S) = \sum_{i=1}^{n} f(x_i^*)\, \Delta_i$ (Definition 11.1). We improve the approximation by refining the partition so that the **mesh** $\|P\| = \max_i \Delta_i$ tends to 0.

- Restricting to **equipartitions** with left endpoints, right endpoints, or midpoints gives three sequences LHS_n, RHS_n, MS_n. For continuous f they have a common limit (Theorem 11.1), which *defines* $\int_a^b f(x)\,dx$ (Definition 11.2).
- More generally, for continuous f **every** sequence of Riemann sums with mesh $\to 0$ converges to the same number (Theorem 11.2). The idea: every Riemann sum is trapped between the **lower and upper sums** $L(P) \le RS(P,S) \le U(P)$; refining pushes these toward each other; and **uniform continuity** makes the gap $U(P) - L(P)$ smaller than $\varepsilon(b-a)$ once the mesh is small (Remark 11.2). Without continuity this can fail, as the Dirichlet function shows (Remark 11.3).
- The definition reproduces known areas — $\int_a^b c\,dx = (b-a)c$ (Example 11.1) and $\int_a^b x\,dx = \frac{b^2-a^2}{2}$ (Example 11.2) — and computes new ones: $\int_0^b x^2\,dx = \frac{b^3}{3}$.

12 Lab: Riemann Sums

Math 140H, Honors Calculus I

In class we approximated the area under a curve by rectangles. This notebook lets you build those rectangles, look at them, and watch the Riemann sums approach the integral as the partition gets finer.

How to use this notebook - Cells marked IN CLASS are the core we will do together. - Cells marked HOMEWORK extend the same ideas further. - Cells marked PREDICT ask you to write down a guess before running the next code cell. Do not skip these; the point is to notice when your intuition is wrong.

START BY RUNNNING THE NEXT CELL, this will load the necessary libraries and define all Python functions needed.

```
#
#
# ALWAYS RUN THIS CELL FIRST!!
#
#

import numpy as np
import sympy as sp
import matplotlib.pyplot as plt
from matplotlib.patches import Patch
from IPython.display import display, Math

plt.rcParams['figure.figsize'] = (8, 4.5)
x = sp.Symbol('x')

_METHODS = {
    'left': 'left', 'l': 'left', 'lefthand': 'left', 'left-hand': 'left',
    'right': 'right', 'r': 'right', 'righthand': 'right', 'right-hand': 'right',
    'midpoint': 'midpoint', 'mid': 'midpoint', 'm': 'midpoint', 'middle': 'midpoint',
}
```

```
def _to_exact(v):
    """Turn a number into an exact sympy number (0.25 -> 1/4, pi stays pi)."""
    v = sp.sympify(v)
    if isinstance(v, sp.Float):
        v = sp.nsimplify(v, rational=True)
    return v


def _fmt(v, digits=6):
    """Format a float compactly, e.g. 0.0625, 1.41421, -3.2e-05."""
    s = f'{float(v):.{digits}g}'
    if 'e' in s:
        mant, ex = s.split('e')
        s = rf'{mant} \cdot 10^{{{int(ex)}}}'
    return s


def _join_terms(terms, n, max_terms):
    """Join LaTeX terms with + and - signs; if n is large, abbreviate with + ... +."""
    if n > max_terms:
        terms = terms[:3] + [r'\cdots'] + terms[-2:]
    out = terms[0]
    for t in terms[1:]:
        if t.startswith('-'):
            out += ' - ' + t[1:].lstrip()
        else:
            out += ' + ' + t
    return out


def riemann_sum(f, a, b, n, method='left', x=None, show=True,
                show_integral=False, max_terms=6):
    """
    Compute (and by default draw) a Riemann sum for f on [a, b].

    Parameters
    ----------
    f : sympy expression in one variable, e.g. x**2 or sp.sin(x)
    a, b : the integration bounds, a < b (numbers or sympy numbers like sp.pi)
    n : number of subintervals of equal length
    method : 'left', 'right', or 'midpoint'
    x : the variable of f (only needed if f has more than one symbol)
```

```
    show : if False, just return the value (no picture, no computation lines)
    show_integral : also display the exact integral and the error of the sum
    max_terms : largest n for which every term of the sum is written out;
                for larger n the sum is abbreviated with "+ ... +"

    Returns
    -------
    The value of the Riemann sum as a float.
    """
    # ---- check and prepare the input ---------------------------------
    method = _METHODS.get(str(method).lower().replace(' ', ''))
    if method is None:
        raise ValueError("method must be 'left', 'right', or 'midpoint'")
    n = int(n)
    if n < 1:
        raise ValueError('n must be a positive integer')
    f = sp.sympify(f)
    if x is None:
        syms = sorted(f.free_symbols, key=str)
        if len(syms) > 1:
            raise ValueError(f'f has several symbols {syms}; pass the variable as x=...'
        x = syms[0] if syms else sp.Symbol('x')
    a, b = _to_exact(a), _to_exact(b)
    if not (sp.N(a) < sp.N(b)):
        raise ValueError('need a < b')

    # ---- the partition and the sample points -------------------------
    delta = (b - a) / n                    # exact length of each subinterval
    shift = {'left': 0, 'right': 1, 'midpoint': sp.Rational(1, 2)}[method]
    offset = {'left': 1, 'right': 0, 'midpoint': sp.Rational(1, 2)}[method]
    # sample point x_i^* = a + (i - offset) * delta, for i = 1, ..., n

    f_num = sp.lambdify(x, f, 'numpy')
    a_f, b_f, d_f = float(a), float(b), float(delta)
    left_edges = a_f + d_f * np.arange(n)
    samples = left_edges + float(shift) * d_f
    with np.errstate(all='ignore'):
        heights = np.broadcast_to(np.asarray(f_num(samples), dtype=float), samples.shape
    if not np.all(np.isfinite(heights)):
        bad = samples[~np.isfinite(heights)][0]
        raise ValueError(f'f is not defined (or infinite) at the sample point {bad:g}; '
                         'try a different method or interval')
```

```
    value = float(np.sum(heights) * d_f)
    if abs(value) < 1e-12 * max(1.0, float(np.sum(np.abs(heights)) * d_f)):
        value = 0.0          # the exact answer is 0; the rest is rounding noise

    if not show:
        return value

    # ---- the picture ----------------------------------------------------
    fig, ax = plt.subplots()
    colors = np.where(heights >= 0, 'tab:blue', 'tab:red')
    edge = 'black' if n <= 100 else 'none'
    ax.bar(left_edges, heights, width=d_f, align='edge', color=colors,
           alpha=0.35, edgecolor=edge, linewidth=0.8)
    xs = np.linspace(a_f, b_f, 1000)
    with np.errstate(all='ignore'):
        ys = np.array(np.broadcast_to(np.asarray(f_num(xs), dtype=float), xs.shape))
    ys[~np.isfinite(ys)] = np.nan
    ax.plot(xs, ys, color='black', linewidth=2, label=f'$f({x}) = {sp.latex(f)}$')
    if n <= 60:
        ax.plot(samples, heights, 'o', color='darkorange', markersize=5,
                label=r'$(x_i^*, f(x_i^*))$')
    ax.axhline(0, color='gray', linewidth=0.8)
    pad = 0.05 * (b_f - a_f)
    ax.set_xlim(a_f - pad, b_f + pad)
    # if f blows up somewhere, keep the rectangles in view instead of the spike
    h_lo, h_hi = min(0.0, heights.min()), max(0.0, heights.max())
    span = h_hi - h_lo
    y_lo, y_hi = np.nanmin(ys), np.nanmax(ys)
    if span > 0 and (y_hi > h_hi + 2 * span or y_lo < h_lo - 2 * span):
        lo, hi = max(y_lo, h_lo - 0.5 * span), min(y_hi, h_hi + 0.5 * span)
        ax.set_ylim(min(lo, h_lo) - 0.05 * span, max(hi, h_hi) + 0.05 * span)
    ax.set_xlabel(f'${x}$')
    ax.set_title(f'{method.capitalize()} Riemann sum with n = {n}')
    handles, labels = ax.get_legend_handles_labels()
    if np.any(heights < 0):
        handles += [Patch(color='tab:blue', alpha=0.35), Patch(color='tab:red', alpha=0.
        labels += [r'$f(x_i^*) \geq 0$', r'$f(x_i^*) < 0$']
    ax.legend(handles, labels, loc='best')
    ax.grid(True, alpha=0.3)
    plt.show()

    # ---- the computation, line by line ----------------------------------
```

```
A, B, D = sp.latex(a), sp.latex(b), sp.latex(delta)
fx = sp.latex(f)
sample_rule = {
    'left':     r'x_i^* = x_{i-1} = a + (i-1)\Delta',
    'right':    r'x_i^* = x_i = a + i\,\Delta',
    'midpoint': r'x_i^* = \frac{x_{i-1} + x_i}{2} = a + \left(i - \tfrac{1}{2}\right
}[method]

display(Math(rf'f({x}) = {fx}, \qquad [a, b] = [{A}, {B}], \qquad n = {n}'))
a_paren = rf'\left({A}\right)' if sp.N(a) < 0 else A
display(Math(rf'\Delta = \frac{{b-a}}{{n}} = \frac{{{B} - {a_paren}}}{{{n}}} = {D}'
            + (rf' \approx {_fmt(d_f)}' if not delta.is_Rational else '')
            + rf', \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, {n})'))
display(Math(rf'\text{{{method} sample points: }} {sample_rule}'))

# which indices get written out
idx = list(range(1, n + 1)) if n <= max_terms else [1, 2, 3, n - 1, n]
pts = [a + (i - offset) * delta for i in idx]
pts = [sp.nsimplify(p) if p.is_Rational else sp.simplify(p) for p in pts]
f_terms = [rf'f\!\left({sp.latex(p)}\right)' for p in pts]
vals = [sp.nsimplify(sp.simplify(f.subs(x, p))) for p in pts]
if all(v.is_Rational and len(sp.latex(v)) <= 14 for v in vals):
    val_terms = [sp.latex(v) for v in vals]
else:
    val_terms = [_fmt(sp.N(v)) for v in vals]

general = rf'RS(P, S) = \sum_{{i=1}}^{{{n}}} f(x_i^*)\,\Delta'
display(Math(general + rf' = \Delta \left[ f(x_1^*) + f(x_2^*) + \cdots + f(x_{{{n}}
display(Math(rf'\phantom{{RS(P, S)}} = {D} \left[ {_join_terms(f_terms, n, max_terms
display(Math(rf'\phantom{{RS(P, S)}} = {D} \left[ {_join_terms(val_terms, n, max_ter

# exact value when it is a short rational number, else a decimal
exact = None
if n <= 200:
    total = sp.nsimplify(sum(f.subs(x, a + (i - offset) * delta) for i in range(1, n
    if total.is_Rational and len(sp.latex(total)) <= 20:
        exact = total
if exact is not None and exact.q != 1:
    result = rf'{sp.latex(exact)} \approx {_fmt(value, 10)}'
else:
    result = _fmt(value, 10)
display(Math(rf'\phantom{{RS(P, S)}} = \mathbf{{{result}}}'))
```

```
    if show_integral:
        I = sp.integrate(f, (x, a, b))
        I_num = float(sp.N(I))
        if isinstance(I, sp.Integral):   # no closed form found
            display(Math(rf'\int_{{{A}}}^{{{B}}} {fx}\, d{x} \approx {_fmt(I_num, 10)}')
        else:
            approx = '' if I.is_Rational else rf' \approx {_fmt(I_num, 10)}'
            display(Math(rf'\int_{{{A}}}^{{{B}}} {fx}\, d{x} = {sp.latex(I)}' + approx))
        err = value - I_num
        err_tex = '0' if abs(err) < 1e-12 else _fmt(err, 4)
        display(Math(rf'\text{{error: }} RS(P, S) - \int_{{{A}}}^{{{B}}} f\,d{x} \approx

    return value
```

12.1 Recap: the setup from lecture

Let $f : [a, b] \to \mathbb{R}$.

- A partition P of $[a, b]$ is a choice of points $a = x_0 < x_1 < \cdots < x_n = b$.
- $\Delta_i = x_i - x_{i-1}$ is the length of the i-th segment $[x_{i-1}, x_i]$.
- Sample points $S = \{x_1^*, \ldots, x_n^*\}$ with $x_i^* \in [x_{i-1}, x_i]$ determine the heights $f(x_i^*)$ of the rectangles.
- The Riemann sum is

$$RS(P, S) = f(x_1^*)\,\Delta_1 + \cdots + f(x_n^*)\,\Delta_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta_i.$$

In this notebook we always split $[a, b]$ into n segments of equal length, so $\Delta_i = \Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$. There are three standard ways to pick the sample points:

method	sample point in $[x_{i-1}, x_i]$
`'left'`	$x_i^* = x_{i-1}$, the left endpoint
`'right'`	$x_i^* = x_i$, the right endpoint
`'midpoint'`	$x_i^* = \frac{x_{i-1}+x_i}{2}$, the midpoint

Run the next cell to define the function `riemann_sum`. You do not need to read its code to use it.

12.1.1 How to call `riemann_sum`

```
riemann_sum(f, a, b, n, method)
```

- `f` is an expression in `x`, for example `x**2`, `sp.sin(x)`, `sp.exp(-x**2)`, `1/x`, `sp.sqrt(1 - x**2)` (Recall: exponentiation in Python is `**`. The prefix `sp.` instructs python to use a library for symbolic computation called `sympy`.)
- `a`, `b` are the bounds with `a < b`; exact values like `sp.pi` or `sp.Rational(1, 3)` are fine
- `n` is the number of subintervals
- `method` is `'left'`, `'right'`, or `'midpoint'`

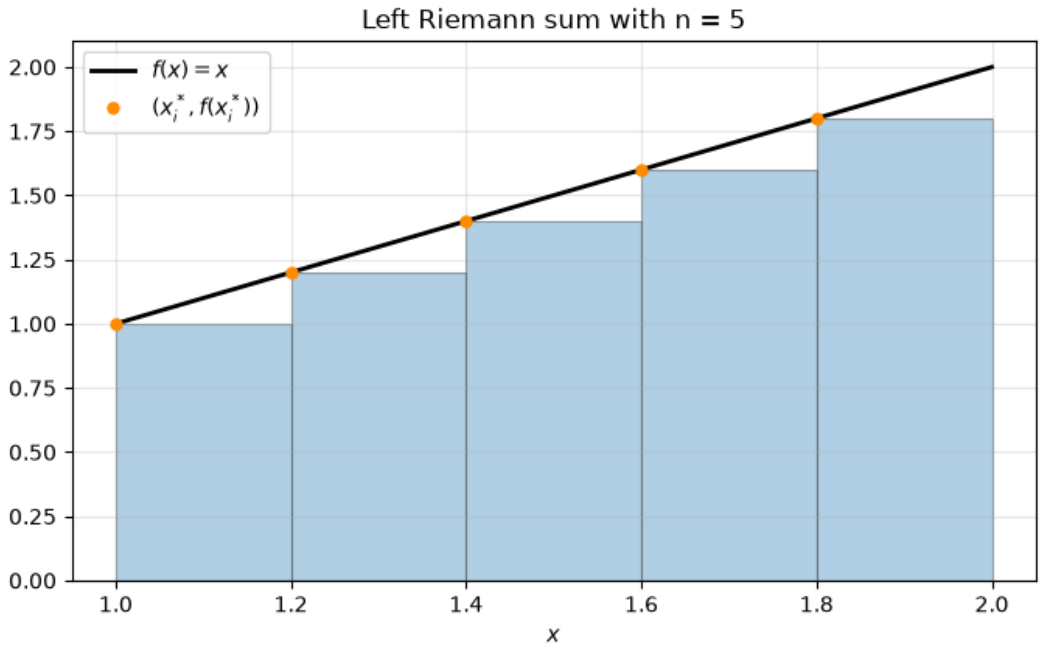
Optional arguments: - `show_integral=True` also displays the exact integral and the error of the Riemann sum - `show=False` skips the picture and just returns the number (useful in loops) - `max_terms=6` controls how many terms are written out before the sum is abbreviated with $+\cdots+$

The function returns the value of the sum. End the line with a semicolon `;` so Jupyter does not print that number a second time.

12.2 Part 1: Start

We start with $f(x) = x$ on $[1, 2]$ and $n = 5$ subintervals.

```
riemann_sum(x, 1, 2, 5, 'left');
```



$f(x) = x, \qquad [a, b] = [1, 2], \qquad n = 5$

$$\Delta = \frac{b-a}{n} = \frac{2-1}{5} = \frac{1}{5}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 5)$$

left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

$$\begin{aligned} RS(P,S) &= \sum_{i=1}^{5} f(x_i^*)\,\Delta = \Delta\,[f(x_1^*) + f(x_2^*) + \cdots + f(x_5^*)] \\ &= \frac{1}{5}\left[f(1) + f\left(\frac{6}{5}\right) + f\left(\frac{7}{5}\right) + f\left(\frac{8}{5}\right) + f\left(\frac{9}{5}\right)\right] \\ &= \frac{1}{5}\left[1 + \frac{6}{5} + \frac{7}{5} + \frac{8}{5} + \frac{9}{5}\right] \\ &= \mathbf{\frac{7}{5} \approx 1.4} \end{aligned}$$

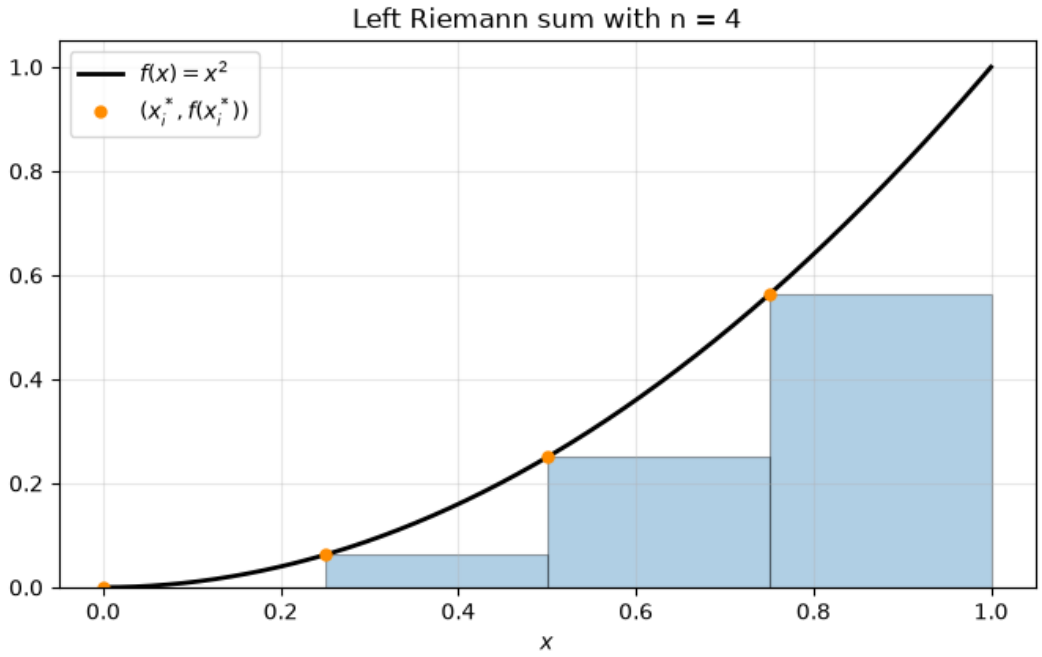
In class, we computed the actual value of the integral $\int_a^b x dx$. How does this value compare? Increase `n` in the cell above and see what happens.

12.3 Part 2: Left, right, and midpoint sums

Now we look at the function $f(x) = x^2$ on $[0, 1]$ and $n = 4$ subintervals.

PREDICT (on paper, before running anything): - Will the left sum overestimate or underestimate the area under x^2? What about the right sum? - Compute the left sum by hand. You should need nothing more than adding four fractions.

```
riemann_sum(x**2, 0, 1, 4, 'left');
```



$f(x) = x^2, \qquad [a, b] = [0, 1], \qquad n = 4$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{4} = \dfrac{1}{4}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 4)$

left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

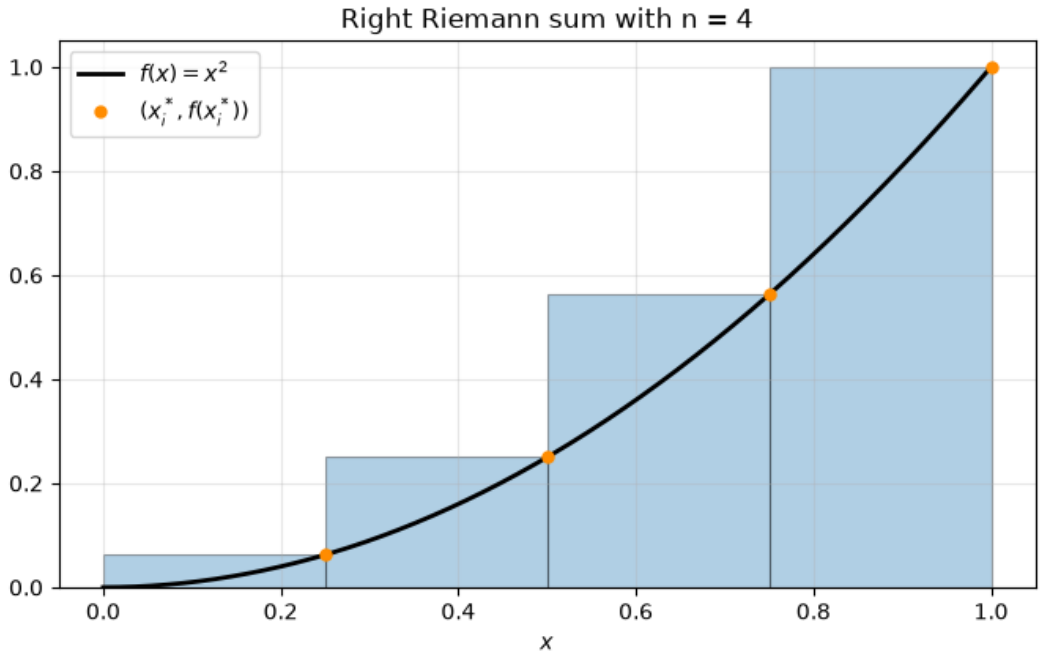
$$RS(P, S) = \sum_{i=1}^{4} f(x_i^*)\,\Delta = \Delta\,[f(x_1^*) + f(x_2^*) + \cdots + f(x_4^*)]$$

$$= \frac{1}{4}\left[f(0) + f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right]$$

$$= \frac{1}{4}\left[0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16}\right]$$

$$= \frac{7}{32} \approx \mathbf{0.21875}$$

```
riemann_sum(x**2, 0, 1, 4, 'right');
```



$f(x) = x^2, \qquad [a,b] = [0,1], \qquad n = 4$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{4} = \dfrac{1}{4}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 4)$

right sample points: $x_i^* = x_i = a + i\,\Delta$

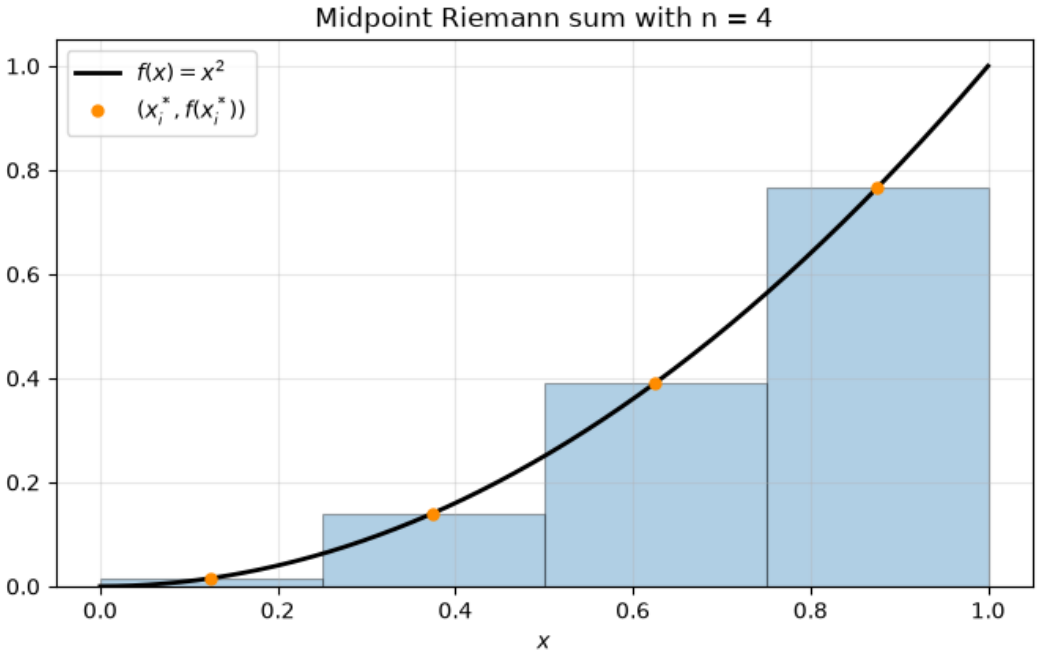
$$RS(P,S) = \sum_{i=1}^{4} f(x_i^*)\,\Delta = \Delta\,[f(x_1^*) + f(x_2^*) + \cdots + f(x_4^*)]$$

$$= \frac{1}{4}\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right) + f(1)\right]$$

$$= \frac{1}{4}\left[\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1\right]$$

$$= \frac{\mathbf{15}}{\mathbf{32}} \approx \mathbf{0.46875}$$

```
riemann_sum(x**2, 0, 1, 4, 'midpoint', show_integral=True);
```



$f(x) = x^2, \qquad [a,b] = [0,1], \qquad n = 4$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{4} = \dfrac{1}{4}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 4)$

midpoint sample points: $x_i^* = \dfrac{x_{i-1} + x_i}{2} = a + \left(i - \frac{1}{2}\right)\Delta$

$$\begin{aligned} RS(P,S) &= \sum_{i=1}^{4} f(x_i^*)\,\Delta = \Delta\left[f(x_1^*) + f(x_2^*) + \cdots + f(x_4^*)\right] \\ &= \frac{1}{4}\left[f\left(\frac{1}{8}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{7}{8}\right)\right] \\ &= \frac{1}{4}\left[\frac{1}{64} + \frac{9}{64} + \frac{25}{64} + \frac{49}{64}\right] \\ &= \mathbf{\frac{21}{64}} \approx \mathbf{0.328125} \end{aligned}$$

$\displaystyle\int_0^1 x^2\,dx = \frac{1}{3}$

error: $\displaystyle RS(P,S) - \int_0^1 f\,dx \approx -0.005208$

VERIFY: The left and right sums share almost all of their terms. Show that for any f and equally spaced partition,

$$RS_{\text{right}} - RS_{\text{left}} = \Delta\,(f(b) - f(a)),$$

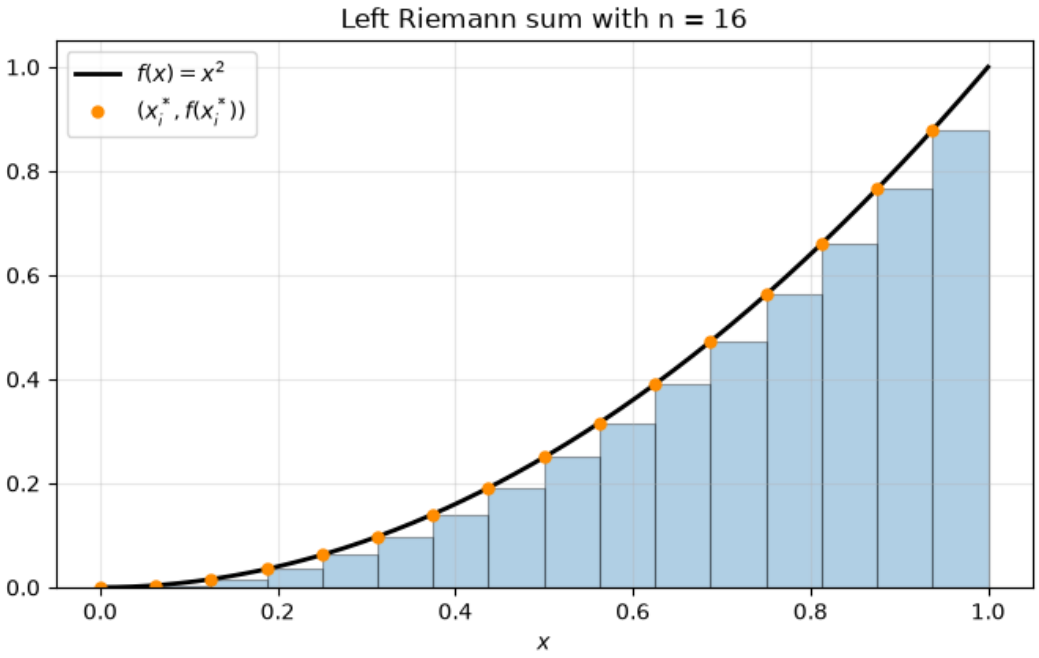
and check this against the two numbers you just computed.

REFLECT: Your prediction about over- and underestimates depended on a property of x^2 on $[0, 1]$. Which property? State a general claim about left and right sums for functions with that property.

12.4 Part 3: Refining the partition

In lecture, the approximation got better by refining the partition. Let us watch that happen. Once n is larger than 6, the written-out sum is abbreviated with $+\cdots+$.

```
riemann_sum(x**2, 0, 1, 16, 'left', show_integral=True);
```



$f(x) = x^2, \qquad [a, b] = [0, 1], \qquad n = 16$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{16} = \dfrac{1}{16}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 16)$

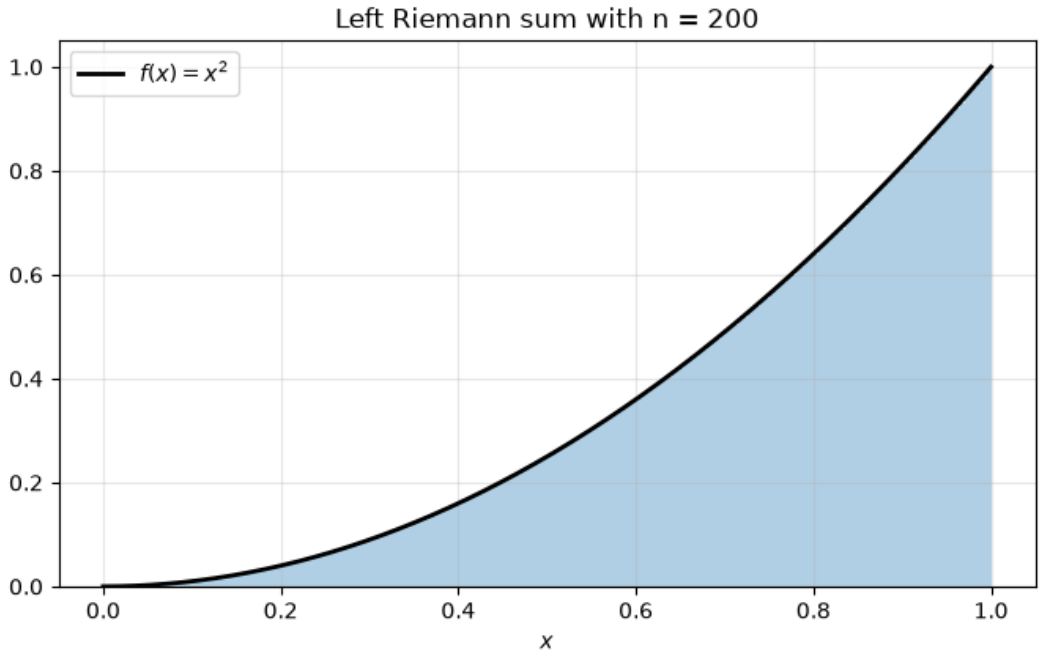
left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

$$
\begin{aligned}
RS(P,S) &= \sum_{i=1}^{16} f(x_i^*)\,\Delta = \Delta\left[f(x_1^*) + f(x_2^*) + \cdots + f(x_{16}^*)\right] \\
&= \frac{1}{16}\left[f(0) + f\left(\frac{1}{16}\right) + f\left(\frac{1}{8}\right) + \cdots + f\left(\frac{7}{8}\right) + f\left(\frac{15}{16}\right)\right] \\
&= \frac{1}{16}\left[0 + 0.00390625 + 0.015625 + \cdots + 0.765625 + 0.878906\right] \\
&= \frac{\mathbf{155}}{\mathbf{512}} \approx \mathbf{0.302734375}
\end{aligned}
$$

$$
\int_0^1 x^2\,dx = \frac{1}{3}
$$

error: $RS(P,S) - \displaystyle\int_0^1 f\,dx \approx -0.0306$

```
riemann_sum(x**2, 0, 1, 200, 'left', show_integral=True);
```



$f(x) = x^2, \qquad [a,b] = [0,1], \qquad n = 200$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{200} = \dfrac{1}{200}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 200)$

left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

$$RS(P,S) = \sum_{i=1}^{200} f(x_i^*)\,\Delta = \Delta\left[f(x_1^*) + f(x_2^*) + \cdots + f(x_{200}^*)\right]$$
$$= \frac{1}{200}\left[f(0) + f\left(\frac{1}{200}\right) + f\left(\frac{1}{100}\right) + \cdots + f\left(\frac{99}{100}\right) + f\left(\frac{199}{200}\right)\right]$$
$$= \frac{1}{200}\left[0 + 2.5 \cdot 10^{-5} + 0.0001 + \cdots + 0.9801 + 0.990025\right]$$
$$= \frac{\mathbf{26467}}{\mathbf{80000}} \approx \mathbf{0.3308375}$$

$$\int_0^1 x^2\,dx = \frac{1}{3}$$

error: $RS(P,S) - \displaystyle\int_0^1 f\,dx \approx -0.002496$

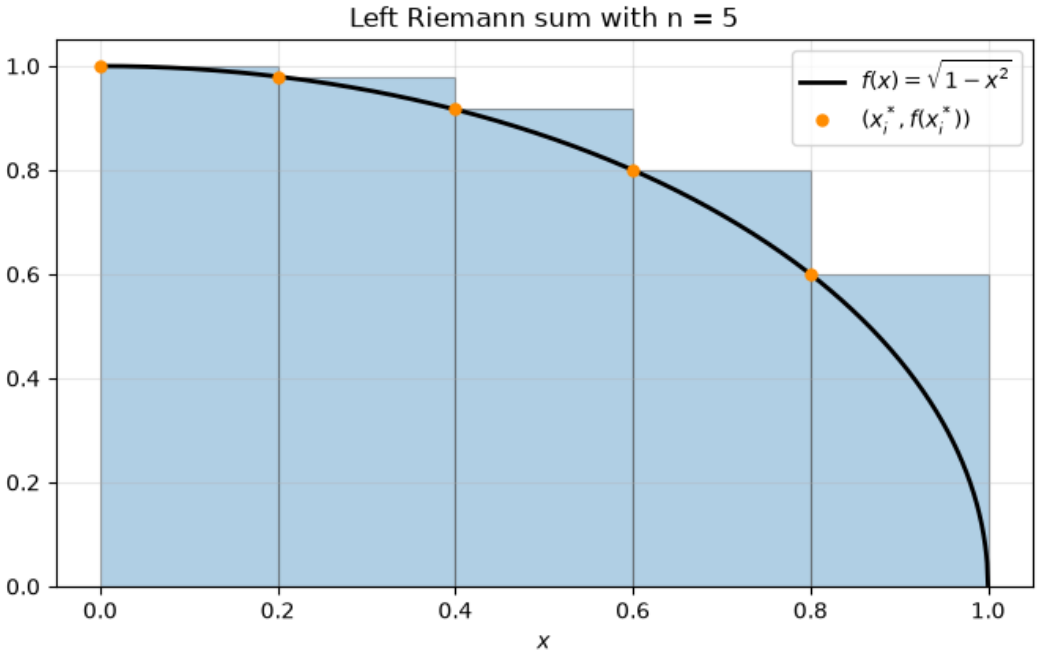
12.5 Part 4: Back to the circle

We started the Riemann integral lecture by approximating the area π of the unit circle with regular n-gons. The quarter of the unit disk in the first quadrant is the region under $f(x) = \sqrt{1-x^2}$ on $[0,1]$, so $4\int_0^1 \sqrt{1-x^2}\,dx = \pi$.

PREDICT: Is the left sum of $\sqrt{1-x^2}$ on $[0,1]$ an overestimate or an underestimate? Why?

```
riemann_sum(sp.sqrt(1 - x**2), 0, 1, 5, 'left', show_integral=True);

# TODO: find the smallest n for which 4 * (left sum) is within 10**(-3) of pi.
#   Use riemann_sum(..., show=False) inside a loop.
# TODO: do the same for the midpoint sum. How much smaller is the n you need?
```



$f(x) = \sqrt{1-x^2}, \qquad [a,b] = [0,1], \qquad n = 5$

$\Delta = \dfrac{b-a}{n} = \dfrac{1-0}{5} = \dfrac{1}{5}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 5)$

left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

$$\begin{aligned} RS(P,S) &= \sum_{i=1}^{5} f(x_i^*)\,\Delta = \Delta\,[f(x_1^*) + f(x_2^*) + \cdots + f(x_5^*)] \\ &= \frac{1}{5}\left[f(0) + f\left(\frac{1}{5}\right) + f\left(\frac{2}{5}\right) + f\left(\frac{3}{5}\right) + f\left(\frac{4}{5}\right)\right] \\ &= \frac{1}{5}\,[1 + 0.979796 + 0.916515 + 0.8 + 0.6] \\ &= \mathbf{0.8592622072} \end{aligned}$$

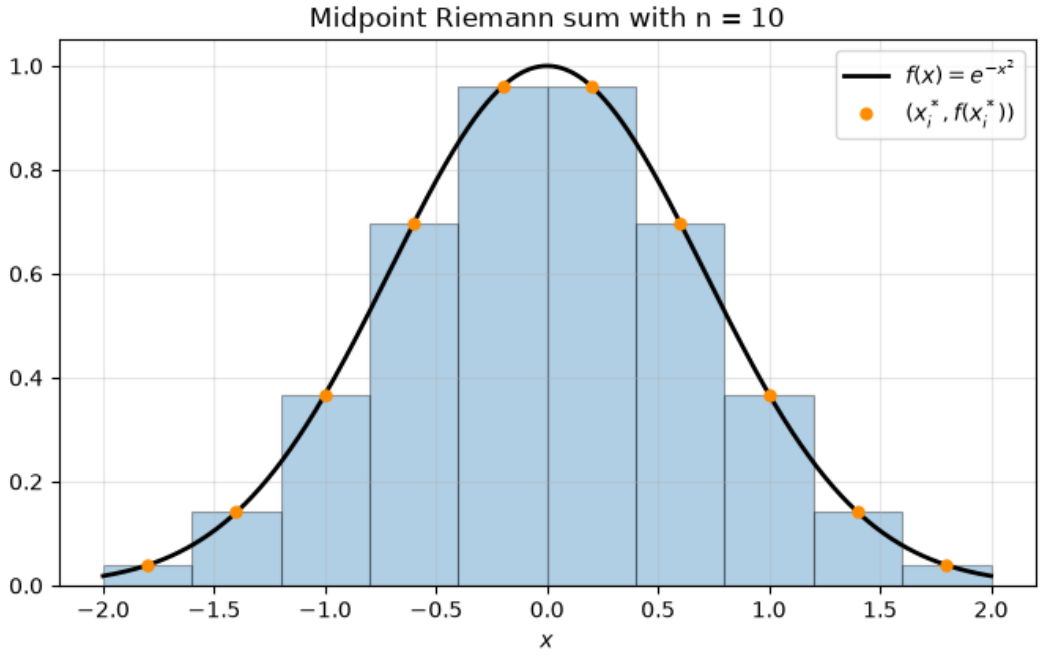
$$\int_0^1 \sqrt{1-x^2}\,dx = \frac{\pi}{4} \approx 0.7853981634$$

error: $RS(P,S) - \displaystyle\int_0^1 f\,dx \approx 0.07386$

12.6 Part 5: Experiment on your own

Change the function, bounds, n, and method in the cell below and run it again. If `ipywidgets` is installed, the cell after it gives you sliders instead.

```
f = sp.exp(-x**2)
riemann_sum(f, -2, 2, 10, 'midpoint', show_integral=True);
```



$f(x) = e^{-x^2}, \qquad [a,b] = [-2,2], \qquad n = 10$

$$\Delta = \frac{b-a}{n} = \frac{2-(-2)}{10} = \frac{2}{5}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 10)$$

midpoint sample points: $x_i^* = \dfrac{x_{i-1} + x_i}{2} = a + \left(i - \frac{1}{2}\right)\Delta$

$$\begin{aligned} RS(P,S) &= \sum_{i=1}^{10} f(x_i^*)\,\Delta = \Delta\left[f(x_1^*) + f(x_2^*) + \cdots + f(x_{10}^*)\right] \\ &= \frac{2}{5}\left[f\left(-\frac{9}{5}\right) + f\left(-\frac{7}{5}\right) + f(-1) + \cdots + f\left(\frac{7}{5}\right) + f\left(\frac{9}{5}\right)\right] \\ &= \frac{2}{5}\left[0.0391639 + 0.140858 + 0.367879 + \cdots + 0.140858 + 0.0391639\right] \\ &= \mathbf{1.765094018} \end{aligned}$$

$$\int_{-2}^{2} e^{-x^2}\,dx = \sqrt{\pi}\,\mathrm{erf}\,(2) \approx 1.764162782$$

error: $RS(P, S) - \displaystyle\int_{-2}^{2} f\,dx \approx 0.0009312$

```
# Interactive version: move the slider or change the method
try:
    from ipywidgets import interact, IntSlider, Dropdown

    def explore(n=4, method='left'):
        riemann_sum(sp.sin(x) + 1.5, 0, 2*sp.pi, n, method, show_integral=True)

    interact(explore,
             n=IntSlider(min=1, max=100, step=1, value=4),
             method=Dropdown(options=['left', 'right', 'midpoint'], value='left'));
except ImportError:
    print('ipywidgets is not installed; use the cell above instead.')
```

```
interactive(children=(IntSlider(value=4, description='n', min=1), Dropdown(description='
```

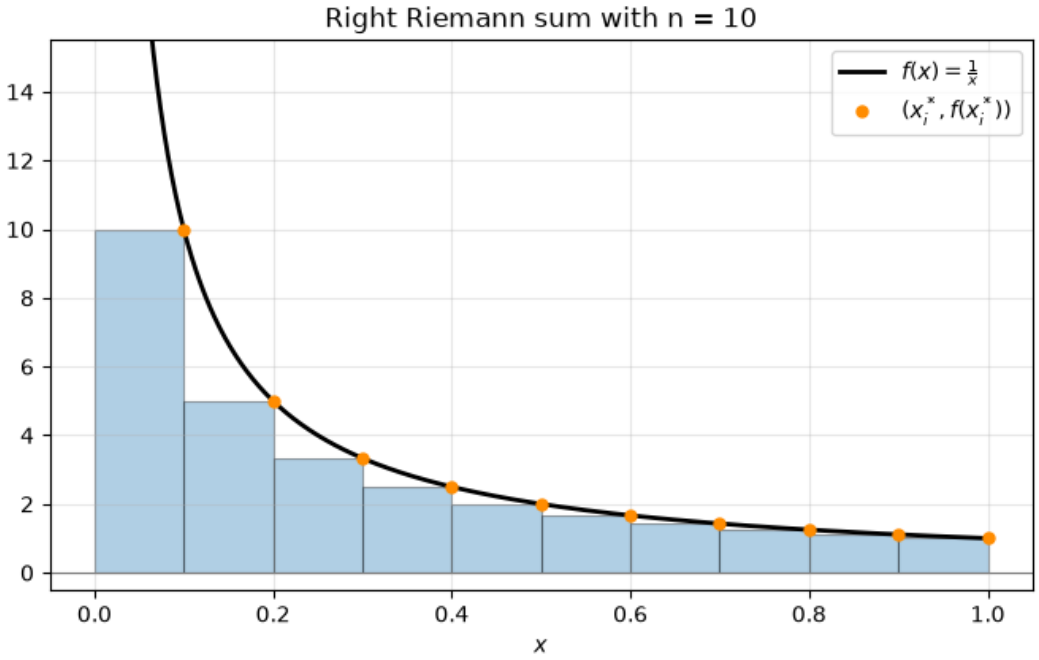
12.7 Part 6: when Riemann sums fail

Riemann sums are defined for bounded functions. Look at what happens for $f(x) = 1/x$ on $[0, 1]$ using right sums (the left sum would need $f(0)$, which does not exist).

```
riemann_sum(1/x, 0, 1, 10, 'right');

for n in [10, 100, 1000, 10000]:
    print(n, riemann_sum(1/x, 0, 1, n, 'right', show=False))

# TODO: what happens to the sums as n grows? Compare with 1/sp.sqrt(x) on [0, 1].
#   The first rectangle is always very tall in both cases,
#    so why do the two behave so differently?
```



$$f(x) = \frac{1}{x}, \qquad [a,b] = [0,1], \qquad n = 10$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{10} = \frac{1}{10}, \qquad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, 10)$$

right sample points: $x_i^* = x_i = a + i\,\Delta$

$$\begin{aligned} RS(P,S) &= \sum_{i=1}^{10} f(x_i^*)\,\Delta = \Delta\,[f(x_1^*) + f(x_2^*) + \cdots + f(x_{10}^*)] \\ &= \frac{1}{10}\left[f\left(\frac{1}{10}\right) + f\left(\frac{1}{5}\right) + f\left(\frac{3}{10}\right) + \cdots + f\left(\frac{9}{10}\right) + f(1)\right] \\ &= \frac{1}{10}\left[10 + 5 + \frac{10}{3} + \cdots + \frac{10}{9} + 1\right] \\ &= \mathbf{\frac{7381}{2520} \approx 2.928968254} \end{aligned}$$

```
10 2.9289682539682538
100 5.187377517639621
1000 7.485470860550345
10000 9.78760603604438
```

13 Properties of the Integral

Companion reading

This lecture corresponds to C&J §2.3.a–d: additivity, the integral of a sum and of a constant multiple, estimating integrals, and the mean value theorem for integrals. The table of basic integrals draws on C&J §2.2.c–e.

13.1 The Integral as Signed Area

Remark 13.1. Nothing in Definition 11.1 or Definition 11.2 requires $f \geq 0$: the integral is defined for every continuous f on $[a, b]$. But the identification

$$\int_a^b f(x)\,dx = \text{area under the graph of } f$$

is only valid when $f \geq 0$. If f takes negative values, the corresponding terms $f(x_i^*)\,\Delta_i$ of a Riemann sum are negative. So in general the integral measures **signed area**: area above the x-axis counts positively, area below the x-axis counts negatively.

Example 13.1.
Let $f(x) = x$ and $b > 0$. By Example 11.2,

$$\int_{-b}^{b} x\,dx = \frac{b^2 - (-b)^2}{2} = 0.$$

Geometrically, the region between the graph and the x-axis consists of two triangles of area $\frac{b^2}{2}$ each: one above the axis over $[0, b]$ (positive area) and one below the axis over $[-b, 0]$ (negative area). They cancel.

If we want the **total** (unsigned) area between the graph of f and the x-axis, we can compute $\int_a^b |f(x)|dx$

13.2 Additivity

Proposition 13.1 (Additivity)**.** *Let f be continuous on $[a,b]$ and $a<c<b$. Then*

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

For $f \geq 0$ this is what the area interpretation predicts: the vertical line $x=c$ cuts the region under the graph into a part A_1 over $[a,c]$ and a part A_2 over $[c,b]$, and the total area is $A_1 + A_2$.

Proof. The idea is that any Riemann sum for $[a,b]$ in which c is a division point splits into a Riemann sum for $[a,c]$ plus one for $[c,b]$. If $c = x_m$ is a point of the partition $a = x_0 < \cdots < x_n = b$, then

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta_i = \sum_{i=1}^{m} f(x_i^*)\,\Delta_i + \sum_{i=m+1}^{n} f(x_i^*)\,\Delta_i,$$

where the first sum on the right is a Riemann sum for a partition of $[a,c]$ into m segments, and the second one is a Riemann sum for a partition of $[c,b]$ into $n-m$ segments.

For equipartitions of $[a,b]$, the point c is usually not a division point. But by Theorem 11.2 we may use *any* partitions whose mesh tends to 0. So let P_n consist of an equipartition of $[a,c]$ into n segments followed by an equipartition of $[c,b]$ into n segments, and let S_n be the right endpoints of its segments. Then $\|P_n\| = \max\left(\frac{c-a}{n}, \frac{b-c}{n}\right) \to 0$, and the splitting above reads

$$RS(P_n, S_n) = RHS_n(f,[a,c]) + RHS_n(f,[c,b]).$$

Letting $n \to \infty$, the left side tends to $\int_a^b f$ by Theorem 11.2, and the right side tends to $\int_a^c f + \int_c^b f$ by Definition 11.2 and the limit laws (Proposition 8.1). □

Rearranging gives a formula that is often helpful for computing areas.

Corollary 13.1.
Let f be continuous on $[a,b]$ and $a<c<b$. Then

$$\int_c^b f(x)\,dx = \int_a^b f(x)\,dx - \int_a^c f(x)\,dx.$$

Example 13.2.
For $0 < a < b$, the computation of $\int_0^b x^2\,dx = \frac{b^3}{3}$ from the previous chapter and Corollary 13.1 give

$$\int_a^b x^2\,dx = \int_0^b x^2\,dx - \int_0^a x^2\,dx = \frac{b^3}{3} - \frac{a^3}{3},$$

without redoing the Riemann sum computation of Exercise 11.3.

13.3 Integrals with $a \geq b$

So far $\int_a^b f(x)\,dx$ was only defined for $a < b$. What if $a = b$ or $a > b$?

In the case $a = b$, we see that $\Delta = \frac{b-a}{n} = 0$ for any n, meaning that all Riemann sums are 0, too, which in turn makes the integral 0.

If $a > b$, Δ becomes negative. The partition points

$$a,\ a + \Delta,\ a + 2\Delta,\ \ldots,\ a + n\Delta = b,$$

now count from *right to left*. And since Δ is negative, the Riemann sum will be (-1) times the sum we get when we count from left to right. This yields $\int_b^a f = -\int_a^b f$.

Definition 13.1.
Let f be continuous on an interval containing a and c.

- $\displaystyle\int_a^a f(x)\,dx = 0$ (this only requires f to be defined at a).
- If $a < b$, then $\displaystyle\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.

Exercise 13.1. Let f be continuous on an interval I. Using Proposition 13.1 and Definition 13.1, show that

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

holds for **all** $a, b, c \in I$, in any order and also when some of them coincide. (Hint: there are six orderings of three distinct points. For $b < c$ with $a < b$, say, apply Proposition 13.1 to $a < b < c$.)

13.4 Sums and Constant Multiples

Proposition 13.2 (Sum rule)**.** *Let f and g be continuous on $[a,b]$. Then*

$$\int_a^b (f(x)+g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx.$$

Proof. This follows directly from the definition via Riemann sums. For the right-hand sums with $\Delta = \frac{b-a}{n}$,

$$\begin{aligned} RHS_n(f+g) &= \sum_{i=1}^{n} \left(f(a+i\Delta)+g(a+i\Delta)\right)\Delta \\ &= \sum_{i=1}^{n} f(a+i\Delta)\,\Delta + \sum_{i=1}^{n} g(a+i\Delta)\,\Delta = RHS_n(f) + RHS_n(g). \end{aligned}$$

By the limit laws (Proposition 8.1),

$$\int_a^b (f+g) = \lim_{n\to\infty} RHS_n(f+g) = \lim_{n\to\infty} \left(RHS_n(f) + RHS_n(g)\right) \tag{13.1}$$

$$= \lim_{n\to\infty} RHS_n(f) + \lim_{n\to\infty} RHS_n(g) = \int_a^b f + \int_a^b g. \tag{13.2}$$

□

Proposition 13.3 (Constant multiple rule)**.** *Let f be continuous on $[a,b]$ and $c \in \mathbb{R}$. Then*

$$\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx.$$

The proof is the same: $RHS_n(cf) = c\cdot RHS_n(f)$, and we let $n\to\infty$. Combining the two rules, $\int_a^b(\alpha f+\beta g) = \alpha\int_a^b f + \beta\int_a^b g$ for all constants α,β; in particular $\int_a^b(f-g) = \int_a^b f - \int_a^b g$. Both rules remain true for $a \geq b$, by Definition 13.1.

Remark 13.2. What about $\int_a^b f(x)\,g(x)\,dx$? There is **no** analogous rule: the integral of a product is in general *not* the product of the integrals. For instance, with $f(x)=g(x)=x$ on $[0,1]$,

$$\int_0^1 x\cdot x\,dx = \frac{1}{3} \qquad \text{but} \qquad \int_0^1 x\,dx\cdot\int_0^1 x\,dx = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}.$$

The trouble is already visible in the Riemann sums: $\sum f(x_i)g(x_i)\Delta$ does not factor into $(\sum f(x_i)\Delta)(\sum g(x_i)\Delta)$.

13.5 Basic Rules for Integrals

We collect the integrals of the elementary functions. Some of them can be computed directly from Riemann sums: we have done x^0, x^1, x^2 (Example 11.1, Example 11.2, and the x^2 computation of the previous chapter), and C&J §2.2.c–e carry out x^α and sin, cos. All of them will become easy consequences of the Fundamental Theorem of Calculus. For now we record them and use them freely.

Proposition 13.4 (Integrals of elementary functions)**.**

$$\int_a^b x^\alpha \, dx = \frac{1}{\alpha+1}(b^{\alpha+1} - a^{\alpha+1}) \qquad (\alpha \neq -1),$$
$$\int_a^b x^{-1} \, dx = \ln b - \ln a,$$
$$\int_a^b \sin x \, dx = -(\cos b - \cos a),$$
$$\int_a^b \cos x \, dx = \sin b - \sin a,$$
$$\int_a^b e^x \, dx = e^b - e^a.$$

Here the interval of integration must lie inside the domain on which the integrand is continuous: $0 < a < b$ *for* x^{-1} *and for general real* α*; any* a, b *if* α *is a nonnegative integer;* a, b *on the same side of* 0 *if* α *is a negative integer.*

What about $\int_a^b \ln x \, dx$? We will come back to it later.

Combining Proposition 13.4 with the sum and constant multiple rules, we can integrate a wide range of functions, for instance every polynomial.

Example 13.3.

$$\begin{aligned}\int_a^b (2x^3 - 4\sin x)\, dx &= 2\int_a^b x^3 \, dx - 4\int_a^b \sin x \, dx \\ &= 2\Big(\frac{1}{4}(b^4 - a^4)\Big) - 4\Big(-(\cos b - \cos a)\Big) \\ &= \frac{1}{2}(b^4 - a^4) + 4(\cos b - \cos a).\end{aligned}$$

Remark 13.3. This leaves open how to integrate products $f(x)\, g(x)$ (Remark 13.2) and compositions $f(g(x))$. Both questions will be answered later, by *integration by parts* and *substitution*.

13.6 Estimating Integrals

Throughout this section, $a < b$.

Proposition 13.5.
If $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\,dx \geq 0.$$

Proof. Every term $f(x_i^*)\,\Delta_i$ of every Riemann sum for f is ≥ 0, so every Riemann sum is ≥ 0. A limit of nonnegative numbers is nonnegative (compare the argument at the end of the proof of Theorem 10.2). □

Proposition 13.6.
If $f(x) \geq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx.$$

Proof. Apply Proposition 13.5 to $f - g \geq 0$ and use $\int_a^b f - \int_a^b g = \int_a^b (f - g)$ (Proposition 13.2, Proposition 13.3). □

Now let $f\colon [a, b] \to \mathbb{R}$ be continuous. As we will show later (this is the *Extreme Value Theorem*, see Remark 11.2), f assumes a maximum and a minimum on $[a, b]$. In particular, there exist m, M such that

$$m \leq f(x) \leq M \qquad \text{for all } x \in [a, b].$$

Interpret m and M as constant functions on $[a, b]$ and apply Proposition 13.6:

$$\int_a^b m\,dx \leq \int_a^b f(x)\,dx \leq \int_a^b M\,dx.$$

For any constant C, $\int_a^b C\,dx = C\int_a^b 1\,dx = C(b - a)$ (Example 11.1). Hence:

Corollary 13.2 (Min–max bound)**.** *Let f be continuous on $[a, b]$, with minimum value m and maximum value M. Then*

$$m(b - a) \leq \int_a^b f(x)\,dx \leq M(b - a).$$

Geometrically (for $f \geq 0$), $m(b-a)$ is the area of the rectangle of height m *inscribed* under the graph, and $M(b-a)$ is the area of the rectangle of height M *circumscribing* it. The argument only used $m \leq f \leq M$, so the bound holds for any lower bound m and upper bound M of f on $[a,b]$; the minimum and maximum give the best bound of this kind.

This bound is simple but immensely useful. Crude on a single interval, it becomes powerful when applied on many small intervals and added up.

Example 13.4 (Harmonic series)**.**
What happens to the sequence

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

(the partial sums of the **harmonic series**) as $n \to \infty$?

Let $f(x) = \frac{1}{x}$ on $[k, k+1]$, $k \in \mathbb{N}$. Since f is decreasing, its minimum and maximum on this interval are $m = \frac{1}{k+1}$ and $M = \frac{1}{k}$, attained at the endpoints. Corollary 13.2 gives

$$\frac{1}{k+1} = \frac{1}{k+1}((k+1)-k) \;\leq\; \int_k^{k+1} \frac{1}{x}\,dx \;\leq\; \frac{1}{k}((k+1)-k) = \frac{1}{k}.$$

Sum all these bounds for $k = 1, \ldots, n$:

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n+1} \;\leq\; \int_1^2 \frac{1}{x}\,dx + \int_2^3 \frac{1}{x}\,dx + \cdots + \int_n^{n+1} \frac{1}{x}\,dx \;\leq\; 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

By repeated use of Proposition 13.1, the integrals in the middle combine into a single integral, and by Proposition 13.4,

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n+1} \;\leq\; \int_1^{n+1} \frac{1}{x}\,dx = \ln(n+1) - \ln 1 = \ln(n+1) \;\leq\; a_n.$$

Since ln is increasing with range $\mathbb{R}$ (Definition 5.5), for every C there is an N with $\ln(N+1) > C$, and then $a_n \geq \ln(n+1) > C$ for all $n \geq N$. So $a_n \to \infty$: the harmonic series diverges.

13.7 The Integral as a Mean Value

Consider the right-hand sum with $\Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$:

$$RHS_n(f) = \sum_{i=1}^{n} f(x_i)\,\Delta = \frac{b-a}{n}(f(x_1) + \cdots + f(x_n)) = (b-a) \cdot \frac{f(x_1) + \cdots + f(x_n)}{n}.$$

The fraction on the right is the **average** (or **arithmetic mean**) of the n values $f(x_1), \dots, f(x_n)$. So

$$\frac{RHS_n(f)}{b-a} = \operatorname{avg}\left(f(x_1), \dots, f(x_n)\right)$$

is the average of f over n equally spaced sample points — a "level-n average". As $n \to \infty$, $RHS_n(f) \to \int_a^b f(x)\,dx$, hence

$$\frac{RHS_n(f)}{b-a} \longrightarrow \frac{1}{b-a}\int_a^b f(x)\,dx.$$

The limit is an "∞-average": the average of *all* values of f on $[a,b]$.

Definition 13.2.
Let f be continuous on $[a,b]$. The **average** or **mean value** of f on $[a,b]$ is

$$\operatorname{avg}\left(f,[a,b]\right) := \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Example 13.5.
Using Proposition 13.4:

$$\operatorname{avg}\left(x,[0,1]\right) = \frac{1}{1}\int_0^1 x\,dx = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2},$$

$$\operatorname{avg}\left(\sin x,[0,2\pi]\right) = \frac{1}{2\pi}\int_0^{2\pi} \sin x\,dx = \frac{1}{2\pi}\Big(-\left(\cos 2\pi - \cos 0\right)\Big) = \frac{1}{2\pi}\left(-(1-1)\right) = 0,$$

$$\operatorname{avg}\left(\sin x,[0,\pi]\right) = \frac{1}{\pi}\Big(-\left(\cos \pi - \cos 0\right)\Big) = \frac{1}{\pi}\left(-(-1-1)\right) = \frac{2}{\pi},$$

$$\operatorname{avg}\left(x^2,[0,2]\right) = \frac{1}{2}\int_0^2 x^2\,dx = \frac{1}{2}\Big(\frac{2^3}{3} - \frac{0^3}{3}\Big) = \frac{4}{3}.$$

Over the full period $[0,2\pi]$ the positive and negative areas of sin cancel (Remark 13.1), so the average is 0. Over $[0,\pi]$, where $\sin \geq 0$, the average $\frac{2}{\pi} \approx 0.64$ lies strictly between the minimum 0 and the maximum 1.

Connection with the min–max bound. Let f be continuous on $[a,b]$. Then f assumes its minimum m and its maximum M on $[a,b]$: there exist $x_m, x_M \in [a,b]$ with

$$f(x_m) = m, \qquad f(x_M) = M, \qquad \text{and} \qquad m \leq f(x) \leq M \ \text{ for all } a \leq x \leq b.$$

Dividing the inequality of Corollary 13.2 by $b-a>0$ gives

$$m \;\le\; \underbrace{\frac{1}{b-a}\int_a^b f(x)\,dx}_{=\operatorname{avg}(f,[a,b])} \;\le\; M.$$

The average of a continuous function lies between its smallest and its largest value. By the Intermediate Value Theorem, f assumes **every** value between m and M — in particular its average.

Theorem 13.1 (Mean Value Theorem for Integrals)**.** *Let f be continuous on $[a,b]$. Then there exists $c\in[a,b]$ with*

$$\frac{1}{b-a}\int_a^b f(x)\,dx = f(c), \qquad \textit{i.e.} \qquad \int_a^b f(x)\,dx = f(c)\,(b-a).$$

Proof. Let $\mu=\operatorname{avg}(f,[a,b])$. As shown above, $f(x_m)=m\le\mu\le M=f(x_M)$. Apply the general form of the Intermediate Value Theorem (Remark 10.4) to f on the closed interval with endpoints x_m and x_M (if $x_m=x_M$, then $m=M$, f is constant, and any c works). It yields a point c between x_m and x_M, hence in $[a,b]$, with $f(c)=\mu$. □

Geometrically (for $f\ge 0$): the area under the graph of f equals the area of the rectangle over $[a,b]$ of height $f(c)$. The parts of the region under the graph that stick out above the rectangle exactly make up for the parts of the rectangle that lie above the graph.

Remark 13.4.

1. The theorem only asserts that *some* c exists; it gives no information about where it is, and c need not be unique. For sin on $[0,2\pi]$ (Example 13.5), each of $c=0,\pi,2\pi$ works.
2. Continuity is essential. Let $f(x)=0$ for $x<1$ and $f(x)=1$ for $x\ge 1$, on $[0,2]$. Computing piece by piece, $\int_0^2 f=0+1=1$, so the average is $\frac{1}{2}$ — a value f never takes.
3. Since both sides of $\int_a^b f=f(c)(b-a)$ change sign when a and b are interchanged (Definition 13.1), the theorem also holds for $a>b$, with c between b and a.

13.8 Summary

- The integral of a continuous function is defined whether or not $f \geq 0$; in general it is a **signed area** (Remark 13.1), and positive and negative contributions can cancel (Example 13.1). The **total area** is $\int_a^b |f|$.
- **Additivity** (Proposition 13.1): $\int_a^b f = \int_a^c f + \int_c^b f$ for $a < c < b$. Requiring that additivity survive forces $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$ (Definition 13.1): integrating from right to left counts area negatively.
- **Linearity**: $\int (f+g) = \int f + \int g$ and $\int cf = c \int f$ (Proposition 13.2, Proposition 13.3), both straight from the Riemann sums and the limit laws. There is no such rule for products (Remark 13.2).
- Together with the **integrals of elementary functions** (Proposition 13.4), linearity lets us integrate all linear combinations of them (Example 13.3).
- **Estimates**: $f \geq 0 \Rightarrow \int f \geq 0$ and $f \geq g \Rightarrow \int f \geq \int g$ (Proposition 13.5, Proposition 13.6). For continuous f with minimum m and maximum M, $m(b-a) \leq \int_a^b f \leq M(b-a)$ (Corollary 13.2). Applied to $\frac{1}{x}$ on the intervals $[k, k+1]$, this shows $1 + \frac{1}{2} + \cdots + \frac{1}{n} \geq \ln(n+1) \to \infty$ (Example 13.4).
- $\frac{1}{b-a}\int_a^b f$ is the **average value** of f on $[a,b]$ (Definition 13.2), the limit of the averages of f over n equally spaced points. By the Intermediate Value Theorem it is actually attained: $\int_a^b f = f(c)(b-a)$ for some $c \in [a,b]$ (Theorem 13.1).

14 The Indefinite Integral

> Companion reading
>
> This lecture corresponds to C&J §2.4 (the integral as a function of the upper limit).

So far we have treated the integral $\int_a^b f(x)\,dx$ (Definition 11.2) as a *number* attached to a function and an interval. But this number depends on the limits of integration a and b. If we hold the lower limit fixed and let the upper limit vary, the integral becomes a *function* of that upper limit — and it is worth asking what kind of function we get.

14.1 Definition and Basic Formula

To make clear that the upper limit is now the variable, we rename it x and use a different letter, say u, for the variable of integration. (Recall that the name of the integration variable is irrelevant: $\int_a^b f(x)\,dx = \int_a^b f(u)\,du$.)

Definition 14.1.
Let f be continuous on an interval I, and let $\alpha \in I$. The function

$$\Phi_\alpha(x) = \int_\alpha^x f(u)\,du \qquad (x \in I)$$

is called **an indefinite integral** of f. For $x < \alpha$ and $x = \alpha$ the integral is understood in the sense of Definition 13.1; in particular $\Phi_\alpha(\alpha) = 0$.

Remark 14.1. Note the article: *an* indefinite integral, not *the* indefinite integral. If we choose a different lower limit β, we obtain a different indefinite integral,

$$\Phi_\beta(x) = \int_\beta^x f(u)\,du.$$

As we will see in a moment, the choice of the lower limit is not that important. So we fix α once and for all and simply write

$$\Phi(x) = \Phi_\alpha(x).$$

Proposition 14.1 (Basic properties of the indefinite integral)**.** *Let f be continuous on I and $\Phi(x) = \int_\alpha^x f(u)\,du$.*

1. *If we know Φ, we can compute **any** definite integral of f: for all $a, b \in I$,*

$$\int_a^b f(x)\,dx = \Phi(b) - \Phi(a).$$

2. *Any other indefinite integral differs from Φ by a constant: for every $\beta \in I$,*

$$\Phi_\beta(x) = \int_\beta^x f(u)\,du = \Phi(x) - \Phi(\beta).$$

Proof.

(1) By additivity of the integral (Proposition 13.1, in the general form of Exercise 13.1, since a, b, α may lie in any order) and by Definition 13.1,

$$\begin{aligned}\int_a^b f(x)\,dx &= \int_a^\alpha f(x)\,dx + \int_\alpha^b f(x)\,dx \\ &= -\int_\alpha^a f(x)\,dx + \int_\alpha^b f(x)\,dx \\ &= \Phi(b) - \Phi(a).\end{aligned}$$

(2) This is the special case $a = \beta$, $b = x$ of (1).

□

Remark 14.2. In (2) the constant $-\Phi(\beta)$ depends on β, but **not** on x. So all the indefinite integrals of f have graphs that are vertical translates of one another; changing the lower limit only shifts the graph up or down. Part (1) generalizes Corollary 13.1: there the lower limit a played the role of α.

14.2 Properties of Φ as a Function

What properties does $\Phi(x)$ have as a function? Geometrically, $\Phi(x)$ is the signed area (Remark 13.1) under the graph of f between α and x, and moving x to the right adds the signed area over the new piece of the interval.

Proposition 14.2.

- *If $f \geq 0$ on I, then Φ is increasing (1): as x gets bigger we add positive area.*
- *If $f \leq 0$ on I, then Φ is decreasing: as x gets bigger we add negative area.*

Proof. Let $x < y$ in I (for $x = y$ there is nothing to show). By Proposition 14.1,

$$\Phi(y) - \Phi(x) = \int_x^y f(u)\,du.$$

If $f \geq 0$, this integral is ≥ 0 by Proposition 13.5, so $\Phi(x) \leq \Phi(y)$. If $f \leq 0$, apply Proposition 13.5 to $-f \geq 0$ and use Proposition 13.3 to get $\int_x^y f(u)\,du \leq 0$, so $\Phi(x) \geq \Phi(y)$. □

14.2.1 Continuity of the indefinite integral

Theorem 14.1.
If f is continuous on I, then Φ is continuous on I.

Proof. Let $x, y \in I$. By Proposition 14.1 and the Mean Value Theorem for Integrals (Theorem 13.1, which requires f continuous; for $y < x$ use Remark 13.4, part 3),

$$\Phi(y) - \Phi(x) = \int_x^y f(u)\,du = f(c)\,(y - x) \qquad (*)$$

for some c between x and y.

What happens if $y \to x$? For Φ to be continuous at x, we need $\Phi(y) \to \Phi(x)$ (Proposition 10.2). Using $(*)$ and the limit laws (Proposition 10.3),

$$\begin{aligned}\lim_{y\to x} \Phi(y) &= \lim_{y\to x} \big[\Phi(x) + f(c)\,(y - x)\big] \\ &= \lim_{y\to x} \Phi(x) + \lim_{y\to x} f(c)\,(y - x) \\ &= \Phi(x) + f(x) \cdot 0 = \Phi(x).\end{aligned}$$

□

Remark 14.3. The point c in $(*)$ depends on y, so a little care is needed in the last step. Since c lies between x and y, we have $c \to x$ as $y \to x$, and hence $f(c) \to f(x)$ by continuity of f. In fact, all we really need is that $f(c)$ stays **bounded** while $y - x \to 0$ — which is exactly what the next result exploits.

Proposition 14.3 (Lipschitz continuity)**.** *Let f be continuous on I, let $[a,b] \subseteq I$, and let $M = \max_{[a,b]} |f|$ (this maximum exists by the Extreme Value Theorem, see Remark 11.2). Then for all $x, y \in [a,b]$,*

$$|\Phi(y) - \Phi(x)| \le M\,|y - x|.$$

A function satisfying an estimate of this kind is called ***Lipschitz continuous****.*

Proof. By $(*)$, $|\Phi(y) - \Phi(x)| = |f(c)|\,|y - x|$ with c between x and y, so $c \in [a,b]$ and $|f(c)| \le M$. □

Remark 14.4. Lipschitz continuity is a stronger property than continuity: given $\varepsilon > 0$, the choice $\delta = \varepsilon/(M+1)$ works at **every** point of $[a,b]$ simultaneously (compare Example 6.3, where the same kind of δ appeared). In other words, Φ is *uniformly continuous* on $[a,b]$ in the sense of Equation 11.1, and here we get this directly, with an explicit δ.

14.3 Examples: Volumes and Dipsticks

Indefinite integrals arise naturally whenever a quantity accumulates as a variable grows. A good example is the volume of liquid in a container, as a function of the fill height.

Suppose a container has a vertical axis, and let $A(u)$ be the area of its horizontal cross-section at height u above the bottom. Slicing the container into thin horizontal layers of thickness Δ_i, the volume up to height x is approximately $\sum_i A(u_i^*)\,\Delta_i$ — a Riemann sum (Definition 11.1) for A on $[0,x]$. Letting the layers get thinner, we obtain the volume as an indefinite integral of A with lower limit $\alpha = 0$:

$$V(x) = \int_0^x A(u)\,du.$$

14.3.1 A cone-shaped glass

Example 14.1 (Constructing a dipstick)**.** A glass has the shape of an (upside-down) cone with rim radius R and height H. At which height h is the glass half full?

The cross-section at height u is a disk. Its radius grows linearly from 0 at the bottom to R at the rim, so it equals $R \cdot \frac{u}{H}$, and

$$A(u) = \pi\Big(\frac{R\,u}{H}\Big)^2 = \pi\,\frac{R^2}{H^2}\,u^2.$$

By Proposition 13.3 and Proposition 13.4 (or directly by the Riemann sum computation of $\int_0^b x^2\,dx$ in the previous chapters),

$$V(x) = \int_0^x A(u)\,du = \pi\,\frac{R^2}{H^2}\int_0^x u^2\,du = \pi\,\frac{R^2}{H^2}\cdot\frac{x^3}{3}.$$

In particular, the total volume is

$$V_T = V(H) = \pi\,\frac{R^2}{H^2}\cdot\frac{H^3}{3} = \frac{\pi}{3}R^2H,$$

the familiar formula for the volume of a cone.

We want to know for which x we have $V(x) = \frac{1}{2}V_T$:

$$\pi\,\frac{R^2}{H^2}\cdot\frac{x^3}{3} = \pi\,\frac{R^2}{H^2}\cdot\frac{H^3}{3}\cdot\frac{1}{2} \quad\Leftrightarrow\quad x^3 = \frac{H^3}{2} \quad\Leftrightarrow\quad x = \frac{H}{\sqrt[3]{2}} \approx 0.79\,H.$$

So the "half full" mark on the dipstick sits at about 79% of the height, not at 50%. Indeed, filling the glass to half its height gives only

$$V(\tfrac{1}{2}H) = \pi\,\frac{R^2}{H^2}\cdot\frac{H^3}{24} = \frac{1}{8}\,V(H),$$

an eighth of the total volume.

Remark 14.5. **Why does a height x with $V(x) = \frac{1}{2}V_T$ exist?** Here we found x explicitly, but there is a reason that works for *any* container. By Theorem 14.1, V is continuous on $[0, H]$. Moreover $V(0) = 0$ and $V(H) = V_T$, so

$$V(0) = 0 < \tfrac{1}{2}V_T < V_T = V(H).$$

By the Intermediate Value Theorem (Theorem 10.2, in the general form of Remark 10.4), there exists $c \in (0, H)$ with $V(c) = \frac{1}{2}V_T$. The same argument gives a mark for every fraction of the total volume — a dipstick can always be calibrated, even if we cannot compute V in closed form.

14.3.2 A horizontal oil tank

Example 14.2 (Oil tank)**.** An oil tank is a cylinder of radius R and length L, buried **horizontally** in the ground. A measuring stick lowered vertically through the top measures the height x of the oil in the tank. Reading the stick, how much oil (volume) is left?

(Why not bury the tank vertically? The math would be easier: then every cross-section is a disk of area πR^2, and $V(x) = \pi R^2 x$ is linear in x.)

Step 1: the area of the cross-section, as a function of u. At height u (with $0 \le u \le 2R$) the oil surface is a rectangle of length L and width $W(u)$, so $A(u) = L \cdot W(u)$. To find $W(u)$, look at the circular end of the tank. The center of the circle lies at height R, so the horizontal chord at height u is at distance $|R-u|$ from the center. By Pythagoras, half the chord has length

$$r(u) = \sqrt{R^2 - (R-u)^2} = \sqrt{2Ru - u^2} = \sqrt{u\,(2R-u)},$$

and therefore

$$W(u) = 2\sqrt{u\,(2R-u)}, \qquad A(u) = 2L\sqrt{u\,(2R-u)}.$$

Step 2: the volume. Hence the volume at height x is

$$V(x) = \int_0^x 2L\sqrt{u\,(2R-u)}\,du, \qquad 0 \le x \le 2R.$$

We do not know how to integrate this function yet — it is not covered by Proposition 13.4. But we can still compute $V(x)$ as accurately as we like using Riemann sums (Theorem 11.1).

The interactive figure below shows the end view of the tank. Move the slider to change the fill height x (shown in units of R). The coral strip is a typical thin slice of a Riemann sum for $V(x)$. The graph on the right plots the volume $V(x)$, as a percentage of the full tank, horizontally against the height x vertically. The orange dashed guides show how the 25%, 50% and 75% marks on the dipstick are found: start at the desired volume, go up to the curve, then across to read off the height. This is solving $V(x) = c$, whose solvability is guaranteed by the argument of Remark 14.5.

An interactive version of this figure, with a slider for the fill height, is included in the HTML version of these notes.

Two things can be seen directly in the picture. First, the 25% and 75% marks sit closer to the 50% mark than to the ends of the stick, because the tank is widest in the middle. Second, the volume changes fastest near mid-height, where the cross-section $A(u)$ is largest (on the graph, the curve is flattest there, since the height is plotted vertically). This is the Lipschitz bound of Remark 14.6 below in action: a misread of the stick costs the most volume where the tank is widest.

Remark 14.6. **Some remarks.**

1. $V(y) - V(x)$ is the **total change** in volume between heights x and y, and by Proposition 14.1,
$$V(y) - V(x) = \int_x^y A(u)\,du.$$

2. Lipschitz continuity of V (Proposition 14.3) is of practical use. Suppose we botch our measurement. How wrong can our calculated volume at most be? Let x_* be our measurement and x the actual height. Then

$$|V(x_*) - V(x)| \leq M\,|x_* - x|,$$

where $M = \max_{[0,2R]} A$. So the error in the volume is at most a constant multiple of the error in the measurement. For the oil tank we can even give M explicitly: since $u(2R-u) = R^2 - (R-u)^2 \leq R^2$, the cross-section is widest at the center height $u = R$, and $M = A(R) = 2LR$. So a reading that is off by 1 cm misjudges the volume by at most $2LR \cdot 1\,\text{cm}$ — the volume of a layer 1 cm thick through the widest part of the tank.

Part V

The Derivative

15 The Derivative

Companion reading

This lecture corresponds to C&J §2.8.a (the derivative and the tangent) and §2.8.c (examples of differentiation). The difference quotient and Leibniz's notation are discussed further in §2.8.g.

15.1 The Tangent Problem

How should we define the **tangent** to a given curve $y = f(x)$ at a point P on the curve? Intuitively, the tangent is the line through P that points in the direction in which the curve is heading at P. Since every candidate line passes through P, a line is completely characterized by its *direction*: the angle α it makes with the line through P parallel to the x-axis, measured counterclockwise (Figure 15.1).

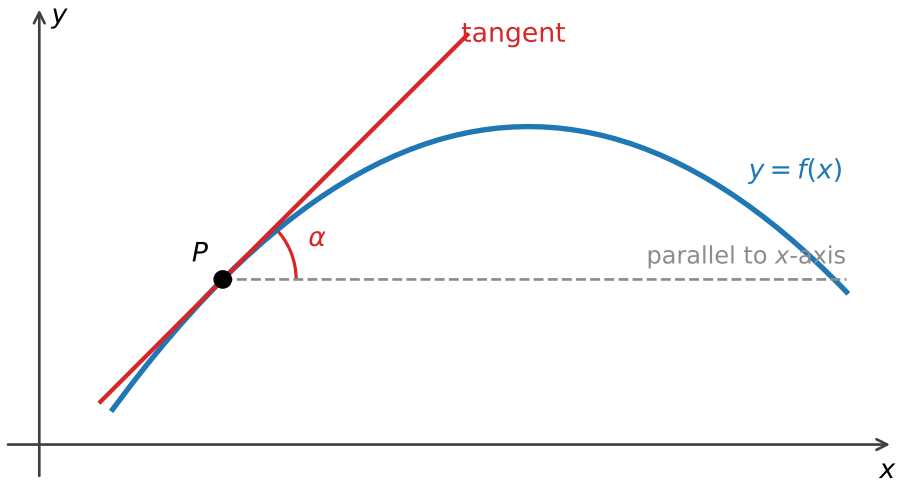


Figure 15.1: The tangent at P is completely characterized by the angle α it makes with the line through P parallel to the x-axis.

Two observations guide the definition.

1. Knowing the point $P = (x, f(x))$ *alone* does not determine the tangent: infinitely many lines pass through P. We need to know the curve at other points, too.
2. On the other hand, only the points *very, very close* — "infinitesimally close" — to P have an influence on the tangent. At any point $Q \neq P$ we can change the curve without changing the tangent at P, as long as we leave the curve alone in some small neighborhood of P (Figure 15.2).

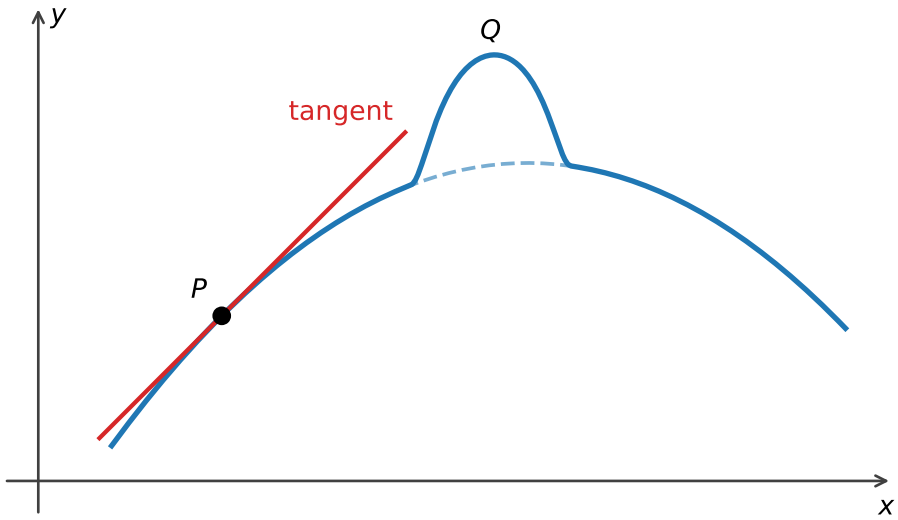


Figure 15.2: Changing the curve away from P (solid) does not change the tangent at P: the original curve (dashed) and the modified one agree near P.

Together, these observations suggest that the tangent should be defined via a **limiting process**.

15.2 Secants

A **secant** is a line through P and a second point P_1 of the curve. Unlike the tangent, a secant is easy to describe, since it is determined by two points. Now move P_1 towards P along the curve. As P_1 approaches P, the secant should approach the tangent (Figure 15.3):

$$\text{secant} \longrightarrow \text{tangent}.$$

This picture rests on an assumption: that the secants actually settle down to a definite limiting position, and that this position is **independent of the side** from which P_1 approaches P. In other words, we are assuming that the tangent *exists*. This is a strong assumption, and it need not be true.

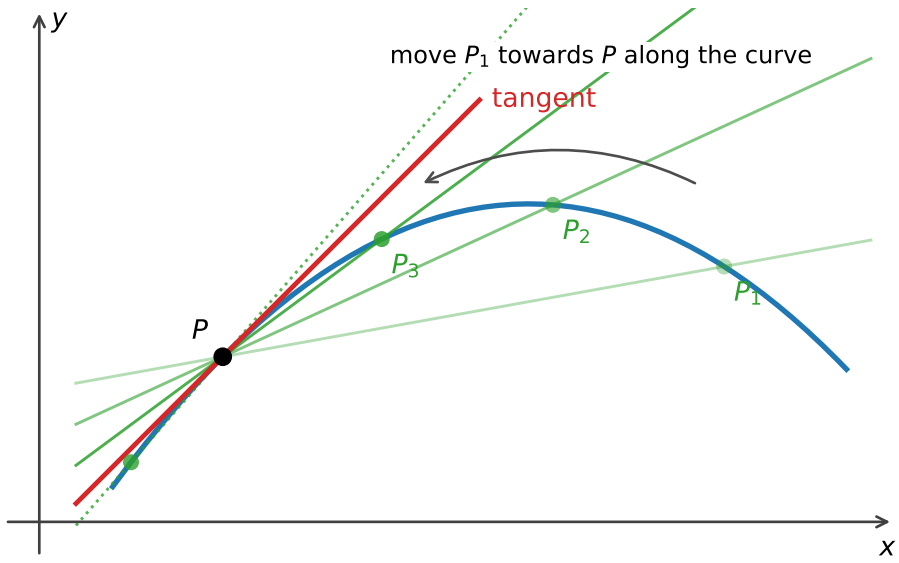


Figure 15.3: As P_1 moves along the curve towards P, the secants through P and P_1 approach the tangent at P. The limiting position should not depend on the side from which P_1 approaches P.

Example 15.1.
Let $f(x) = |x|$ and $P = (0, 0)$. For $x_1 > 0$, the secant through P and $P_1 = (x_1, x_1)$ is the line $y = x$, no matter how close x_1 is to 0. For $x_1 < 0$, the secant through P and $P_1 = (x_1, -x_1)$ is the line $y = -x$. Each side produces a perfectly definite limiting position, but the two sides disagree, just as one-sided limits can (Remark 10.2). So $|x|$ has no tangent at $(0, 0)$: the graph has a corner there (Figure 15.4, left).

Exercise 15.1. Let $f(x) = \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ (Remark 6.5). Does the curve $y = f(x)$ have a tangent at $P = (0, 0)$?

Solution. No. The secant through P and $P_1 = (x_1, \sin(1/x_1))$ has slope $\sin(1/x_1)/x_1$.

- For $x_1 = \dfrac{1}{n\pi}$ the slope is 0: the secant is the x-axis.
- For $x_1 = \dfrac{1}{2\pi n + \pi/2}$ the slope is $2\pi n + \pi/2$, which grows without bound: these secants become steeper and steeper.

Points of both kinds come arbitrarily close to P as $n \to \infty$, so the secants do not settle down to any single limiting position. (In fact f is not even continuous at 0, so the graph does not pass through P in any reasonable sense.)

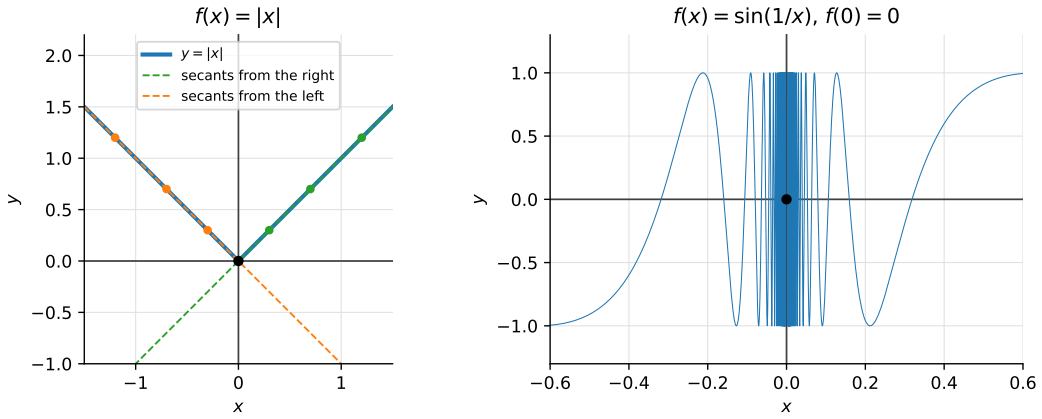


Figure 15.4: Two curves without a tangent at the origin. Left: for $|x|$, secants from the right are all the line $y = x$, secants from the left are all the line $y = -x$. Right: the graph of $\sin(1/x)$ (with value 0 at $x = 0$) near the origin.

Damping the oscillation does not help. For $g(x) = x\sin(1/x)$, $g(0) = 0$, which *is* continuous at 0 (Example 6.4), the secant through $(0,0)$ and $(x_1, g(x_1))$ has slope $\sin(1/x_1)$, which keeps oscillating between -1 and 1 as $x_1 \to 0$. So continuity alone does not guarantee a tangent.

15.3 From Angles to Slopes

If the tangent at P exists, how do we formulate it *analytically*, that is, in terms of f? We use angles. Let α_1 be the angle between the line through P parallel to the x-axis and the secant PP_1, again measured counterclockwise (Figure 15.5). Informally,

$$\lim_{P_1 \to P} \alpha_1 = \alpha.$$

Write $P = (x, y)$ and $P_1 = (x_1, y_1)$, where $y = f(x)$ and $y_1 = f(x_1)$. The secant is the hypotenuse of a right triangle with horizontal leg $x_1 - x$ and vertical leg $y_1 - y$, so by trigonometry

$$\tan(\alpha_1) = \frac{y_1 - y}{x_1 - x} = \frac{f(x_1) - f(x)}{x_1 - x}.$$

(If $x_1 < x$, both the numerator and the denominator change sign, and the formula remains valid.) Hence we can express the limiting process of finding the tangent at P as

$$\lim_{x_1 \to x} \frac{f(x_1) - f(x)}{x_1 - x} = \lim_{x_1 \to x} \tan(\alpha_1) = \tan(\alpha).$$

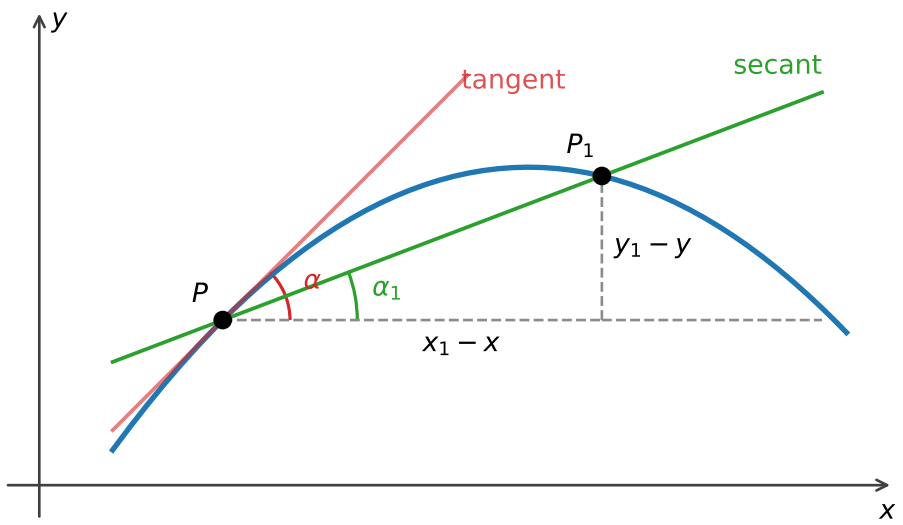


Figure 15.5: The secant through $P = (x, y)$ and $P_1 = (x_1, y_1)$ forms a right triangle with legs $x_1 - x$ and $y_1 - y$, so $\tan(\alpha_1) = (y_1 - y)/(x_1 - x)$.

Remark 15.1. Measuring counterclockwise from the horizontal, we take angles in $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ (clockwise counts as negative). These angles cover every non-vertical line through P, and tan maps this interval one-to-one onto all of $\mathbb{R}$. So a non-vertical line through P is described equally well by its angle α or by its **slope** $m = \tan(\alpha)$, and since tan is continuous (Remark 10.3), $\alpha_1 \to \alpha$ gives $\tan(\alpha_1) \to \tan(\alpha)$ (Proposition 10.4). From now on we work with slopes: they come directly from f and need no trigonometry. (C&J measure angles in $0 \le \alpha < \pi$ instead; with slopes the convention does not matter.)

A **vertical** tangent ($\alpha = \frac{\pi}{2}$) has no slope, and this approach excludes it. We will meet one in Example 15.4.

Definition 15.1.
The quotient

$$\frac{f(x_1) - f(x)}{x_1 - x} = \frac{y_1 - y}{x_1 - x} = \frac{\Delta y}{\Delta x}$$

is called the **difference quotient** of f at x. Here $\Delta y = y_1 - y$ and $\Delta x = x_1 - x$ denote the differences of the function values and of the independent variable. The symbol Δ is an abbreviation for "difference", not a factor.

15.4 The Derivative

Definition 15.2.
Let f be defined on an open interval containing x. If the limit

$$f'(x) = \lim_{x_1 \to x} \frac{f(x_1) - f(x)}{x_1 - x}$$

exists, we say f is **differentiable** at x, and the limit is called the **derivative of f at x**.

The tangent to $y = f(x)$ at $P = (x, f(x))$ is then the line through P with slope $f'(x)$, that is, the graph of $t \mapsto f(x) + f'(x)\,(t - x)$.

Remark 15.2. The difference quotient, as a function of x_1, is *not defined* at $x_1 = x$: there it would read $\frac{0}{0}$. This is no obstacle. Definition 10.1 only tests sequences $x_n \to x$ with $x_n \neq x$, so the limit depends only on the values of the difference quotient *near* x, never *at* x. This is exactly what observation (2) above asks for.

Notation. There are two common notations for the derivative:

$$f'(x) \quad \text{(Lagrange)}, \qquad \frac{df(x)}{dx} \quad \text{or} \quad \frac{d}{dx} f(x) \quad \text{(Leibniz)}.$$

If $y = f(x)$, one also writes y' or $\dfrac{dy}{dx}$. Leibniz's notation is meant to recall the difference quotient $\frac{\Delta y}{\Delta x}$, as a quotient of "infinitely small" differences. For us, however, $\frac{dy}{dx}$ is a single symbol standing for the limit in Definition 15.2, not a quotient of two numbers dy and dx.

Remark 15.3. **Important:** the derivative is itself a function of x. For each x at which f is differentiable we get a number $f'(x)$, so f' is a new function $x \mapsto f'(x)$, defined at (at least some of) the points of the domain of f. Figure 15.6 shows this for $f(x) = x^2$.

Substituting $x_1 = x + h$ gives an alternative form of the difference quotient, which is often more convenient for calculations:

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}.$$

The two limits are the same: sequences $x_n \to x$ with $x_n \neq x$ correspond exactly to sequences $h_n = x_n - x \to 0$ with $h_n \neq 0$, and the corresponding difference quotients agree term by term. In the notation of Definition 15.1, $h = \Delta x$.

15.5 Examples

Example 15.2.
Let $f(x) = x^2$. For $x_1 \neq x$,

$$\frac{f(x_1) - f(x)}{x_1 - x} = \frac{x_1^2 - x^2}{x_1 - x} = \frac{(x_1 - x)(x_1 + x)}{x_1 - x} = x_1 + x.$$

Cancelling the factor $x_1 - x$ is allowed precisely because $x_1 \neq x$ (Remark 15.2). Since $x_1 \mapsto x_1 + x$ is continuous,

$$\lim_{x_1 \to x} \frac{f(x_1) - f(x)}{x_1 - x} = \lim_{x_1 \to x} (x_1 + x) = 2x$$

by Proposition 10.2. Hence f is differentiable at every $x \in \mathbb{R}$, with

$$f'(x) = 2x.$$

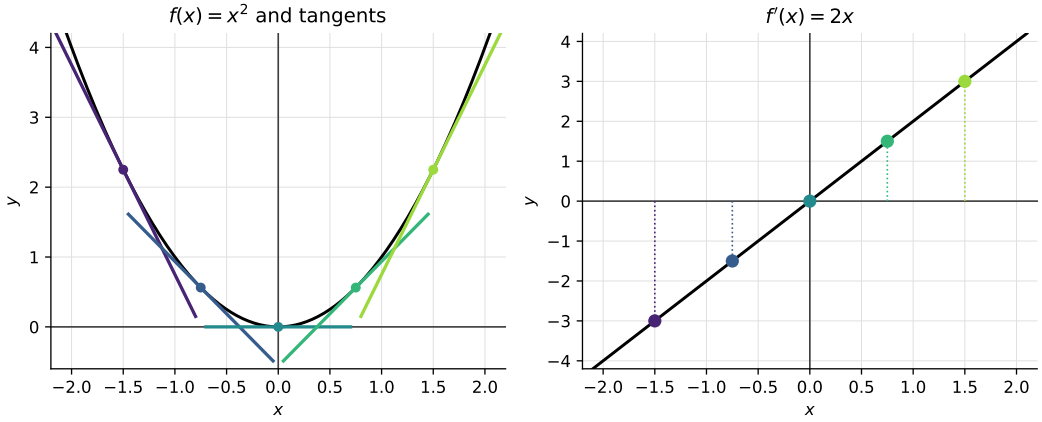


Figure 15.6: Left: $f(x) = x^2$ with its tangents at several points. Right: the derivative $f'(x) = 2x$. The height of each dot on the right is the slope of the tangent of the same color on the left.

Example 15.3.
Let $f(x) = x^3$. This time we use the h-form. For $h \neq 0$,

$$\frac{f(x+h) - f(x)}{h} = \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = 3x^2 + 3xh + h^2.$$

Hence

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} (3x^2 + 3xh + h^2) = 3x^2,$$

so

$$f'(x) = 3x^2.$$

Alternatively, we can work with x_1 directly and factor:

$$\frac{x_1^3 - x^3}{x_1 - x} = \frac{(x_1 - x)(x_1^2 + x_1 x + x^2)}{x_1 - x} = x_1^2 + x_1 x + x^2,$$

and therefore

$$\lim_{x_1 \to x} \frac{x_1^3 - x^3}{x_1 - x} = \lim_{x_1 \to x} (x_1^2 + x_1 x + x^2) = 3x^2.$$

In both versions, the limit is evaluated by plugging in, since the expression is a polynomial in h (respectively x_1) and polynomials are continuous.

Example 15.4.
Let $f(x) = \sqrt{x}$ and $x > 0$. For $x_1 \geq 0$, $x_1 \neq x$, we rationalize the numerator:

$$\frac{f(x_1) - f(x)}{x_1 - x} = \frac{\sqrt{x_1} - \sqrt{x}}{x_1 - x} = \frac{(\sqrt{x_1} - \sqrt{x})(\sqrt{x_1} + \sqrt{x})}{(x_1 - x)(\sqrt{x_1} + \sqrt{x})} = \frac{x_1 - x}{(x_1 - x)(\sqrt{x_1} + \sqrt{x})} = \frac{1}{\sqrt{x_1} + \sqrt{x}}.$$

Since $\sqrt{\ }$ is continuous (Remark 10.3) and the denominator tends to $2\sqrt{x} \neq 0$, the limit laws (Proposition 10.3) give

$$\lim_{x_1 \to x} \frac{f(x_1) - f(x)}{x_1 - x} = \lim_{x_1 \to x} \frac{1}{\sqrt{x_1} + \sqrt{x}} = \frac{1}{2\sqrt{x}}, \qquad \text{so} \qquad f'(x) = \frac{1}{2\sqrt{x}} \quad (x > 0).$$

Singularity at $x = 0$. The formula $\frac{1}{2\sqrt{x}}$ breaks down at $x = 0$, and so does differentiability. Since $\sqrt{x}$ is only defined for $x \geq 0$, P_1 can only approach $P = (0, 0)$ from the right, and the difference quotient there is

$$\frac{\sqrt{x_1} - 0}{x_1 - 0} = \frac{1}{\sqrt{x_1}} \longrightarrow \infty \qquad (x_1 \to 0^+).$$

So even the one-sided limit (Definition 10.2) does not exist as a real number: the secants become steeper and steeper, and their limiting position is the y-axis. The curve has a *vertical* tangent at the origin, the case excluded in Remark 15.1 (Figure 15.7).

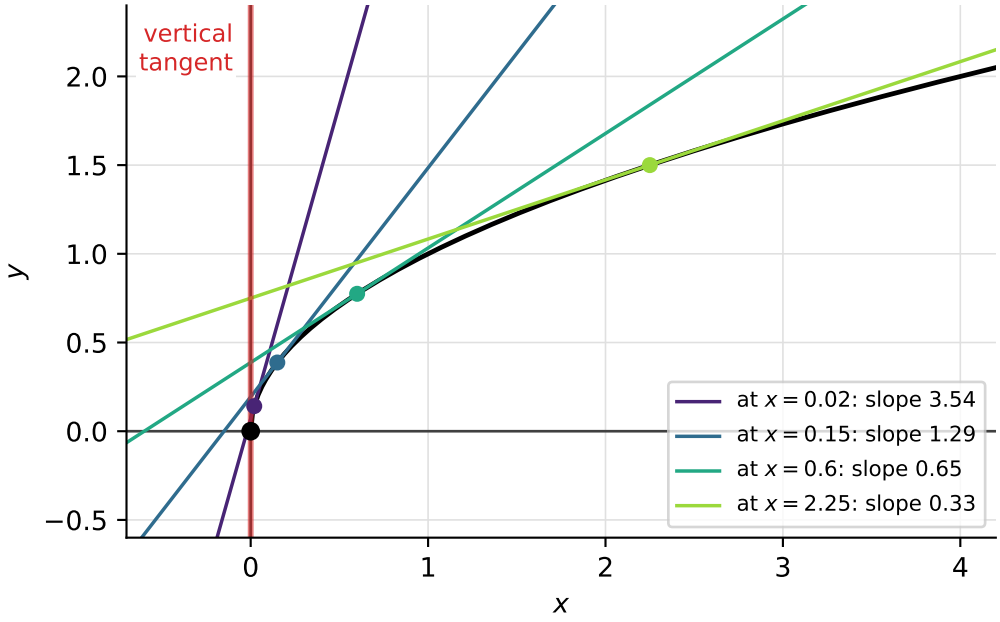


Figure 15.7: Tangents to $y = \sqrt{x}$ with slopes $1/(2\sqrt{x})$. As the point of tangency approaches 0 the tangents become steeper without bound; at the origin the tangent is vertical and $\sqrt{x}$ is not differentiable.

15.6 Summary

- The **tangent** to $y = f(x)$ at P is determined by its angle with the horizontal. That angle cannot be read off from P alone, but it depends only on the curve arbitrarily close to P. This calls for a limiting process (Figure 15.1, Figure 15.2).
- The tangent is the limiting position of the **secants** PP_1 as $P_1 \to P$ along the curve, from either side (Figure 15.3). Such a limiting position need not exist: $|x|$ has a corner at the origin (Example 15.1), and $\sin(1/x)$ oscillates too wildly (Exercise 15.1).
- The slope of the secant is the **difference quotient** $\frac{\Delta y}{\Delta x} = \frac{f(x_1)-f(x)}{x_1-x} = \tan(\alpha_1)$ (Definition 15.1). Working with slopes instead of angles loses nothing except vertical tangents (Remark 15.1).
- The **derivative** is the limit of the difference quotient, $f'(x) = \lim_{x_1 \to x} \frac{f(x_1)-f(x)}{x_1-x} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$ (Definition 15.2), written $f'(x)$ (Lagrange) or $\frac{df}{dx}$ (Leibniz). It is itself a **function** of x (Remark 15.3).
- First examples: $(x^2)' = 2x$, $(x^3)' = 3x^2$, and $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$ for $x > 0$ (Example 15.2, Example 15.3, Example 15.4). In each case the trick is to cancel the factor $x_1 - x$ (or h), which is legitimate because $x_1 \neq x$. The square root is not differentiable at 0, where its graph has a vertical tangent.

16 Working with Derivatives

16.1 Interpretations of the Derivative

16.1.1 Rate of change

If $y = f(x)$, the difference quotient (Definition 15.1)

$$\frac{\Delta y}{\Delta x} = \frac{f(x_1) - f(x)}{x_1 - x}$$

measures how much y changes per unit change of x between x and x_1: it is the **average rate of change** of f on the interval between x and x_1. Its limit as $x_1 \to x$, the derivative $f'(x)$, is the **instantaneous rate of change** of f at x.

16.1.2 Velocity

Consider a point moving along a straight line, which we take to be the y-axis, so that y is the (signed) distance of the point from the origin. The motion is described by giving the position as a function of time:

$$y = f(t).$$

Uniform motion. If f is linear, $f(t) = mt+b$, the motion is **uniform** with **velocity** m: in t units of time, the point moves $m \cdot t$ units of distance. In this case we recover m from any two distinct times $t_1 \neq t$ as

$$\frac{f(t_1) - f(t)}{t_1 - t} = \frac{mt_1 + b - mt - b}{t_1 - t} = m \cdot \frac{t_1 - t}{t_1 - t} = m.$$

Non-uniform motion. If f is not linear, the same quotient

$$\frac{f(t_1) - f(t)}{t_1 - t}$$

depends on t_1. It is the **average velocity** of the point in the time interval between t and t_1: the velocity of a uniform motion that covers the same distance in the

same time. Letting $t_1 \to t$, the average velocity converges, *if the limit exists*, to the **velocity at time** t,

$$f'(t) = \lim_{t_1 \to t} \frac{f(t_1) - f(t)}{t_1 - t}.$$

In other words, velocity is the instantaneous rate of change of position. Geometrically, the average velocity is the slope of a secant of the graph of f, and the velocity at time t is the slope of the tangent (Figure 16.1).

Notation. Newton, who thought of derivatives primarily as velocities, wrote $\dot{y}$ or $\dot{f}(t)$ for the derivative with respect to time. This notation is still common in physics.

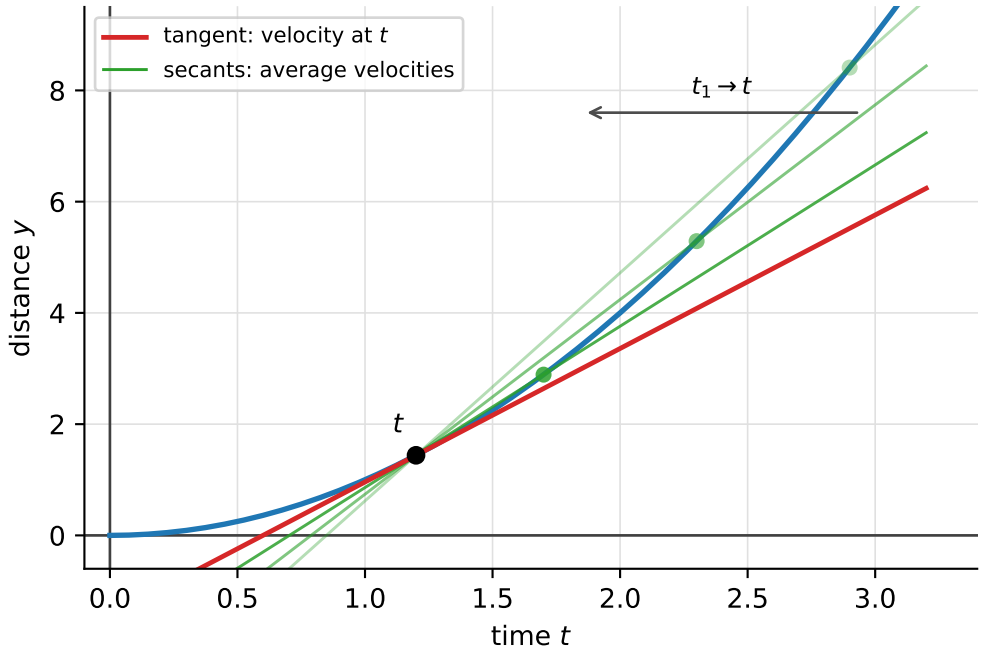


Figure 16.1: Position $y = f(t)$ of a freely falling body. The average velocity on $[t, t_1]$ is the slope of the secant; as $t_1 \to t$ it approaches the velocity at time t, the slope of the tangent.

Example 16.1 (Free fall)**.**
Experiments (going back to Galileo) show that the distance traversed in time t by a freely falling body, starting at rest, is proportional to t^2:

$$y = f(t) = at^2$$

for some constant $a > 0$. For $t_1 \neq t$,

$$\frac{f(t_1) - f(t)}{t_1 - t} = a\,\frac{t_1^2 - t^2}{t_1 - t} = a\,(t_1 + t) \;\longrightarrow\; 2at \qquad (t_1 \to t),$$

exactly as in Example 15.2. Hence

$$f'(t) = 2at\,:$$

the velocity of a freely falling body **increases proportionally to the time** t. (In physics one writes $a = \frac{g}{2}$, where $g \approx 9.81\,\mathrm{m/s^2}$ is the gravitational acceleration, so that $f'(t) = gt$.)

16.2 Computing Derivatives

16.2.1 Linear functions

Example 16.2.
Let $f(x) = mx + b$. For $h \neq 0$,

$$f'(x) = \lim_{h\to 0} \frac{m(x+h) + b - mx - b}{h} = \lim_{h\to 0} \frac{mh}{h} = \lim_{h\to 0} m = m.$$

The derivative of a linear function is the constant m, its slope, as it should be: the tangent to a line is the line itself. In particular ($m = 0$), the derivative of a **constant** function is 0.

Remark 16.1. **Writing limits.** In a chain of equalities such as the one above, every "=" must be a true statement. So the symbol $\lim_{h\to 0}$ stays in front of the expression until the limit is actually evaluated. Writing

$$f'(x) = \frac{m(x+h) + b - mx - b}{h} \qquad \text{(wrong)}$$

is false: the left side is a number that does not depend on h, the right side is a function of h (and undefined at $h = 0$). On the other hand, the step $\frac{mh}{h} = m$ inside the limit is legitimate, because the limit only involves $h \neq 0$ (Remark 15.2).

16.2.2 Power functions with integer exponents

We need an algebraic identity that generalizes $x_1^2 - x^2 = (x_1 - x)(x_1 + x)$ and $x_1^3 - x^3 = (x_1 - x)(x_1^2 + x_1 x + x^2)$, which we used in Example 15.2 and Example 15.3. For every integer $n \geq 1$,

$$x_1^n - x^n = (x_1 - x)(x_1^{n-1} + x_1^{n-2}x + \cdots + x_1 x^{n-2} + x^{n-1}). \tag{16.1}$$

To check it, multiply out the right-hand side: x_1 times the bracket gives $x_1^n + x_1^{n-1}x + \cdots + x_1 x^{n-1}$, and $-x$ times the bracket gives $-x_1^{n-1}x - \cdots - x_1 x^{n-1} - x^n$. All terms cancel except x_1^n and $-x^n$.

Proposition 16.1 (Power rule, positive integers)**.** *Let $n \geq 1$ be an integer and $f(x) = x^n$. Then f is differentiable at every $x \in \mathbb{R}$, and*

$$f'(x) = n\,x^{n-1}.$$

Proof. For $x_1 \neq x$, Equation 16.1 gives

$$\frac{f(x_1) - f(x)}{x_1 - x} = \frac{x_1^n - x^n}{x_1 - x} = x_1^{n-1} + x_1^{n-2}x + \cdots + x_1 x^{n-2} + x^{n-1}.$$

The right-hand side is a polynomial in x_1, hence continuous, so we can take the limit $x_1 \to x$ by plugging in $x_1 = x$ (Proposition 10.2). Each of the n terms becomes x^{n-1}, so

$$f'(x) = \lim_{x_1 \to x} \left(x_1^{n-1} + x_1^{n-2}x + \cdots + x_1 x^{n-2} + x^{n-1}\right) = n\,x^{n-1}.$$

□

For negative exponents we must assume $x \neq 0$, since x^{-n} is not defined at 0 (Definition 5.3).

Proposition 16.2 (Power rule, negative integers)**.** *Let $n \geq 1$ be an integer and $f(x) = x^{-n}$. Then f is differentiable at every $x \neq 0$, and*

$$f'(x) = -n\,x^{-n-1}.$$

In other words, with $\beta = -n$: $(x^\beta)' = \beta\,x^{\beta-1}$, the same formula as for positive exponents.

Proof. Let $x \neq 0$, and let $x_1 \neq x$ with $x_1 \neq 0$. Then

$$\frac{f(x_1) - f(x)}{x_1 - x} = \frac{\dfrac{1}{x_1^n} - \dfrac{1}{x^n}}{x_1 - x} = \frac{x^n - x_1^n}{x_1^n x^n\,(x_1 - x)} = -\frac{1}{x_1^n x^n} \cdot \frac{x_1^n - x^n}{x_1 - x}.$$

By Equation 16.1,

$$\frac{f(x_1) - f(x)}{x_1 - x} = -\frac{x_1^{n-1} + x_1^{n-2}x + \cdots + x_1 x^{n-2} + x^{n-1}}{x_1^n x^n}.$$

As $x_1 \to x$, the numerator tends to nx^{n-1} (as in the previous proof) and the denominator to $x^{2n} \neq 0$. By the limit laws (Proposition 10.3),

$$f'(x) = -\frac{n\,x^{n-1}}{x^{2n}} = -n\,x^{-n-1}.$$

□

So the **power rule**

$$\frac{d}{dx}\,x^{\alpha} = \alpha\, x^{\alpha-1}$$

holds for every integer α (for $\alpha = 0$ both sides are 0, if $x \neq 0$). We will extend it to arbitrary real exponents α later. We have already seen one instance: $(\sqrt{x})' = \frac{1}{2\sqrt{x}} = \frac{1}{2}x^{-1/2}$ (Example 15.4).

16.2.3 Trigonometric functions

To differentiate sin and cos we need two limits. Here, as always in calculus, angles are measured in **radians**.

Lemma 16.1.

$$\lim_{h\to 0}\frac{\sin h}{h} = 1, \qquad \lim_{h\to 0}\frac{\cos h - 1}{h} = 0.$$

Proof. Let $0 < h < \frac{\pi}{2}$, and consider the unit circle with $O = (0,0)$, $A = (1,0)$, $P = (\cos h, \sin h)$ and $T = (1, \tan h)$ (Figure 16.2). The triangle OAP lies inside the circular sector OAP, which lies inside the triangle OAT. Comparing areas,

$$\frac{1}{2}\sin h \;<\; \frac{h}{2} \;<\; \frac{1}{2}\tan h.$$

(The sector is the fraction $\frac{h}{2\pi}$ of the disk, whose area is π, so its area is $\frac{h}{2}$. This is where radian measure enters.) Dividing by $\frac{1}{2}\sin h > 0$ gives $1 < \frac{h}{\sin h} < \frac{1}{\cos h}$, that is,

$$\cos h \;<\; \frac{\sin h}{h} \;<\; 1.$$

Since $\cos(-h) = \cos h$ and $\frac{\sin(-h)}{-h} = \frac{\sin h}{h}$, these inequalities hold for $0 < |h| < \frac{\pi}{2}$. Now let $h_n \to 0$ with $h_n \neq 0$. Eventually $|h_n| < \frac{\pi}{2}$, and $\cos h_n \to \cos 0 = 1$ by continuity of cos (Remark 10.3). By the squeeze theorem (Proposition 8.3), $\frac{\sin h_n}{h_n} \to 1$. Since this holds for every such sequence, the first limit follows (Definition 10.1).

For the second limit, multiply numerator and denominator by $\cos h + 1$ and use $\cos^2 h - 1 = -\sin^2 h$:

$$\frac{\cos h - 1}{h} = \frac{\cos^2 h - 1}{h\,(\cos h + 1)} = -\frac{\sin h}{h}\cdot\frac{\sin h}{\cos h + 1} \;\longrightarrow\; -1\cdot\frac{0}{2} = 0$$

by the first limit, continuity of sin and cos, and the limit laws (Proposition 10.3). □

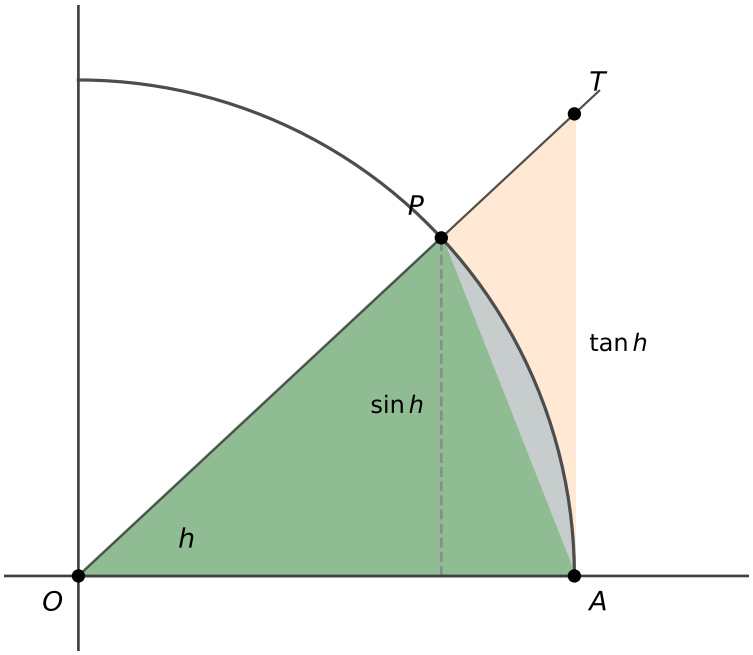


Figure 16.2: For $0 < h < \pi/2$: triangle $OAP \subset$ sector $OAP \subset$ triangle OAT, so $\frac{1}{2}\sin h < \frac{h}{2} < \frac{1}{2}\tan h$.

Proposition 16.3 (Derivatives of sine and cosine)**.** *For all* $x \in \mathbb{R}$*,*

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

Proof. By the addition formula $\sin(x+y) = \sin x \cos y + \cos x \sin y$, for $h \neq 0$,

$$\begin{aligned}\frac{\sin(x+h) - \sin x}{h} &= \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.\end{aligned}$$

Here x is fixed, so $\sin x$ and $\cos x$ are constants. By Lemma 16.1 and the limit laws (Proposition 10.3),

$$\lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h} = \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

The derivative of cos is Exercise 16.1. □

Exercise 16.1. Use the addition formula $\cos(x+y) = \cos x \cos y - \sin x \sin y$ and Lemma 16.1 to show that $(\cos x)' = -\sin x$.

Solution. For $h \neq 0$,

$$\frac{\cos(x+h) - \cos x}{h} = \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \tag{16.2}$$

$$= \cos x \cdot \frac{\cos h - 1}{h} - \sin x \cdot \frac{\sin h}{h} \longrightarrow \cos x \cdot 0 - \sin x \cdot 1 = -\sin x. \tag{16.3}$$

16.3 Sum and Constant Multiple Rules

As for the integral (Proposition 13.2, Proposition 13.3), there are basic rules that follow directly from the definition and the limit laws.

Proposition 16.4 (Sum rule)**.** *If* $f'(x)$ *and* $g'(x)$ *exist and* $\varphi(x) = f(x) + g(x)$*, then* $\varphi'(x)$ *exists and*

$$\varphi'(x) = f'(x) + g'(x).$$

Proof. For $h \neq 0$,

$$\frac{\varphi(x+h) - \varphi(x)}{h} = \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h} = \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}.$$

As $h \to 0$, the first quotient on the right tends to $f'(x)$ and the second to $g'(x)$. By the limit law for sums (Proposition 10.3), the left side tends to $f'(x) + g'(x)$. □

Proposition 16.5 (Constant multiple rule)**.** *If $f'(x)$ exists, $c \in \mathbb{R}$, and $\psi(x) = c\,f(x)$, then $\psi'(x)$ exists and*

$$\psi'(x) = c\,f'(x).$$

Proof. For $h \neq 0$,

$$\frac{\psi(x+h)-\psi(x)}{h} = c \cdot \frac{f(x+h)-f(x)}{h} \longrightarrow c\,f'(x) \qquad (h \to 0),$$

by the limit law for products, applied to the constant function c. □

Combining the two rules, $(\alpha f + \beta g)' = \alpha f' + \beta g'$ for all constants α, β; in particular $(f-g)' = f' - g'$. Together with the power rule, this lets us differentiate every polynomial.

Example 16.3.
For a polynomial $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, Proposition 16.1 and the two rules give

$$p'(x) = n a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$

For example, $(x^3 - 3x)' = 3x^2 - 3$. This is negative for $|x| < 1$ and positive for $|x| > 1$, so by Proposition 16.6, $x^3 - 3x$ is strictly increasing on $(-\infty, -1)$, strictly decreasing on $(-1, 1)$, and strictly increasing on $(1, \infty)$, with critical points ± 1 (Figure 16.5). Similarly,

$$\left(5x^{-2} + 2\sin x - 4\cos x\right)' = -10x^{-3} + 2\cos x + 4\sin x \qquad (x \neq 0).$$

Remark 16.2. As for integrals (Remark 13.2), there is **no** rule of the form $(fg)' = f'g'$. For $f(x) = g(x) = x$, we have $(x \cdot x)' = (x^2)' = 2x$, but $f'(x)\,g'(x) = 1 \cdot 1 = 1$. The correct *product rule* will come next, together with rules for quotients and compositions.

16.4 Differentiability vs. Continuity

Theorem 16.1.
If f is differentiable at x, then f is continuous at x.

Proof. Assume

$$\lim_{h \to 0} \frac{f(x+h)-f(x)}{h} = f'(x), \qquad -\infty < f'(x) < \infty.$$

For $h \neq 0$ we can write $f(x+h) - f(x) = \dfrac{f(x+h)-f(x)}{h} \cdot h$. By the limit law for products (Proposition 10.3),

$$\lim_{h\to 0} \big(f(x+h) - f(x)\big) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \cdot \lim_{h\to 0} h = f'(x) \cdot 0 = 0.$$

Hence $f(x+h) \to f(x)$ as $h \to 0$. Substituting $x_1 = x+h$ as in the previous chapter, this says $\lim_{x_1 \to x} f(x_1) = f(x)$, so f is continuous at x (Proposition 10.2). □

Remark 16.3. The proof uses that $f'(x)$ is a (finite) real number: the limit law gives "$f'(x) \cdot 0 = 0$", and "$\infty \cdot 0$" has no meaning. This is not a technicality. For the sign function, $\operatorname{sgn}(x) = 1$ for $x > 0$, $\operatorname{sgn}(0) = 0$, $\operatorname{sgn}(x) = -1$ for $x < 0$, the difference quotient at 0 is

$$\frac{\operatorname{sgn}(h) - \operatorname{sgn}(0)}{h} = \frac{1}{|h|} \longrightarrow \infty \qquad (h \to 0),$$

yet sgn is not continuous at 0.

As a consequence, every function we have differentiated so far (polynomials, x^{-n}, $\sqrt{x}$, sin, cos) is continuous wherever it is differentiable, without a separate continuity proof (compare Remark 10.3).

The converse is false. There are functions that are continuous everywhere but not differentiable at some point. The simplest is $f(x) = |x|$ at $x = 0$ (Example 15.1). For $h \neq 0$,

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The one-sided limits (Definition 10.2) exist but disagree: the **right-hand derivative** at 0 is 1 and the **left-hand derivative** is -1. So $\lim_{h\to 0} \frac{|h|}{h}$ does not exist, and $|x|$ is not differentiable at 0; its graph has a corner. Another example is $x \sin(1/x)$ (with value 0 at 0), which is continuous at 0 but has no tangent there (Exercise 15.1).

Differentiability thus expresses precisely what we would intuitively call *smoothness* of a graph. How badly can it fail?

Remark 16.4. **Question.** Does there exist a function that is continuous everywhere but differentiable *nowhere*?

Answer. Yes, surprisingly. The first published example is due to Weierstrass (1872):

$$W(x) = \sum_{k=0}^{\infty} a^k \cos\left(b^k \pi x\right),$$

for suitable constants $0 < a < 1$ and $b > 1$, for instance $a = \frac{1}{2}$, $b = 3$. Each term is a smooth wave, and the terms get smaller ($a^k \to 0$) but oscillate faster and

faster ($b^k \to \infty$). The series converges fast enough for W to be continuous, but the added oscillations create corners at every scale, so that no tangent exists anywhere. Zooming in on the graph does not make it look any straighter (Figure 16.3). Proving this takes tools beyond this course.

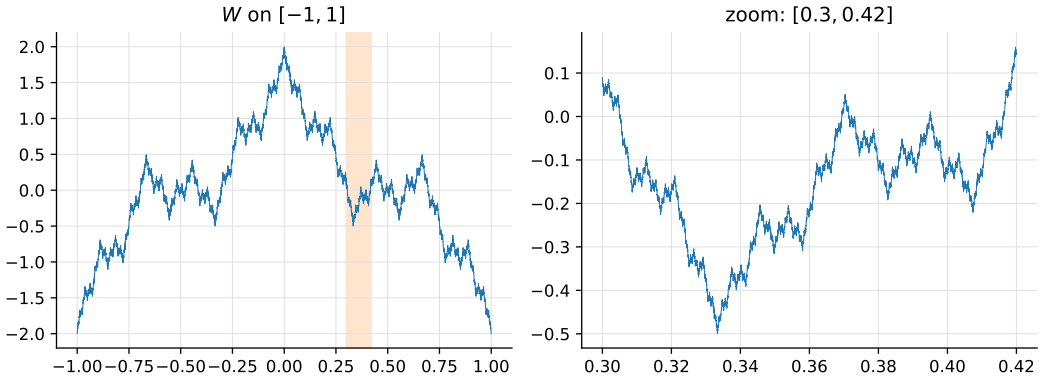


Figure 16.3: The Weierstrass function $W(x) = \sum_k 2^{-k} \cos(3^k \pi x)$ (partial sum with 13 terms). Right: zooming in on the shaded window reveals the same kind of roughness.

16.5 Infinite Derivatives

By the power rule, $\frac{d}{dx} x^\alpha = \alpha x^{\alpha-1}$. For some exponents the derivative is not defined at a point where the original function is.

Example 16.4.
Let $f(x) = x^{1/3} = \sqrt[3]{x}$, with domain $\mathbb{R}$ (Remark 5.2). For $x \neq 0$, the power rule predicts

$$f'(x) = \frac{1}{3}\, x^{-2/3},$$

which you can verify directly in Exercise 16.2. This formula is undefined at $x = 0$.

What happens at 0? The graph has a **vertical tangent** there (Figure 16.4). Analytically, for $h \neq 0$,

$$\frac{f(0+h) - f(0)}{h} = \frac{h^{1/3} - 0}{h} = h^{-2/3} = \frac{1}{\left(h^{1/3}\right)^2} \longrightarrow \infty \qquad (h \to 0).$$

The difference quotient grows beyond all bounds, from both sides, so f is not differentiable at 0 even though it is continuous there.

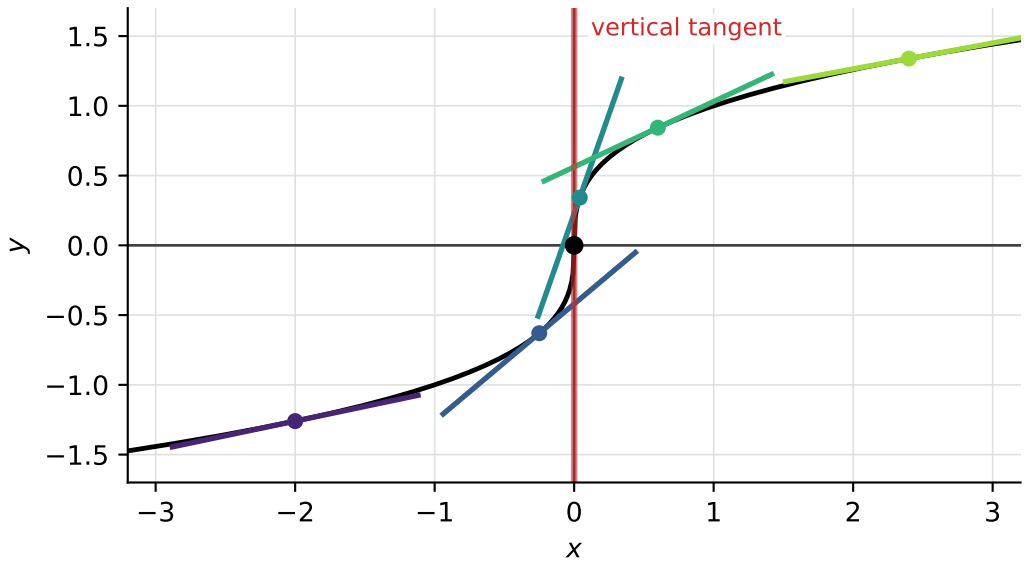


Figure 16.4: The graph of $y = \sqrt[3]{x}$, with tangents of slope $\frac{1}{3}x^{-2/3}$. As $x \to 0$ from either side the tangents become steeper without bound; at the origin the tangent is vertical.

One says that f has an **infinite derivative** at such a point. Keep in mind that this is only a manner of speaking: the derivative in the sense of Definition 15.2 does *not* exist, and the difference quotient merely increases beyond all bounds. We met the same phenomenon, from one side only, for $\sqrt{x}$ at 0 (Example 15.4).

Exercise 16.2. Let $x \neq 0$. Using the identity $u^3 - v^3 = (u - v)(u^2 + uv + v^2)$ with $u = x_1^{1/3}$ and $v = x^{1/3}$, show directly from Definition 15.2 that $(x^{1/3})' = \frac{1}{3}x^{-2/3}$. Where does your argument break down for $x = 0$?

Solution. For $x_1 \neq x$, with $u = x_1^{1/3}$, $v = x^{1/3}$ we have $u \neq v$ and $x_1 - x = u^3 - v^3$, so

$$\frac{x_1^{1/3} - x^{1/3}}{x_1 - x} = \frac{u - v}{(u - v)(u^2 + uv + v^2)} = \frac{1}{u^2 + uv + v^2}.$$

As $x_1 \to x$, $u \to v$ by continuity of the cube root, so the denominator tends to $3v^2 = 3x^{2/3}$. If $x \neq 0$ this is nonzero, and the limit laws give $\frac{1}{3x^{2/3}} = \frac{1}{3}x^{-2/3}$. If $x = 0$, then $v = 0$ and the denominator u^2 tends to 0: the quotient $\frac{1}{u^2}$ is unbounded.

Exercise 16.3. Let $g(x) = x^{2/3} = \left(\sqrt[3]{x}\right)^2$. Compute the difference quotient of g at 0 and describe what happens as $h \to 0^+$ and as $h \to 0^-$. Sketch the graph near the origin.

Solution. For $h \neq 0$, $\dfrac{g(h)-g(0)}{h} = \dfrac{h^{2/3}}{h} = h^{-1/3} = \dfrac{1}{\sqrt[3]{h}}$. As $h \to 0^+$ this tends to $+\infty$, and as $h \to 0^-$ it tends to $-\infty$. The graph has a **cusp** at the origin: both branches meet the y-axis vertically, coming down from above on either side. (C&J call this curve the *semicubical parabola*.)

16.6 What the Derivative Tells Us About a Function

If $f'(x) > 0$, the tangent at $(x, f(x))$ has positive slope, and we expect f to be "increasing at x". If $f'(x) < 0$, we expect f to be decreasing there. To make this precise, we have to talk about intervals.

Proposition 16.6 (Monotonicity and the sign of f')**.** *Let f be differentiable on an open interval I.*

1. *If f is increasing on I (1), then $f'(x) \geq 0$ for all $x \in I$. If f is decreasing on I, then $f'(x) \leq 0$ for all $x \in I$.*
2. *If $f'(x) > 0$ for all $x \in I$, then f is strictly increasing on I. If $f'(x) < 0$ for all $x \in I$, then f is strictly decreasing on I.*

Proof.

(1) Let f be increasing and $x \in I$. For $h > 0$ (with $x + h \in I$), $f(x+h) \geq f(x)$, so the difference quotient $\frac{f(x+h)-f(x)}{h}$ is ≥ 0. For $h < 0$, $f(x+h) \leq f(x)$: numerator and denominator are both ≤ 0, and again the quotient is ≥ 0. So every difference quotient is ≥ 0, and so is their limit $f'(x)$ (compare the argument at the end of the proof of Theorem 10.2). The decreasing case is the same with the inequalities reversed.

(2) This direction is deeper. It needs the **Mean Value Theorem** of differential calculus, and we will prove it then (C&J §2.8.h–i).

□

Remark 16.5.

1. Statement (2) cannot be reversed: a strictly increasing function can have $f'(x) = 0$ at some points. For $f(x) = x^3$, which is strictly increasing on $\mathbb{R}$ (Example 4.2), we have $f'(0) = 0$. This is why (1) only gives $f' \geq 0$, even for *strictly* increasing f.

2. What does $f'(x) > 0$ at a *single* point x tell us? Since the difference quotient tends to $f'(x) > 0$, it is positive for all h with $0 < |h| < \delta$, for some $\delta > 0$. That is,

$$f(x+h) > f(x) \text{ for } 0 < h < \delta, \qquad f(x+h) < f(x) \text{ for } -\delta < h < 0.$$

This is the precise meaning of "f is increasing at x". It compares values of f only with the value *at* x, and does not mean that f is increasing on any interval around x. (The function $x + 2x^2 \sin(1/x)$, with value 0 at 0, has derivative 1 at 0 but is not increasing on any interval around 0; we will be able to check this once we have the product and chain rules.)

Definition 16.1.
Let f be differentiable at x. If $f'(x) = 0$, then f has a **horizontal tangent** at x, and x is called a **critical point** of f.

For example, 0 is a critical point of x^2, where the function has its minimum, and of x^3, where it has neither a minimum nor a maximum.

Next, consider how the derivative itself changes. If f' is increasing on an interval, the slope of f increases from left to right: the graph **bends upward**. If f' is decreasing, the graph bends downward (Figure 16.5).

Definition 16.2.
Let f be differentiable on an interval I. Then f is

- **concave up** on I if f' is increasing on I;
- **concave down** on I if f' is decreasing on I.

For instance, x^2 is concave up on $\mathbb{R}$, since $f'(x) = 2x$ is increasing. The function x^3 has $f'(x) = 3x^2$, which is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$: so x^3 is concave down on $(-\infty, 0]$ and concave up on $[0, \infty)$.

16.7 Summary

- The difference quotient $\frac{\Delta y}{\Delta x}$ is the **average rate of change** of f; the derivative is the **instantaneous rate of change**. For a position $y = f(t)$, the difference quotient is the **average velocity** and $f'(t) = \dot{f}(t)$ is the **velocity** at time t (Figure 16.1). A freely falling body, $f(t) = at^2$, has velocity $2at$ (Example 16.1).

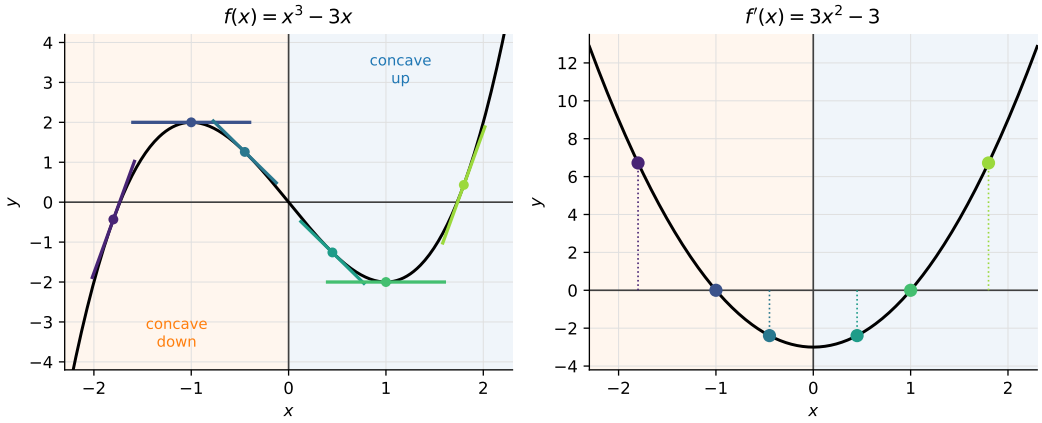


Figure 16.5: Left: $f(x) = x^3 - 3x$ with tangents. Right: $f'(x) = 3x^2 - 3$ (Example 16.3). Where $f' > 0$ ($|x| > 1$), f increases; where $f' < 0$, f decreases; at the critical points $x = \pm 1$ the tangent is horizontal. Where f' decreases ($x < 0$), f is concave down; where f' increases ($x > 0$), f is concave up.

- **Basic derivatives**: $(mx + b)' = m$ (Example 16.2); $(x^n)' = nx^{n-1}$ for all integers n (with $x \neq 0$ if $n < 0$), by the factorization Equation 16.1 (Proposition 16.1, Proposition 16.2); $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$ (Proposition 16.3), using $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$ (Lemma 16.1).
- When computing derivatives, keep lim in every line until the limit is taken (Remark 16.1).
- **Linearity**: $(f + g)' = f' + g'$ and $(cf)' = cf'$ (Proposition 16.4, Proposition 16.5), so every polynomial can be differentiated term by term (Example 16.3). There is no such rule for products (Remark 16.2).
- **Differentiable implies continuous** (Theorem 16.1), as long as the derivative is finite (Remark 16.3). The converse fails: $|x|$ has a corner at 0, with one-sided derivatives ± 1, and there are even continuous functions that are nowhere differentiable (Remark 16.4).
- A continuous function can have an **infinite derivative**, where the difference quotient is unbounded and the tangent is vertical, as for $\sqrt[3]{x}$ at 0 (Example 16.4).
- **Shape from the derivative**: increasing f has $f' \geq 0$, and $f' > 0$ on an interval makes f strictly increasing (proof later, via the Mean Value Theorem) (Proposition 16.6); the converse fails for x^3 at 0. Points with $f'(x) = 0$ are **critical points** (Definition 16.1). f is **concave up/down** where f' is increasing/decreasing (Definition 16.2).

17 Activity: Reading Derivatives from Graphs

17.1 Background

For now we use the geometric meaning of the derivative: $f'(x_0)$ is the **slope of the tangent line** to the graph of f at the point $(x_0, f(x_0))$. Reading off a slope from a graph means laying a ruler along the curve at x_0 and measuring rise over run, using the grid squares. Doing this at every x produces a new function, $x \mapsto f'(x)$, with its own graph.

17.2 Warm-up: a dictionary

Exercise 17.1. Complete the right-hand column. Each row translates a feature of the graph of f into a feature of the graph of f'.

If the graph of f …	… then the graph of f' …
is increasing on an interval	
is decreasing on an interval	
has a horizontal tangent at x_0	
is a straight line of slope m on an interval	
bends upward (is concave up) on an interval	
bends downward (is concave down) on an interval	
changes from bending down to bending up (inflection point) at x_0	
has a corner at x_0	

Solution.

If the graph of f …	… then the graph of f' …
is increasing on an interval	lies on or above the x-axis there
is decreasing on an interval	lies on or below the x-axis there
has a horizontal tangent at x_0	meets the x-axis at x_0: $f'(x_0) = 0$
is a straight line of slope m on an interval	is the horizontal line at height m there
bends upward (is concave up) on an interval	is increasing there
bends downward (is concave down) on an interval	is decreasing there
changes from bending down to bending up at x_0	has a local minimum at x_0
has a corner at x_0	has no value at x_0; it typically jumps there

Note "on or above" in the first row, not "above": $x \mapsto x^3$ is strictly increasing, yet its tangent at 0 is horizontal. A zero of f' does *not* by itself signal a maximum or minimum. Problems 2 and 4 contain exactly this situation.

17.3 Part I: From f to f'

In each problem, the top panel shows the graph of f. Sketch the graph of f' on the empty axes below it; the x-axes are aligned, so you can transfer x-coordinates straight down.

17.3.1 Problem 1

Exercise 17.2.

1. Find all points where the tangent to f is horizontal. Mark the corresponding zeros of f' on the lower axes.
2. Between these points, decide whether f is increasing or decreasing, and hence whether the graph of f' lies above or below the x-axis.
3. Where is f *steepest* while going down? What does that tell you about f' there?
4. Estimate slopes with a ruler and fill in the table. Plot the points on the lower axes.

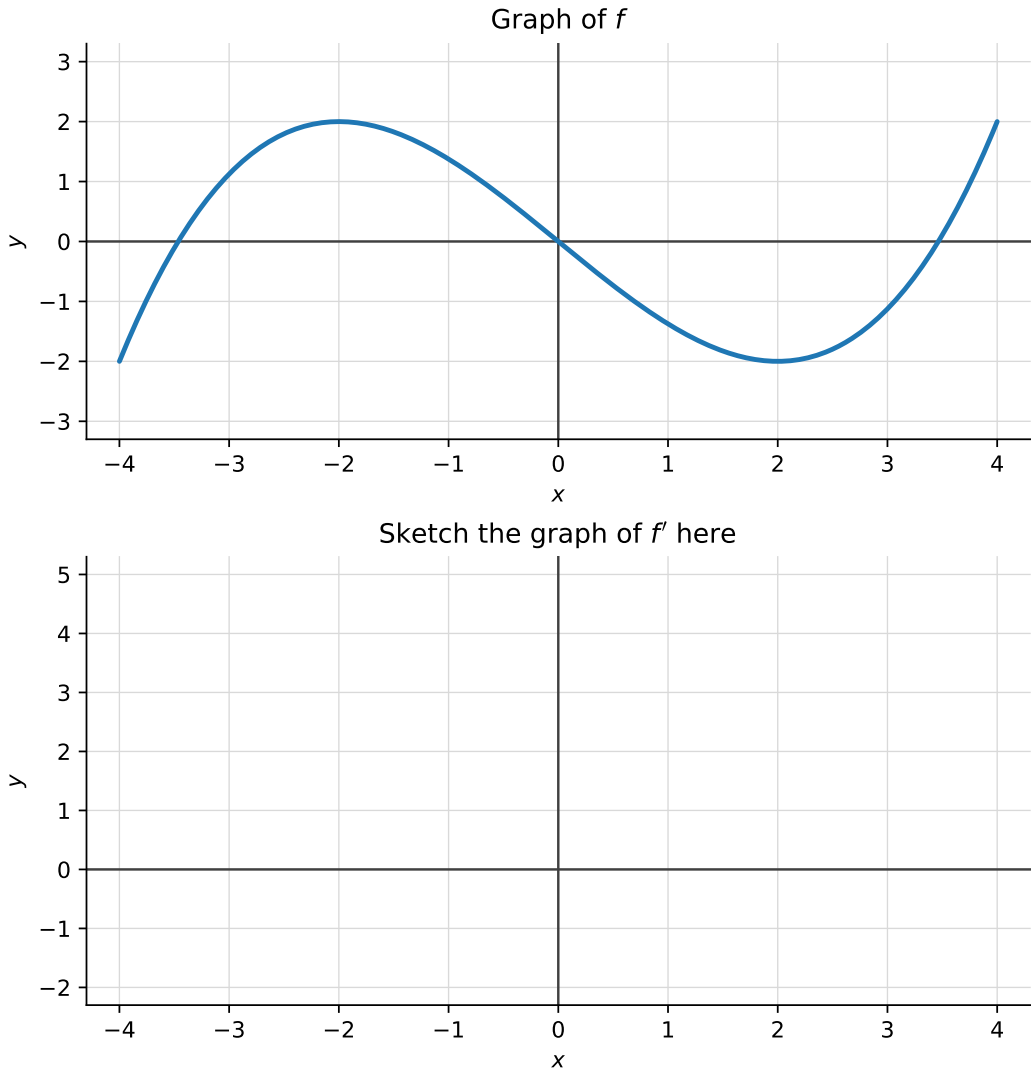


Figure 17.1: Problem 1: the graph of f (top) and axes for your sketch of f' (bottom).

x	-4	-3	-2	-1	0	1	2	3	4
$f'(x) \approx$									

5. Connect your points to a smooth sketch of f'. Check it against the dictionary: where f bends up, is your f' increasing?

Solution. The function is $f(x) = \frac{1}{8}(x^3 - 12x)$, so $f'(x) = \frac{3}{8}(x^2 - 4)$, an upward parabola.

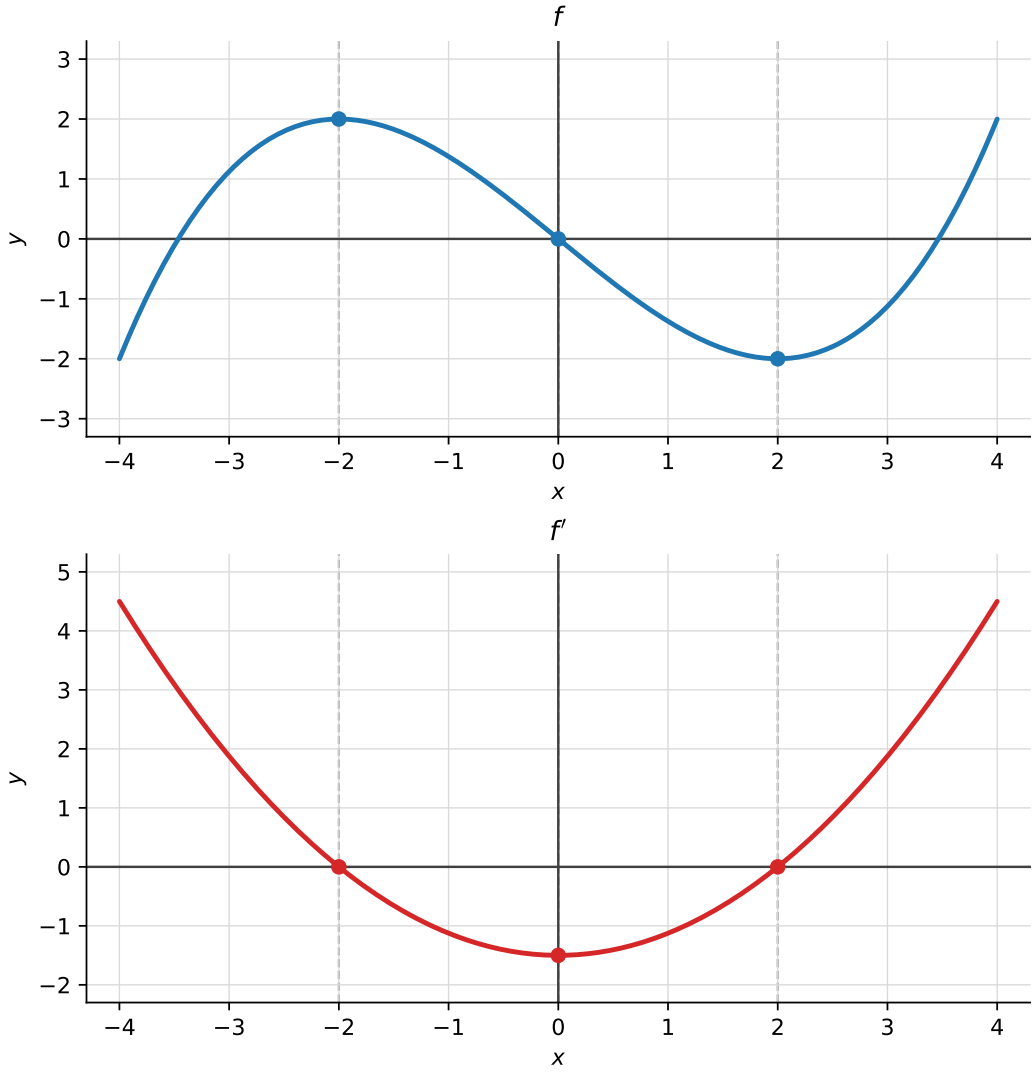


Figure 17.2: Solution to Problem 1.

x	-4	-3	-2	-1	0	1	2	3	4
$f'(x)$	4.5	1.875	0	-1.125	-1.5	-1.125	0	1.875	4.5

- Horizontal tangents at $x = \pm 2$ (local max at -2, local min at 2), so $f'(\pm 2) = 0$.
- f increases on $[-4, -2]$ and $[2, 4]$ ($f' \geq 0$) and decreases on $[-2, 2]$ ($f' \leq 0$).
- The steepest descent is at the inflection point $x = 0$, where f' has its minimum -1.5.
- f bends down on $[-4, 0]$ and up on $[0, 4]$, matching f' decreasing, then increasing.

17.3.2 Problem 2

Exercise 17.3.

1. At $x = -1$ the tangent to f is horizontal. Does f have a maximum or a minimum there? What does the graph of f' do near $x = -1$ — does it cross the x-axis or only touch it?
2. To the right of $x = 1$ the graph of f is a straight line. What is its slope, and what does f' look like there?
3. What happens at $x = 1$? Compare the slopes just to the left and just to the right of 1. Is there a sensible value for $f'(1)$? How do you show this in your sketch?
4. Sketch f' on the whole interval $[-3, 3]$.

Solution. On $[-3, 1]$, $f(x) = \frac{1}{4}(x+1)^3 + 1$, so $f'(x) = \frac{3}{4}(x+1)^2$; on $[1, 3]$, $f(x) = 4 - x$, so $f'(x) = -1$.

- At $x = -1$ the tangent is horizontal, yet f keeps increasing on both sides: there is no extremum. Accordingly the graph of f' **touches** the x-axis at -1 without crossing it. (A "flat" inflection point, as for x^3 at 0.)
- On $(1, 3]$ the graph is a line of slope -1, so f' is constantly -1.
- Just left of 1 the slope is close to 3; just right of 1 it is -1. No single tangent line fits at the corner, so $f'(1)$ is undefined, and the graph of f' **jumps** from 3 to -1. We draw open circles at both ends. Note that f itself is continuous at 1: continuity does not guarantee a derivative (C&J §2.8.e).

17.4 Part II: From f' back to f

Now the roles are reversed: the top panel shows the graph of the **derivative** f', and your task is to sketch f itself.

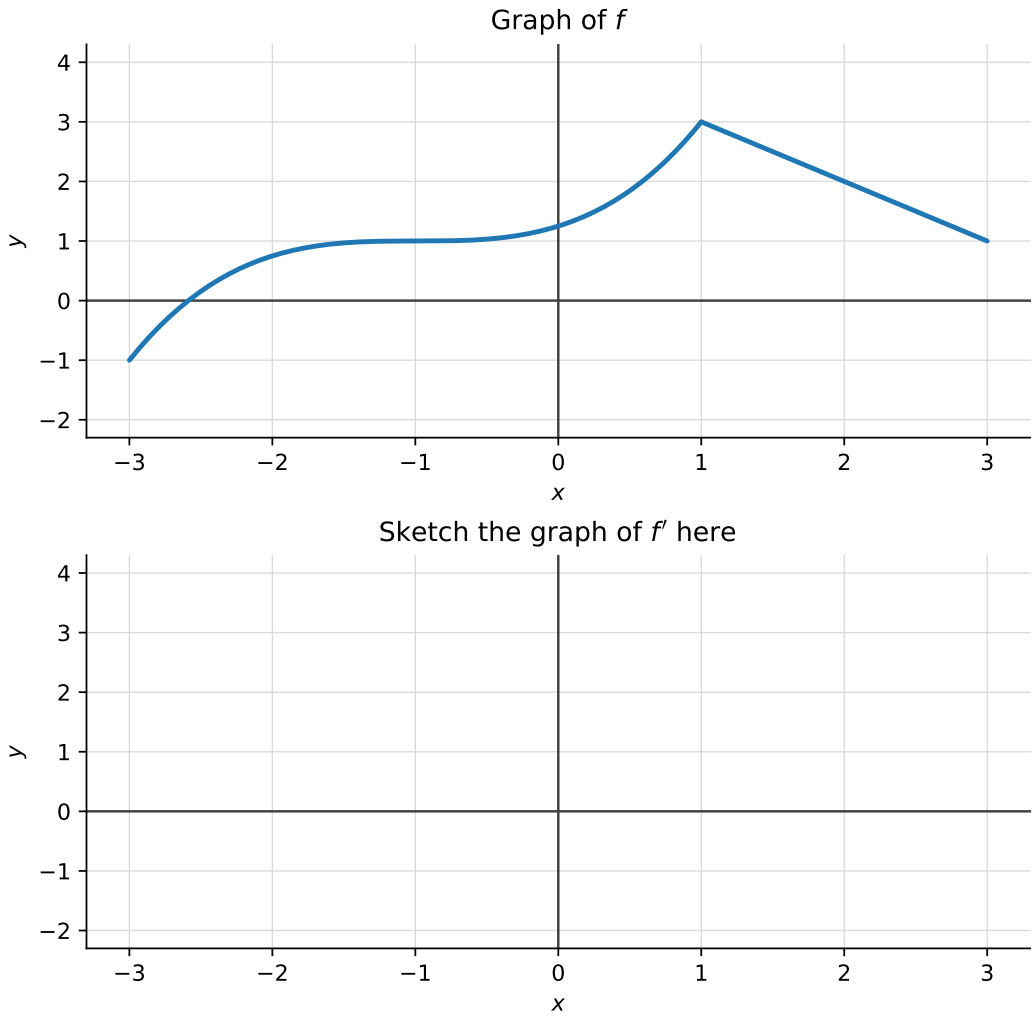


Figure 17.3: Problem 2: the graph of f (top) and axes for your sketch of f' (bottom).

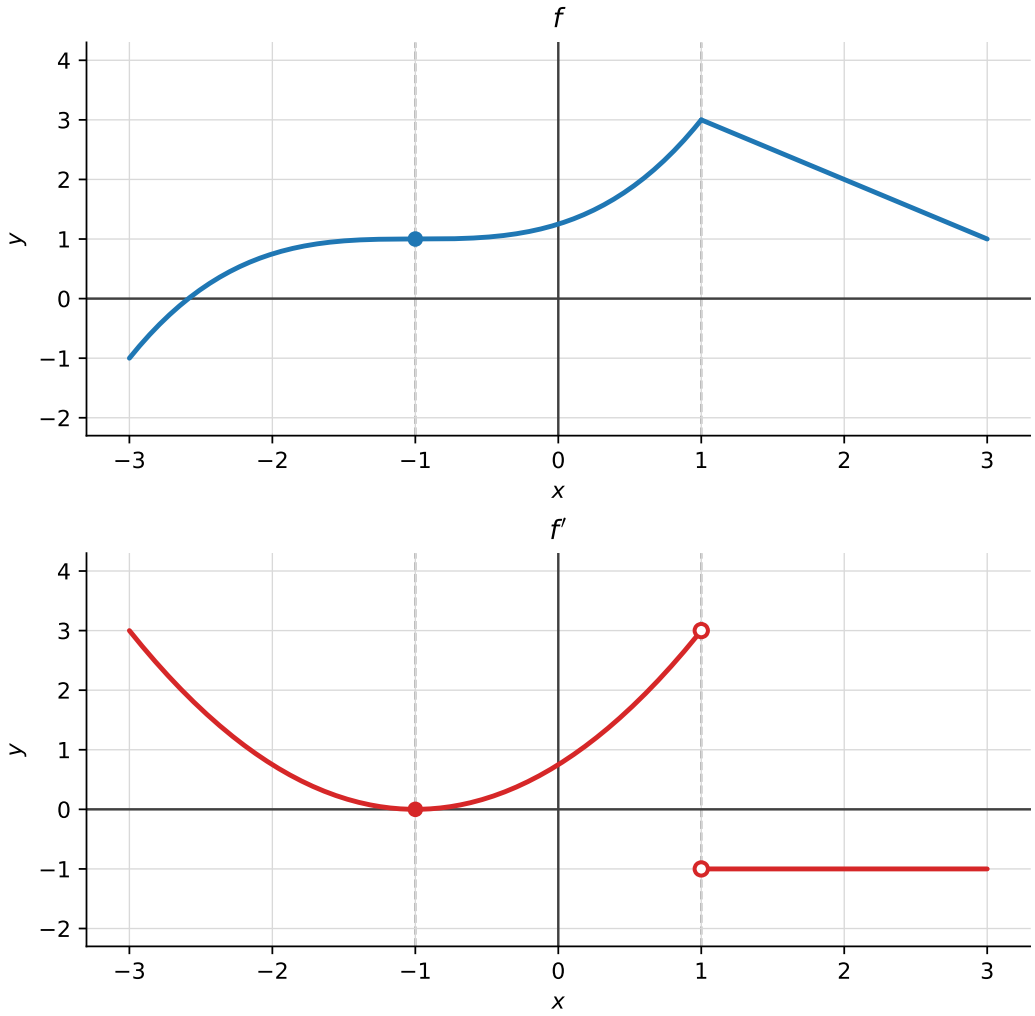


Figure 17.4: Solution to Problem 2. The open circles mark that $f'(1)$ is undefined.

17.4.1 Problem 3

Exercise 17.4.

1. Read off from the graph of f': on which intervals is f increasing or decreasing? Where does f have a local maximum or minimum? Where does f bend up, where does it bend down, and where is it a straight line (with what slope)?
2. Sketch **a** function f with this derivative on axes A.
3. Compare your sketch with your neighbor's. Are they the same? On axes A, sketch a *second* function that also has the derivative shown, but is different from your first one. How are the two graphs related to each other?
4. Describe all functions whose derivative is the one shown. How much information about f is the graph of f' missing?
5. Now suppose you are also told that

   $$f(-2) = -1.$$

   Sketch f on axes B. Is there now only one possible graph? Read off approximate values of $f(-4)$, $f(0)$, $f(2)$, $f(3)$, and $f(4)$ from your sketch.

Solution.

1. $f' < 0$ on $[-4, -2)$ and $(3, 4]$, $f' > 0$ on $(-2, 3)$: f decreases, then increases, then decreases, with a local minimum at $x = -2$ and a local maximum at $x = 3$. Since f' increases on $[-4, 0]$, f bends up there; on $[0, 2]$, $f' = 2$ is constant, so f is a straight line of slope 2; on $[2, 4]$, f' decreases, so f bends down.
2. A correct sketch has all the features from part 1; where it sits vertically is up to you.
3. Any two correct sketches differ by a **vertical shift**: the derivative only records slopes, and sliding a graph up or down changes no slope.
4. If f is one such function, the others are exactly the functions $f + C$, C a constant. That every function with the same derivative arises this way is a real theorem, not just a picture: it says that a function with derivative 0 on an interval is constant, which follows from the mean value theorem (C&J §2.8.h, §2.9.b). The graph of f' determines f only up to this one unknown number.
5. The condition $f(-2) = -1$ pins down the constant: of all the vertical shifts, exactly one passes through the point $(-2, -1)$. The red graph is the unique answer, with $f(-4) = 1$, $f(0) = 1$, $f(2) = 5$, $f(3) = 6$, $f(4) = 5$. (Explicitly: $f(x) = \frac{1}{2}x^2 + 2x + 1$ on $[-4, 0]$, $f(x) = 2x + 1$ on $[0, 2]$, $f(x) = -x^2 + 6x - 3$ on $[2, 4]$.)

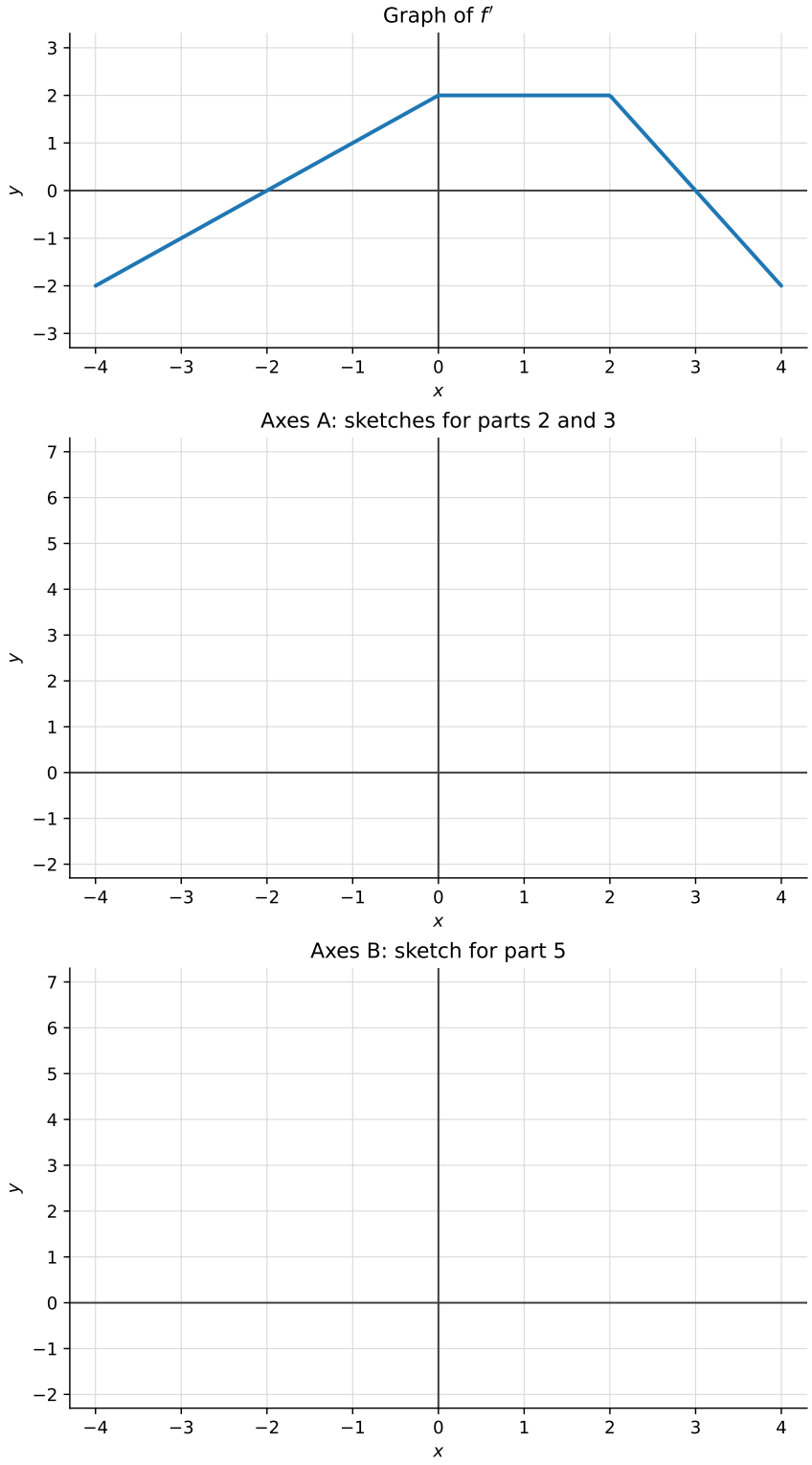


Figure 17.5: Problem 3: the graph of f' (top) and two sets of axes for your sketches of f.

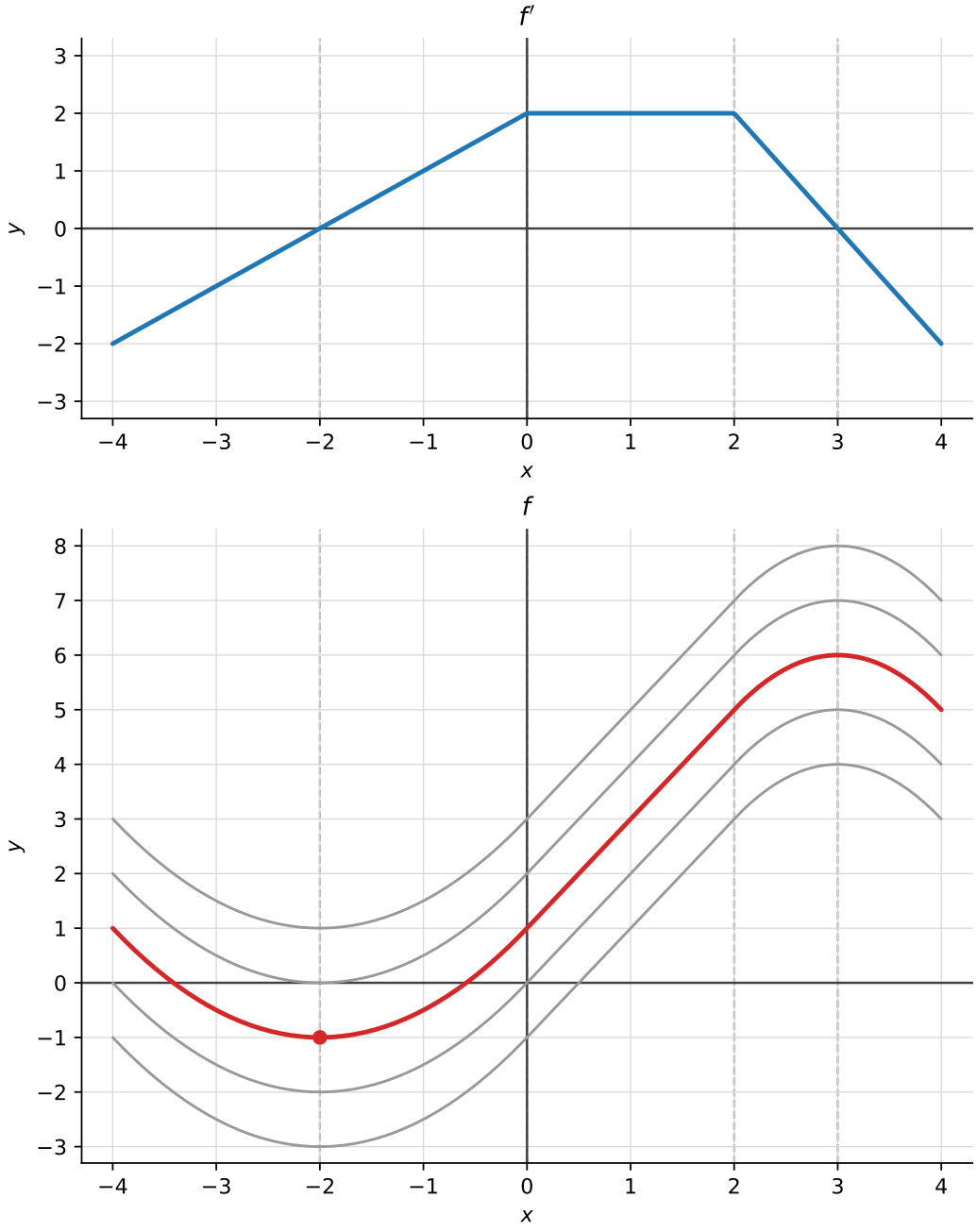


Figure 17.6: Solution to Problem 3. Gray: several functions with the given derivative. Red: the one with $f(-2) = -1$.

17.4.2 Problem 4

Here f is continuous on $[-3, 3]$ and differentiable everywhere except at $x = 0$; its derivative is shown below.

Exercise 17.5.

1. Read off the main features of f. In particular: what kind of point is $x = -1$? What does the graph of f look like at $x = 0$? Does f have a maximum or minimum at $x = 2$?
2. On axes A, sketch two different functions with the derivative shown.
3. Now you are told that
$$f(2) = 1.$$
Sketch f on axes B. Read off approximate values of $f(-3)$, $f(-1)$, $f(0)$, and $f(3)$.

Solution.

1. f' changes sign from $-$ to $+$ at $x = -1$: a local minimum. At $x = 0$ the slope jumps from 1 to 3: f has a **corner** there (the reverse of Problem 2). At $x = 2$, f' touches zero without changing sign: f has a horizontal tangent but keeps increasing — no extremum. f bends up on $[-3, 0]$ and on $[2, 3]$, and bends down on $[0, 2]$.
2. Again, any two answers differ by a vertical shift.
3. Only one shift passes through $(2, 1)$: $f(-3) = 0.5$, $f(-1) = -1.5$, $f(0) = -1$, $f(3) = 1.25$. To reach the left half from the given point you must pass *through* the corner, and this is where the continuity of f at 0 is used (see Going Further, question 3).

17.5 Going further

Exercise 17.6.

1. **Change as area.** In Problem 3, compute the signed area between the graph of f' and the x-axis from $x = -2$ to $x = 3$ (it is made of triangles and a rectangle). Compare it with $f(3) - f(-2)$ from your sketch. Repeat for another pair of points. What do you conjecture?
2. **Why the real numbers matter.** Define h on the *rational* numbers by $h(x) = 0$ if $x < \sqrt{2}$ and $h(x) = 1$ if $x > \sqrt{2}$. Convince yourself that at every rational x, the graph of h is flat nearby, so its "derivative" is 0 everywhere on $\mathbb{Q}$. Yet h is not constant. Which property of $\mathbb{R}$ (from the first weeks of the course) rules this out for functions on a real interval?

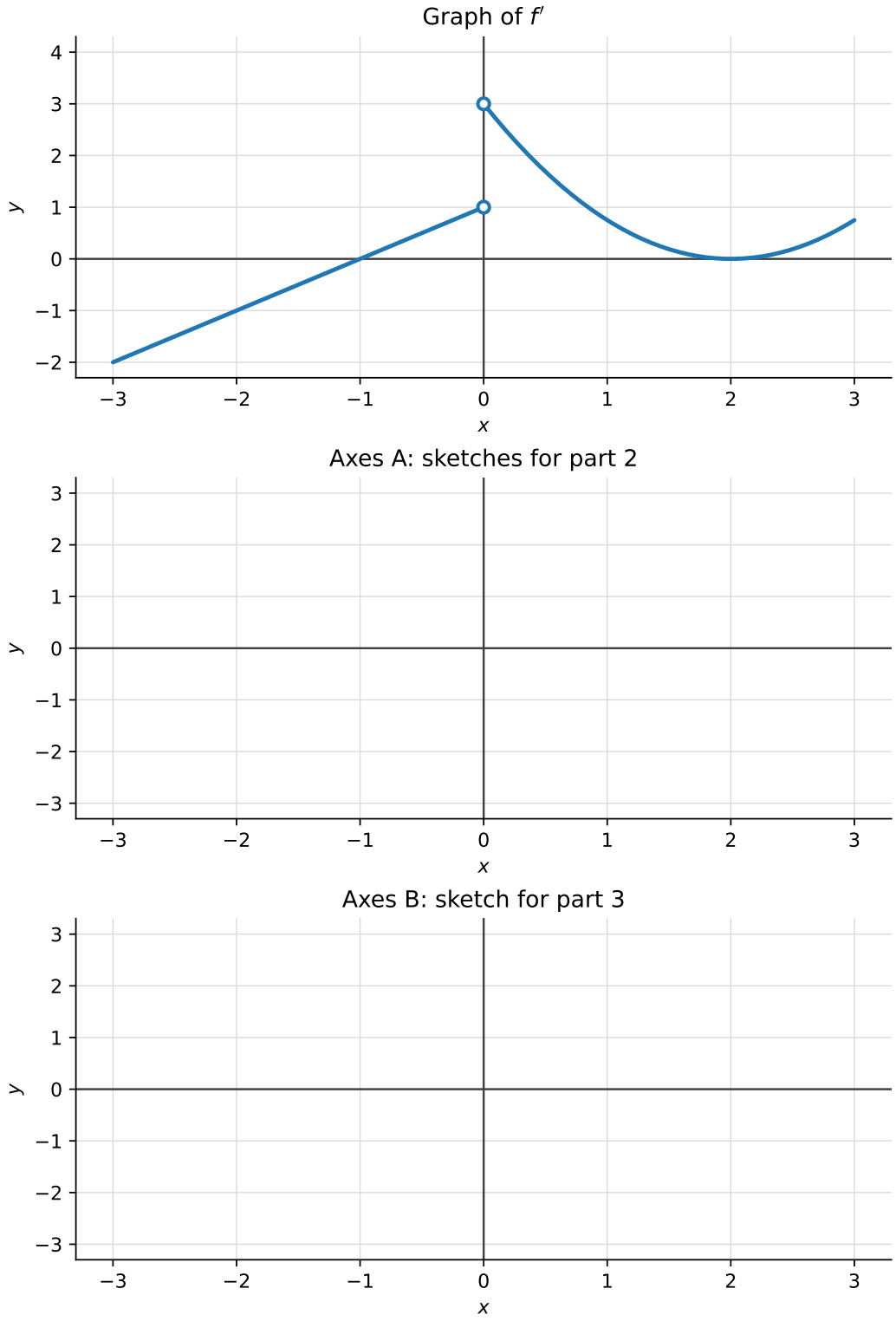


Figure 17.7: Problem 4: the graph of f' (top) and two sets of axes for your sketches of f. The open circles indicate that $f'(0)$ is undefined.

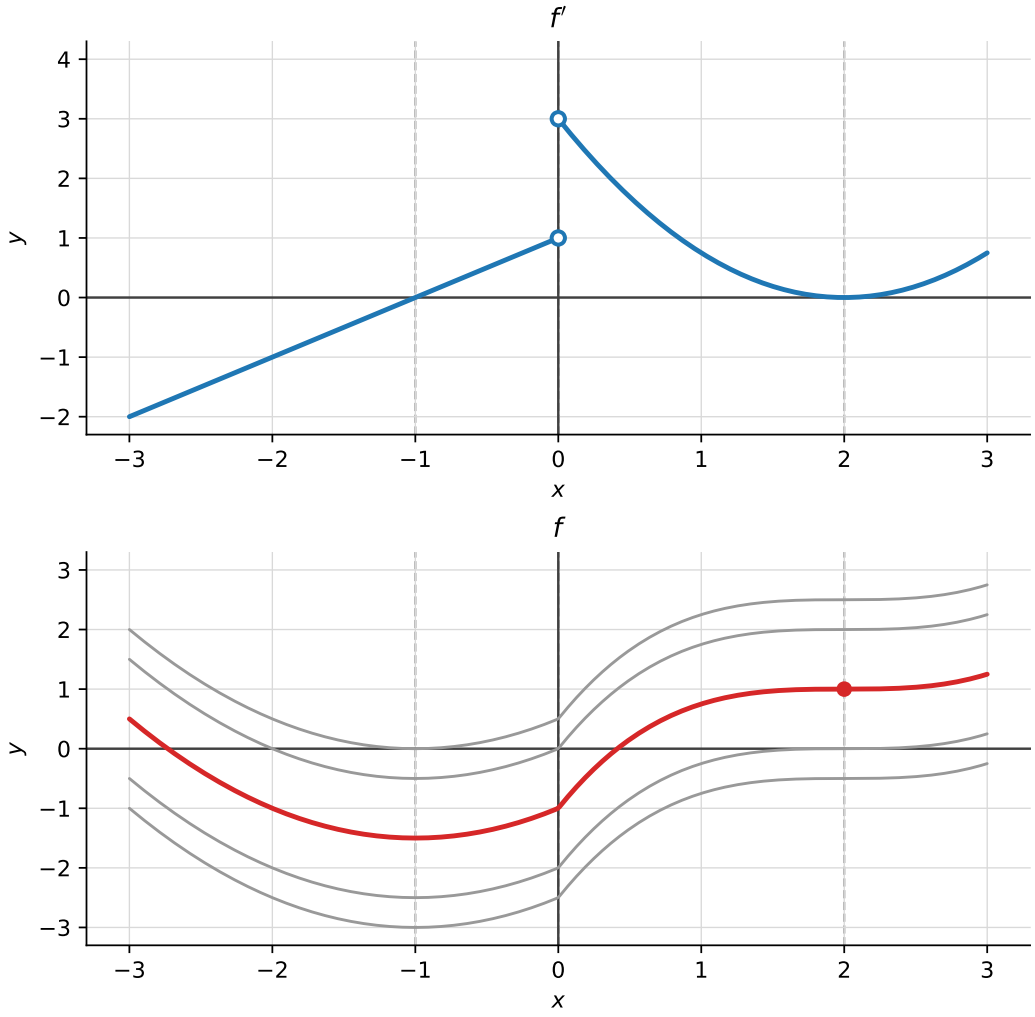


Figure 17.8: Solution to Problem 4. Gray: several functions with the given derivative. Red: the one with $f(2) = 1$.

3. **Is one value always enough?** In Problem 4, drop the assumption that f is continuous at 0, so f is only required to have the given derivative on $[-3, 0)$ and on $(0, 3]$. Does $f(2) = 1$ still determine f? How many function values would you need?

Solution.

1. The area is 2 (triangle on $[-2, 0]$) + 4 (rectangle on $[0, 2]$) + 1 (triangle on $[2, 3]$) = 7, and indeed $f(3) - f(-2) = 6 - (-1) = 7$. The conjecture

$$f(b) - f(a) = \int_a^b f'(x)\,dx$$

   is the **fundamental theorem of calculus** (C&J §2.9). It explains the ambiguity in a second way: f' determines every *difference* $f(b) - f(a)$, but not the starting height.
2. On $\mathbb{Q}$ the jump of h happens across a "hole" at $\sqrt{2}$ that contains no input of h. On $\mathbb{R}$ there are no such holes: this is completeness (the nested interval principle). The proof that $f' = 0$ forces f constant goes through the mean value theorem, which in turn rests on completeness.
3. No. On each of the two pieces f is determined up to its own constant, and without continuity the two constants are independent: the left half of the graph can be shifted up or down on its own. You need one value from each side of 0 — for instance $f(2)$ and $f(-1)$. The principle is: one value per *interval* on which f is differentiable.

18 The Mean Value Theorem

18.1 Local and Global Information

The derivative

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

is a **local** quantity: it only depends on the values of f in an arbitrarily small neighborhood of x. A difference quotient

$$\frac{f(b) - f(a)}{b - a},$$

on the other hand, compares the values of f at two points that may be far apart, so it reflects a **global** property of f. The Mean Value Theorem connects the two. It is the most important tool for deriving *global* properties of a function from *local* properties of its derivative.

Geometric picture. The difference quotient $\frac{f(b)-f(a)}{b-a}$ is the slope of the **secant** through the points $(a, f(a))$ and $(b, f(b))$ of the graph of f. Imagine shifting this secant parallel to itself, away from the graph. At some point ξ between a and b the shifted line will just touch the graph without crossing it: it becomes the **tangent** at ξ. (One such point is the point of the graph that is farthest from the secant.) Since the tangent is parallel to the secant, the two have the same slope:

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Exercise 18.1. Try to draw the graph of a function on $[a, b]$ for which there is **no** point $\xi \in (a, b)$ at which the tangent is parallel to the secant. Is your function continuous? Does it have a tangent at every point?

18.2 The Mean Value Theorem

Theorem 18.1 (Mean Value Theorem)**.** *Suppose f is continuous on $[a, b]$ and differentiable on (a, b). Then there exists ξ with $a < \xi < b$ and*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Equivalently, $f(b)-f(a) = f'(\xi)\,(b-a)$. The theorem asserts only that such a ξ *exists*; it says nothing about where in (a,b) it lies, and there may be several such points. Note also that f must be continuous on the *closed* interval, but differentiability is only required on the *open* interval: the derivative need not exist at the endpoints.

Why do we need to assume that $f'(x)$ exists for **all** $x \in (a,b)$? A single point of non-differentiability is enough to break the theorem.

Example 18.1.
Let $f(x) = |x|$ on $[-1,1]$. Then f is continuous on $[-1,1]$, and the secant has slope

$$\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0.$$

But $f'(x) = -1$ for $x < 0$ and $f'(x) = 1$ for $x > 0$, while $f'(0)$ does not exist (the left- and right-sided limits of the difference quotient are -1 and 1). So there is no $\xi \in (-1,1)$ with $f'(\xi) = 0$. The graph has a corner at 0, and no tangent is parallel to the secant.

Example 18.2 (Driving)**.**
Suppose you drive for 2 hours and end up 120 miles from where you started. Let $s(t)$ be your distance from the starting point at time $t \in [0,2]$. Assume that your movement is continuous and that your velocity $s'(t)$ exists at all times. By Theorem 18.1, there is a moment $\xi \in (0,2)$ at which your velocity was

$$s'(\xi) = \frac{s(2)-s(0)}{2-0} = \frac{120-0}{2-0} = 60 \text{ mph}.$$

Here the numerator is a distance and the denominator a time. In particular, you cannot claim to have driven at less than 50 mph the whole time. On the other hand, the theorem does not tell us *when* you were going exactly 60 mph.

18.3 Rolle's Theorem

We prove the Mean Value Theorem by first establishing the special case of a horizontal secant.

Theorem 18.2 (Rolle's Theorem)**.** *Suppose f is continuous on $[a,b]$ and differentiable on (a,b), and $f(a) = f(b) = 0$. Then there exists ξ with $a < \xi < b$ and*

$$f'(\xi) = 0.$$

This is the Mean Value Theorem with a horizontal tangent: if the graph of f meets the x-axis at a and at b, it must have a horizontal tangent somewhere in between.

The proof rests on the following key fact, which we have used before (see Remark 11.2) and will prove later.

Theorem 18.3 (Extreme Value Theorem)**.** *A continuous function on a closed interval $[a, b]$ assumes a global maximum and a global minimum. That is, there exist M and m such that*

$$m \le f(x) \le M \qquad \text{for all } x \in [a, b],$$

and there are points $x_m, x_M \in [a, b]$ with $f(x_m) = m$ and $f(x_M) = M$.

With this, we can prove Theorem 18.2.

Proof. Let m and M be the minimum and maximum of f on $[a, b]$ (Theorem 18.3). Since $f(a) = f(b) = 0$, we must have

$$m \le 0 \le M.$$

Case $m = M = 0$. Then $f(x) = 0$ for all $x \in [a, b]$. Every difference quotient of f is 0, so $f'(\xi) = 0$ for every $\xi \in (a, b)$, and we are done.

Case $M > 0$. There exists $\xi \in [a, b]$ with $f(\xi) = M$. Since $f(a) = f(b) = 0 < M$, ξ is an **interior point**: $\xi \in (a, b)$. Since $f(x) \le M$ for all $x \in [a, b]$, we have

$$f(\xi + h) - f(\xi) \le 0$$

for all h with $\xi + h \in [a, b]$, in particular for all small enough $|h|$. Hence

$$\frac{f(\xi + h) - f(\xi)}{h} \le 0 \quad \text{for } h > 0, \qquad \frac{f(\xi + h) - f(\xi)}{h} \ge 0 \quad \text{for } h < 0.$$

Since f is differentiable at ξ, the limit

$$\lim_{h \to 0} \frac{f(\xi + h) - f(\xi)}{h}$$

exists, no matter from which direction $h \to 0$. Letting $h \to 0$ through positive values, the limit is ≤ 0; letting $h \to 0$ through negative values, it is ≥ 0 (a limit of nonpositive numbers is nonpositive; compare the argument at the end of the proof of Theorem 10.2). Since the left-sided and the right-sided limit agree (Definition 10.2), we must have

$$f'(\xi) = \lim_{h \to 0} \frac{f(\xi + h) - f(\xi)}{h} = 0.$$

Case $M = 0$, $m < 0$. The same argument, applied at a point ξ where f attains its minimum m (or the previous case applied to $-f$), gives $f'(\xi) = 0$. □

Remark 18.1. The proof shows more than Rolle's theorem: if f attains a maximum or a minimum at an **interior** point ξ of an interval, and f is differentiable at ξ, then $f'(\xi) = 0$. The assumption $f(a) = f(b) = 0$ was only used to guarantee that an extremum occurs in the interior.

18.4 Proof of the Mean Value Theorem

To deduce Theorem 18.1 from Theorem 18.2, we subtract the secant from f. Define

$$\varphi(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}\,(x - a).$$

The secant through $(a, f(a))$ and $(b, f(b))$ is the line $y = f(a) + \frac{f(b)-f(a)}{b-a}(x - a)$, so $\varphi(x)$ is the **vertical distance** between the point $(x, f(x))$ on the graph and the point of the secant above (or below) x.

Proof. Plugging in the endpoints,

$$\varphi(a) = f(a) - f(a) - 0 = 0, \qquad \varphi(b) = f(b) - f(a) - \frac{f(b) - f(a)}{b - a}(b - a) = 0.$$

The function φ is of the form

$$\varphi(x) = f(x) + c\,x + d, \qquad \text{with constants} \quad c = -\frac{f(b) - f(a)}{b - a}, \quad d = -f(a) + \frac{f(b) - f(a)}{b - a}\,a.$$

So φ is continuous on $[a, b]$ (Corollary 10.1), and since

$$\frac{\varphi(x + h) - \varphi(x)}{h} = \frac{f(x + h) - f(x)}{h} + c,$$

it is differentiable on (a, b) with

$$\varphi'(x) = f'(x) + c.$$

By Theorem 18.2 there exists ξ with $a < \xi < b$ and $\varphi'(\xi) = 0$. Hence

$$f'(\xi) = -c = \frac{f(b) - f(a)}{b - a}.$$

□

Remark 18.2. Applying Theorem 18.1 to a function with $f(a) = f(b)$ gives $f'(\xi) = 0$. So Rolle's theorem holds more generally whenever $f(a) = f(b)$, not only when both values are 0.

18.5 Application: Lipschitz Continuity

Definition 18.1.
A function f is **Lipschitz continuous** on an interval I if there is a constant $L \geq 0$ such that

$$|f(x) - f(y)| \leq L\,|x - y| \qquad \text{for all } x, y \in I.$$

In other words, all difference quotients $\frac{f(x)-f(y)}{x-y}$ of f on I are bounded in absolute value by L: the secant slopes of f cannot become arbitrarily steep. A Lipschitz continuous function is continuous: given $\varepsilon > 0$, the choice $\delta = \varepsilon/L$ (any δ if $L = 0$) works in Definition 6.2 at every point.

We say f is **continuously differentiable** on $[a, b]$ if f' exists on $[a, b]$ (as a one-sided derivative at the endpoints) and f' is continuous on $[a, b]$.

Proposition 18.1.
Any function f that is continuously differentiable on $[a, b]$ is Lipschitz continuous on $[a, b]$.

Proof. Since f' is continuous on $[a, b]$, it is bounded there by Theorem 18.3: there is L with

$$|f'(x)| \leq L \qquad \text{for all } x \in [a, b].$$

Now let $a \leq x < y \leq b$. Then f is continuous on $[x, y]$ and differentiable on (x, y), so by Theorem 18.1 there is $\xi \in (x, y)$ with

$$|f(y) - f(x)| = |f'(\xi)\,(y - x)| = |f'(\xi)|\,|y - x| \leq L\,|y - x|.$$

□

The proof only used that f' is bounded on (a, b) (and that f is continuous on $[a, b]$), so this weaker assumption is already enough.

Example 18.3 (Not every continuous function is Lipschitz continuous)**.**
Let $f(x) = \sqrt[3]{x}$ on $[-1, 1]$. This function is continuous. But for $x \neq 0$,

$$\frac{f(x) - f(0)}{x - 0} = \frac{x^{1/3}}{x} = x^{-2/3},$$

which is **not bounded** near 0: for $x = \frac{1}{n^3}$ it equals n^2. So no constant L can satisfy $|f(x) - f(0)| \leq L|x - 0|$ for all $x \in [-1, 1]$. This does not contradict Proposition 18.1: f is not differentiable at 0 — the graph has a vertical tangent there, and $f'(x) = \frac{1}{3}x^{-2/3}$ is unbounded near 0.

18.6 Summary

- The **Mean Value Theorem** (Theorem 18.1): if f is continuous on $[a, b]$ and differentiable on (a, b), then $f'(\xi) = \frac{f(b)-f(a)}{b-a}$ for some $\xi \in (a, b)$. Geometrically, some tangent is parallel to the secant. It turns local information (the derivative) into global information (difference quotients). For example, an average speed of 60 mph forces a moment of speed exactly 60 mph (Example 18.2).
- Differentiability at **every** interior point is needed: $|x|$ on $[-1, 1]$ has a corner and no horizontal tangent (Example 18.1).
- **Rolle's Theorem** (Theorem 18.2) is the special case $f(a) = f(b) = 0$. Its proof uses the **Extreme Value Theorem** (Theorem 18.3): f attains a maximum or minimum at an interior point ξ, and there the right-sided difference quotients are ≤ 0 and the left-sided ones ≥ 0 (or vice versa), so $f'(\xi) = 0$ (Remark 18.1).
- The Mean Value Theorem follows by applying Rolle's theorem to $\varphi(x) = f(x) - (\text{secant at } x)$, which vanishes at a and b and has $\varphi' = f' - \frac{f(b)-f(a)}{b-a}$.
- **Application:** a continuously differentiable function on $[a, b]$ is **Lipschitz continuous**, $|f(x)-f(y)| \leq L|x-y|$ with L a bound for $|f'|$ (Proposition 18.1). Continuous functions need not be Lipschitz: $\sqrt[3]{x}$ has unbounded difference quotients at 0 (Example 18.3).

19 The Derivative as Linear Approximation

19.1 The Tangent Line as an Approximation

Near the point $(a, f(a))$, the graph of a differentiable function is very close to its tangent line. Since lines are easy to evaluate, we can use the tangent to **approximate** the values $f(a+h)$ for small h (Figure 19.1).

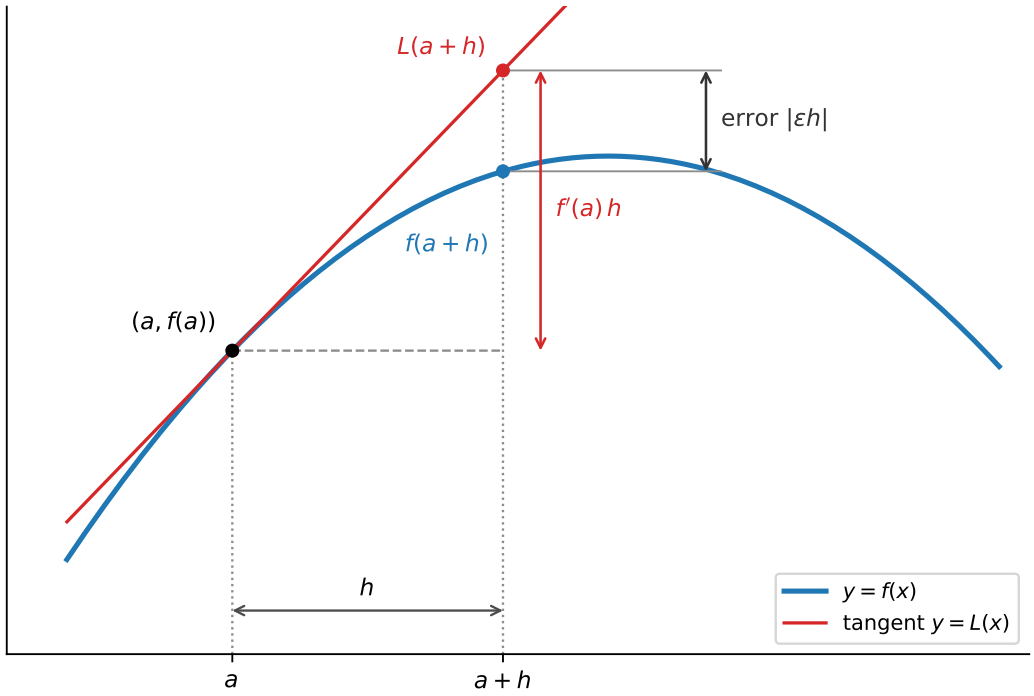


Figure 19.1: The tangent at $(a, f(a))$ approximates the graph near a. Moving from a to $a+h$, the tangent rises by $f'(a)\,h$. The value $L(a+h)$ on the tangent differs from the actual value $f(a+h)$ by the error εh.

To use the tangent, we need its equation. The tangent at $(a, f(a))$ is the line through

the **point** $(a, f(a))$ with **slope** $f'(a)$ (Definition 15.2). In point-slope form,

$$y - f(a) = f'(a)\,(x - a).$$

Definition 19.1.
Let f be differentiable at a. The **linear approximation** (or **linearization**) of f about $x = a$ is the function

$$L(x) = f(a) + f'(a)\,(x - a).$$

To approximate f close to a, plug in $a + h$ for x:

$$f(a + h) \approx L(a + h) = f(a) + f'(a)\,(a + h - a) = f(a) + f'(a) \cdot h.$$

So to estimate f near a, we only need two numbers: the value $f(a)$ and the derivative $f'(a)$.

Example 19.1.
Let $f(x) = x^2 + 1$ and $a = 2$. Then $f'(x) = 2x$, so

$$f(2) = 5, \qquad f'(2) = 4.$$

With $h = 0.1$,

$$f(2.1) \approx L(2.1) = 5 + 4 \cdot 0.1 = 5.4.$$

The true value is $f(2.1) = 2.1^2 + 1 = 4.41 + 1 = 5.41$. The approximation is off by 0.01, and it is an **underestimate**.

Exercise 19.1. Let $f(x) = x^3 - 2x + 4$. Approximate $f(1.1)$ using $f(1)$ and $f'(1)$, and compare with the true value.

Solution. We have $f(1) = 1 - 2 + 4 = 3$ and $f'(x) = 3x^2 - 2$, so $f'(1) = 1$. The linear approximation about 1 is

$$L(x) = f(1) + f'(1)\,(x - 1) = 3 + (x - 1),$$

and

$$f(1.1) \approx L(1.1) = 3 + (1.1 - 1) = 3 + 0.1 = 3.1.$$

The true value is $f(1.1) = 1.331 - 2.2 + 4 = 3.131$. Again the approximation is an underestimate, this time by 0.031.

Exercise 19.2. In Example 19.1 and Exercise 19.1, the linear approximation *underestimates* the true value. Why? Sketch the graph of f near a together with its tangent.

Solution. In both cases f' is increasing near a: $f'(x) = 2x$ for the first function and $f'(x) = 3x^2 - 2$ (increasing for $x \geq 0$) for the second. So f is **concave up** near a (Definition 16.2): the graph bends upward, away from the tangent, and therefore lies **above** the tangent line. The tangent values $L(a+h)$ are then too small. In Figure 19.1 the function is concave down, the graph lies below the tangent, and the approximation is an overestimate. We prove this precisely in Proposition 19.1.

19.2 Estimating the Error

How far off is $L(a+h)$, relative to h? We write the error as a multiple of h:

$$f(a+h) = L(a+h) + \underbrace{\varepsilon h}_{\text{error term}} = f(a) + f'(a)\,h + \varepsilon h,$$

where ε depends on h. Solving for ε,

$$\frac{f(a+h) - f(a)}{h} = f'(a) + \varepsilon.$$

Remark 19.1. The left-hand side is the difference quotient of f at a, which tends to $f'(a)$ as $h \to 0$. So

$$\varepsilon = \frac{f(a+h) - f(a)}{h} - f'(a) \longrightarrow 0 \qquad (h \to 0),$$

Conversely, suppose $f(a+h) = f(a) + c\,h + \varepsilon h$ for some number c, with $\varepsilon \to 0$ as $h \to 0$. Then $\frac{f(a+h)-f(a)}{h} = c + \varepsilon \to c$, so f is differentiable at a with $f'(a) = c$. In other words, f is differentiable at a exactly when f can be approximated near a by a linear function with an error that is small *compared to* h. This is how C&J phrase the definition of the derivative in §2.8.j.

This tells us that the error goes to 0 faster than h, but not *how much* faster. To estimate ε, suppose $h > 0$ and f is differentiable on an open interval containing $[a, a+h]$. By the Mean Value Theorem (Theorem 18.1), there is ξ with $a < \xi < a+h$ and

$$\frac{f(a+h) - f(a)}{h} = f'(\xi).$$

Comparing with the expression above gives

$$\varepsilon = f'(\xi) - f'(a). \tag{19.1}$$

So the error depends on how much the derivative changes between a and ξ. This settles Exercise 19.2.

Proposition 19.1 (Tangent lines of concave-up functions)**.** *Let f be differentiable on an open interval I containing a, and concave up on I (Definition 16.2). Then the graph of f lies on or above its tangent at a:*

$$f(x) \geq L(x) = f(a) + f'(a)(x-a) \qquad \text{for all } x \in I.$$

Proof. Let $x = a + h \in I$. For $h = 0$ both sides equal $f(a)$. For $h > 0$, Equation 19.1 gives $f(a+h) - L(a+h) = \varepsilon h = (f'(\xi) - f'(a))h$ with $a < \xi < a + h$. Since f' is increasing, $f'(\xi) \geq f'(a)$, so this is ≥ 0.

For $h < 0$, apply Theorem 18.1 on $[a+h, a]$ instead: $\frac{f(a+h)-f(a)}{h} = \frac{f(a)-f(a+h)}{a-(a+h)} = f'(\xi)$ for some ξ with $a + h < \xi < a$, and again $f(a+h) - L(a+h) = (f'(\xi) - f'(a))h$. Now $f'(\xi) \leq f'(a)$ and $h < 0$, so the product is again ≥ 0. □

For a concave-down function the inequality is reversed, and the linear approximation overestimates.

19.2.1 Bounding the error with the second derivative

How can we bound $f'(\xi) - f'(a)$? If f' is itself differentiable, we can use the Mean Value Theorem again, this time for f'. The derivative of f' is the **second derivative** of f,

$$f''(x) = (f')'(x).$$

For example, $f(x) = x^2 + 1$ has $f'(x) = 2x$ and $f''(x) = 2$, and $f(x) = x^3 - 2x + 4$ has $f'(x) = 3x^2 - 2$ and $f''(x) = 6x$.

Applying Theorem 18.1 to f' on $[a, \xi]$, we find η with $a < \eta < \xi$ and

$$f'(\xi) - f'(a) = f''(\eta)\,(\xi - a).$$

Since $0 < \xi - a < h$, it follows that

$$|\varepsilon| = \underbrace{|\xi - a|}_{\leq h} \cdot \underbrace{|f''(\eta)|}_{\leq M} \leq h \cdot M,$$

where M is any bound for $|f''|$ on the interval $[a, a+h]$. Multiplying by h, we obtain a bound for the error term.

Theorem 19.1 (Error of the linear approximation)**.** *Let f be twice differentiable on an open interval containing a and $a + h$, and let M be a bound for $|f''|$ on the closed interval between a and $a + h$:*

$$|f''(x)| \leq M \qquad \text{for all } x \text{ between } a \text{ and } a + h.$$

Then

$$|f(a+h) - L(a+h)| = |\varepsilon h| \leq M\,h^2.$$

Proof. For $h > 0$ this is the computation above. (The second application of Theorem 18.1 is allowed because f' is differentiable, hence continuous by Theorem 16.1, on an open interval containing $[a, \xi]$.) For $h < 0$, the same argument on the interval $[a + h, a]$ gives $|\varepsilon| \le |h| \cdot M$, hence $|\varepsilon h| \le Mh^2$. For $h = 0$ there is nothing to prove. □

For small h, the bound Mh^2 is **much smaller** than the linear term $f'(a)\,h$, unless $f'(a) = 0$: their ratio is $\frac{Mh^2}{|f'(a)|\,|h|} = \frac{M}{|f'(a)|}|h|$, which goes to 0 with h.

Example 19.2.
We check Theorem 19.1 against the two earlier examples, with $h = 0.1$.

- For $f(x) = x^2 + 1$ and $a = 2$, we have $|f''| = 2$, so $M = 2$ and the error is at most $2 \cdot (0.1)^2 = 0.02$. The actual error is 0.01. In fact, $f(2 + h) = 5 + 4h + h^2$, so the error is exactly h^2.
- For $f(x) = x^3 - 2x + 4$ and $a = 1$, we have $f''(x) = 6x$, so $|f''| \le 6.6$ on $[1, 1.1]$. The error is at most $6.6 \cdot (0.1)^2 = 0.066$. The actual error is 0.031. In fact, $f(1 + h) = 3 + h + 3h^2 + h^3$, so the error is exactly $3h^2 + h^3$.

In both cases, the bound is about twice the actual error.

Remark 19.2. The examples suggest that the constant M in Theorem 19.1 is not optimal. Indeed, one can show the sharper bound $|\varepsilon h| \le \frac{1}{2}Mh^2$, which is attained for $x^2 + 1$ (where $M = 2$ and the error is h^2). C&J mention this improvement in §2.8.j and derive it later, as part of Taylor's theorem.

Exercise 19.3. Why is Theorem 19.1 important?

Solution. A few reasons:

1. **The error is quadratic in h.** Making h ten times smaller makes the error bound a hundred times smaller. For $f(x) = x^3 - 2x + 4$ at $a = 1$, the actual errors $3h^2 + h^3$ are

   $$0.031 \ (h = 0.1), \qquad 0.000301 \ (h = 0.01), \qquad 0.000003001 \ (h = 0.001).$$

   The linear term $f'(1)\,h = h$, on the other hand, only shrinks by a factor of ten each time. So for small h, almost all of the change $f(a + h) - f(a)$ is captured by the linear term.
2. **It is a guarantee.** We usually use the linear approximation when we *cannot* easily compute $f(a + h)$. The theorem tells us how good the approximation is without knowing the true value: we only need a bound for f''. For example, for $f(x) = \sin x$ and $a = 0$ we get $\sin(0.1) \approx \sin 0 + \cos 0 \cdot 0.1 = 0.1$. Since $f''(x) = -\sin x$ and $|\sin x| \le x \le 0.1$ on $[0, 0.1]$, the error is at most $0.1 \cdot (0.1)^2 = 0.001$. (The true value is $\sin(0.1) = 0.09983\ldots$, an error of about 0.00017.)

3. **It makes step-by-step methods work.** Roughly speaking: in *Euler's method*, a function is approximated by following tangent lines over many small steps of size h. Over an interval of fixed length we need about $1/h$ steps, each with an error of order h^2. So the total error is of order $\frac{1}{h} \cdot h^2 = h$, which goes to 0 as $h \to 0$. If the error per step were only *proportional* to h, the errors would add up to something that does not shrink.

19.3 Summary

- The **linear approximation** of f about a is the tangent line $L(x) = f(a) + f'(a)(x-a)$ (Definition 19.1). For $x = a+h$ it gives $f(a+h) \approx f(a) + f'(a)\,h$, using only $f(a)$ and $f'(a)$. For example, $2.1^2 + 1 \approx 5.4$ (true value 5.41) (Example 19.1).
- Writing the error as εh, so that $f(a+h) = f(a) + f'(a)\,h + \varepsilon h$, differentiability of f at a means exactly that $\varepsilon \to 0$ as $h \to 0$ (Remark 19.1).
- By the **Mean Value Theorem**, $\varepsilon = f'(\xi) - f'(a)$ for some ξ between a and $a+h$ (Equation 19.1). Consequently, the tangent of a **concave-up** function lies below its graph, and the linear approximation underestimates (Proposition 19.1). For a concave-down function it overestimates.
- Applying the Mean Value Theorem a second time, to f', gives the **error bound** $|f(a+h) - L(a+h)| \le Mh^2$, where M bounds $|f''|$ between a and $a+h$ (Theorem 19.1). The error is quadratic in h, much smaller than the linear term $f'(a)\,h$ for small h (unless $f'(a) = 0$). The sharper bound $\frac{1}{2}Mh^2$ will come later (Remark 19.2).

20 Activity: Linear Approximation and Euler's Method

About this activity

An in-class, pencil-and-paper worksheet. It builds on the linear approximation from lecture,

$$L(x) = f(a) + f'(a)(x - a),$$

the tangent line to the graph of f at $x = a$. Here we use it in a new way: to estimate values of a function we do *not* have a formula for, but whose derivative we know through a **differential equation**.

20.1 Background: Differential Equations

Consider the equation

$$\frac{dy}{dx} = -2y.$$

This is an example of a **differential equation**. A **solution** of this differential equation is a function $y = f(x)$ that makes the equation true, i.e. $f'(x) = -2f(x)$ for all x.

Notice that the equation does not tell us what f is. It tells us something more indirect: at any point (x, y) on the graph of a solution, the slope of the tangent line is $-2y$. If we know *one* point on the graph, that is exactly what we need to build a linear approximation there.

20.2 Step 1: A worked example

Suppose we want a solution $y = f(x)$ with the **initial data** $f(0) = 10$, and we wish to estimate $f(0.1)$.

1. Since we know $f(0)$, we use the linear approximation about $a = 0$:

$$L(x) = f(0) + f'(0)(x - 0).$$

2. Use the initial data and the differential equation to compute $f'(0)$. At $x = 0$ we have $y = f(0) = 10$, so

$$f'(0) = \left.\frac{dy}{dx}\right|_{x=0} = -2(10) = -20.$$

3. Use $f(0)$ and $f'(0)$ to build the linear approximation about $a = 0$:

$$L(x) = 10 - 20x.$$

4. Use it to estimate $f(0.1)$:

$$f(0.1) \approx L(0.1) = 10 - 20(0.1) = 8.$$

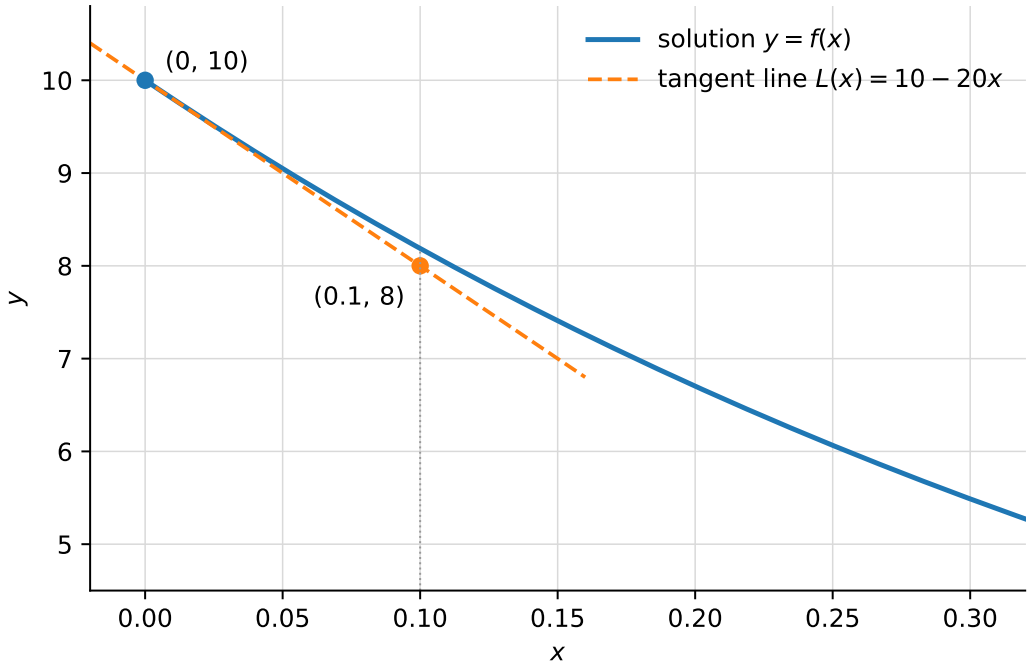


Figure 20.1: One step of Euler's method for $dy/dx = -2y$, $f(0) = 10$: follow the tangent line at $(0, 10)$ for a horizontal distance 0.1.

We now know $f(0) = 10$ and (approximately) $f(0.1) = 8$. Nothing stops us from repeating the process: use the differential equation to get the slope at $x = 0.1$, build the linear approximation about $a = 0.1$, and use it to estimate $f(0.2)$. And so on. This method of solving a differential equation numerically, one small tangent-line step at a time, is called **Euler's method**.

Exercise 20.1.

1. Using $f(0.1) \approx 8$, find the linear approximation about $a = 0.1$ and use it to estimate $f(0.2)$.
2. Take one more step to estimate $f(0.3)$.
3. Do you notice a pattern? Find a formula for the Euler estimate of $f(x)$ if we go from 0 to x in n equal steps of size $h = x/n$.

20.3 Step 2: Exponential growth

Let $N(t)$ denote the mass of a population of bacteria, in milligrams (mg), at time t (in days). Assume that the per capita growth rate is 10% per day:

$$\frac{1}{N}\frac{dN}{dt} = 0.1, \qquad \text{or} \qquad \frac{dN}{dt} = 0.1\,N.$$

Suppose we observe that at time $t = 10$ days the mass of the population is 300 mg, so that $N(10) = 300$.

Exercise 20.2.

1. Find the linear approximation for $N(t)$ about $a = 10$:
$$L(t) = N(10) + N'(10)(t - 10).$$
2. Use your linear approximation to estimate the mass of the population when $t = 10.5$.

20.4 Step 3: Logistic growth

Let $N(t)$ again denote the mass of a population of bacteria, in mg, at time t (in days). This time the population is modeled by a **logistic growth model**

$$\frac{dN}{dt} = \frac{1}{2}\,N\left(1 - \frac{N}{1000}\right).$$

Suppose we observe that at time $t = 20$ days the mass of the population is 500 mg, so that $N(20) = 500$.

Exercise 20.3.

1. Find the linear approximation for $N(t)$ about $a = 20$:
$$L(t) = N(20) + N'(20)(t - 20).$$
2. Use your linear approximation to estimate the mass of the population when $t = 20.1$.

20.5 Step 4: Pollution in a lake

Let $P(t)$ denote the amount of phosphorus pollution in a lake, in tons, t months after monitoring has begun. Assume that the amount of pollution is modeled by the differential equation

$$\frac{dP}{dt} = 20 - \frac{1}{5}P.$$

This says that pollution flows into the lake at a constant rate of 20 tons per month and flows out at a rate proportional to the current amount P, with proportionality constant $\frac{1}{5}$, i.e. 20% per month. Suppose we observe that at time $t = 3$ months the amount of phosphorus pollution is 40 tons, so that $P(3) = 40$.

Exercise 20.4.

1. Find the linear approximation for $P(t)$ about $a = 3$:

$$L(t) = P(3) + P'(3)(t - 3).$$

2. Use your linear approximation to estimate the amount of phosphorus pollution in the lake one month later, at $t = 4$.

20.6 Over or under?

In each of the three models, you can decide whether your estimate is too high or too low *without* knowing a formula for the solution, just by reading the differential equation.

Exercise 20.5. For each of Steps 2–4: as t increases past the base point a, does the solution increase or decrease? And does the **slope** given by the differential equation then get larger or smaller? Use this to decide whether your linear approximation is an **overestimate** or an **underestimate**. Which of the three estimates do you expect to be most accurate, and why?

21 Lab: The Mean Value Theorem and Linear Approximation

Math 140H, Honors Calculus I

Two ideas from class, seen through computation: the Mean Value Theorem, and the fact that near a point, a differentiable function is almost its tangent line. At the end you will learn how to bound the error of the tangent-line approximation.

How to use this notebook

- **On paper:** work this out by hand first (you will hand these pages in), then run the next cell to check.
- **Explore:** change the numbers in the cell, or move the slider, and run the cell again.
- You never need to write code. Only change the inputs you are told to change, then press Shift+Enter.

Functions are typed like this: x^3 is `x**3`, $\sin x$ is `sp.sin(x)`, $\frac{1}{x}$ is `1/x`, $|x|$ is `abs(x)`, $\sqrt[3]{x}$ is `cbrt(x)`, π is `sp.pi`.

START BY RUNNING THE NEXT CELL.

```
#
#
# ALWAYS RUN THIS CELL FIRST!!
#
#

import numpy as np
import sympy as sp
import mpmath as mp
import matplotlib.pyplot as plt
from scipy.optimize import brentq
from IPython.display import display, Math

plt.rcParams['figure.figsize'] = (7, 4.5)
WIDGET_DPI = 72          # on-screen size of the slider pictures: smaller number = small
```

```
x = sp.Symbol('x', real=True)
mp.mp.dps = 40

import io
try:
    import ipywidgets as widgets
    _WIDGETS = True
except ImportError:
    _WIDGETS = False


class cbrt(sp.Function):
    """Real cube root: cbrt(x) = x^(1/3), also for negative x."""
    @classmethod
    def eval(cls, arg):
        if arg.is_Number:
            return sp.sign(arg) * sp.Abs(arg) ** sp.Rational(1, 3)

    def fdiff(self, argindex=1):
        return 1 / (3 * cbrt(self.args[0]) ** 2)

    def _numpycode(self, printer):
        return f"{printer._module_format('numpy.cbrt')}({printer._print(self.args[0])})"

    def _mpmathcode(self, printer):
        a = printer._print(self.args[0])
        return (f"({printer._module_format('mpmath.sign')}({a})"
                f"*{printer._module_format('mpmath.cbrt')}(abs({a})))")

    def _latex(self, printer, exp=None):
        s = r"\sqrt[3]{%s}" % printer._print(self.args[0])
        return s if exp is None else r"\left(%s\right)^{%s}" % (s, exp)


# ---------------------------------------------------------------- helpers
def _exact(v):
    v = sp.sympify(v)
    return sp.nsimplify(v, rational=True) if isinstance(v, sp.Float) else v


def _fmt(v, digits=8):
    return f'{float(v):.{digits}g}'
```

```
def _np(f):
    F = sp.lambdify(x, f, 'numpy')
    def G(t):
        with np.errstate(all='ignore'):
            return np.asarray(F(t), dtype=float) * np.ones_like(t, dtype=float)
    return G


def _mp(f):
    return sp.lambdify(x, f, 'mpmath')


def _value(f, a):
    """f(a), using a limit if the formula is undefined at a (as for x*sin(1/x) at 0)."""
    v = f.subs(x, a)
    if v.has(sp.nan, sp.zoo, sp.oo, -sp.oo):
        v = sp.limit(f, x, a)
    return v


def _slope_at(f, a):
    """f'(a) if f is differentiable at a, otherwise None (checked with one-sided differe
    F, A, fa = _mp(f), mp.mpf(sp.N(a, 50)), mp.mpf(sp.N(_value(f, a), 50))
    qs = []
    for h in [mp.mpf('1e-15'), mp.mpf('1e-20')]:
        for s in (1, -1):
            try:
                qs.append((F(A + s * h) - fa) / (s * h))
            except Exception:
                return None
    if any((not mp.isfinite(q)) or abs(q) > 1e8 for q in qs) or max(qs) - min(qs) > 1e-6
        return None
    d = sp.diff(f, x).subs(x, a)
    return sp.nsimplify(d) if d.is_finite else None


def _p(v):
    """LaTeX of a number, in parentheses if negative."""
    v = sp.nsimplify(v)
    return rf"\left({sp.latex(v)}\right)" if (v.is_real and v < 0) else sp.latex(v)
```

```
def _show(label, expr):
    display(Math(label + sp.latex(sp.nsimplify(expr)) +
                 (r"\approx " + _fmt(expr) if not sp.sympify(expr).is_Integer else "")))


def _slider_view(draw, sliders):
    """Show draw(**slider values) -> (figure, text) and update it in place when a slider

    The figure is rendered to a PNG and swapped into an Image widget of fixed on-screen
    so the output area is never cleared and nothing on the page moves."""
    if not _WIDGETS:
        print('(No sliders available here: showing the starting values.)')
        fig, text = draw(**{k: sl.value if hasattr(sl, 'value') else sl for k, sl in sli
        plt.show(); print(text)
        return
    img = widgets.Image(format='png')
    txt = widgets.HTML()
    def update(change=None):
        fig, text = draw(**{k: sl.value for k, sl in sliders.items()})
        w_in, h_in = fig.get_size_inches()
        # on-screen size in CSS pixels; rendered at twice that resolution so it stays sh
        img.layout.width, img.layout.height = f'{round(w_in * WIDGET_DPI)}px', f'{round(
        buf = io.BytesIO()
        fig.savefig(buf, format='png', dpi=2 * WIDGET_DPI)   # fixed size: no 'tight' cr
        plt.close(fig)
        img.value = buf.getvalue()
        txt.value = ('<pre style="font-family: monospace; font-size: 12px; margin: 0">'
                     + text + '</pre>')
    for sl in sliders.values():
        sl.observe(update, names='value')
    update()
    # align_items='flex-start' stops the box from stretching the picture to the full cel
    display(widgets.VBox(list(sliders.values()) + [img, txt],
                         layout=widgets.Layout(align_items='flex-start')))


def _slider(description, lo, hi, step, value):
    return widgets.FloatSlider(min=lo, max=hi, step=step, value=value, description=descr
                               readout_format='.2f', continuous_update=True,
                               layout=widgets.Layout(width='400px')) if _WIDGETS else va
```

```
# ------------------------------------------------------------------ Part 1
def _witnesses(f, a, b, m):
    dF = _np(sp.diff(f, x))
    xs = np.linspace(float(a), float(b), 20001)[1:-1]
    g = dF(xs) - float(m)
    out = []
    for i in range(len(xs) - 1):
        if g[i] == 0:
            out.append(xs[i])
        elif np.isfinite(g[i]) and np.isfinite(g[i + 1]) and g[i] * g[i + 1] < 0:
            c = brentq(lambda t: float(dF(np.array([t]))[0]) - float(m), xs[i], xs[i + 1
            if abs(float(dF(np.array([c]))[0]) - float(m)) < 1e-7:   # skip jumps of f'
                out.append(c)
    return out


def _plot_mvt(f, a, b, xis, title):
    F, dF = _np(f), _np(sp.diff(f, x))
    A, B = float(a), float(b)
    pad = 0.15 * (B - A)
    xs = np.linspace(A - pad, B + pad, 1500)
    m = (F(np.array([B]))[0] - F(np.array([A]))[0]) / (B - A)
    plt.plot(xs, F(xs), lw=2, label='$y=f(x)$')
    plt.plot([A, B], [F(np.array([A]))[0], F(np.array([B]))[0]], 'ko-', label=f'secant,
    for k, c in enumerate(xis):
        t = np.linspace(c - 0.3 * (B - A), c + 0.3 * (B - A), 2)
        fc = F(np.array([c]))[0]
        plt.plot(t, fc + m * (t - c), 'r--', label='tangent at $\\xi$' if k == 0 else No
        plt.plot(c, fc, 'ro')
    plt.axvline(A, color='0.6', lw=0.8); plt.axvline(B, color='0.6', lw=0.8)
    plt.title(title); plt.legend(); plt.show()


def mvt(f, a, b):
    """Secant slope of f on [a, b], every xi in (a, b) with f'(xi) = slope, and a pictur
    f, a, b = sp.sympify(f), _exact(a), _exact(b)
    m = sp.nsimplify((f.subs(x, b) - f.subs(x, a)) / (b - a))
    display(Math(rf"f(x) = {sp.latex(f)}, \qquad f'(x) = {sp.latex(sp.diff(f, x))}, \qqu
    _show(r"\text{secant slope } \frac{f(b)-f(a)}{b-a} = ", m)
    xis = _witnesses(f, a, b, m)
    if not xis:
        print("There is NO xi in (a, b) with f'(xi) = secant slope.")
```

```
    for c in xis:
        print(f"xi = {c:.8g}   (relative position (xi - a)/(b - a) = {(c - float(a)) / f
    _plot_mvt(f, a, b, xis, f'Mean Value Theorem on [{a}, {b}]')


def mvt_explorer(f, lo=-3, hi=3):
    """Sliders for a and b; shows the secant, the tangents at every xi, and where xi sit
    f = sp.sympify(f)
    F, dF = _np(f), _np(sp.diff(f, x))
    # fixed axes for every slider position: the graph over the whole slider range
    pad = 0.05 * (hi - lo)
    xs = np.linspace(lo - pad, hi + pad, 1500)
    ys = F(xs)
    ymin, ymax = np.nanmin(ys), np.nanmax(ys)
    ymargin = 0.08 * (ymax - ymin) + 1e-9

    def draw(a, b):
        fig = plt.figure(figsize=(7, 4.5))
        fig.subplots_adjust(left=0.1, right=0.97, bottom=0.1, top=0.92)
        ax = fig.add_subplot()
        ax.plot(xs, ys, lw=2, label='$y=f(x)$')
        ax.set_xlim(lo - pad, hi + pad); ax.set_ylim(ymin - ymargin, ymax + ymargin)
        ax.set_title(f'$f(x) = {sp.latex(f)}$')
        lines = []
        if b <= a + 1e-9:
            ax.legend(loc='upper left')
            return fig, 'Choose b > a.\n\n\n'
        A, B = _exact(round(a, 4)), _exact(round(b, 4))
        m = float((f.subs(x, B) - f.subs(x, A)) / (B - A))
        xis = _witnesses(f, A, B, m)
        fa, fb = F(np.array([a]))[0], F(np.array([b]))[0]
        ax.plot([a, b], [fa, fb], 'ko-', label='secant')
        ax.axvline(a, color='0.6', lw=0.8); ax.axvline(b, color='0.6', lw=0.8)
        for k, c in enumerate(xis):
            t = np.linspace(c - 0.3 * (b - a), c + 0.3 * (b - a), 2)
            fc = F(np.array([c]))[0]
            ax.plot(t, fc + m * (t - c), 'r--', label='tangent at $\\xi$' if k == 0 else
            ax.plot(c, fc, 'ro')
        ax.legend(loc='upper left')
        lines.append(f"secant slope = {m:+10.5f}      midpoint (a+b)/2 = {(a + b) / 2:+8
        if not xis:
            lines.append("no xi in (a, b)")
```

```
        for c in xis[:3]:
            lines.append(f"xi = {c:+8.4f}     relative position (xi - a)/(b - a) = {(c -
        lines += [''] * (4 - len(lines))
        return fig, '\n'.join(lines)

    _slider_view(draw, {'a': _slider('a', lo, hi, (hi - lo) / 120, -1.0),
                        'b': _slider('b', lo, hi, (hi - lo) / 120, 2.0)})


# ------------------------------------------------------------------ Part 2
def zoom(f, a):
    """Zoom in on the graph of f at (a, f(a)) with a slider. The window stays square."""
    f, a = sp.sympify(f), _exact(a)
    fa = float(_value(f, a)); A = float(a)
    d = _slope_at(f, a)
    F = _np(f)

    def draw(zoom_level):
        w = 10.0 ** (-zoom_level)
        u = np.linspace(-w, w, 4001)               # offsets from a
        fig = plt.figure(figsize=(5.8, 5.6))
        fig.subplots_adjust(left=0.19, right=0.87, bottom=0.12, top=0.9)
        ax = fig.add_subplot()
        ax.plot(u, F(A + u) - fa, lw=1.5, label='$y=f(x)$')
        if d is not None:
            ax.plot(u, float(d) * u, 'r--', lw=1.2, label='tangent')
        ax.plot(0, 0, 'ko', ms=4)
        ax.set_xlim(-w, w); ax.set_ylim(-w, w); ax.set_aspect('equal')
        # tick labels are offsets from the point (a, f(a)), always in the same format
        ticks = [-w, 0, w]
        labels = [f'{-w:+.1e}', '0', f'{w:+.1e}']
        ax.set_xticks(ticks, labels); ax.set_yticks(ticks, labels)
        ax.set_xlabel('$x - a$'); ax.set_ylabel('$y - f(a)$')
        ax.set_title(f'$f(x)={sp.latex(f)}$ near $a={sp.latex(a)}$')
        ax.legend(loc='upper left')
        if d is None:
            text = (f"window: x within {w:.1e} of a\n"
                    "f is not differentiable at a: there is no tangent line to draw.")
        else:
            gap = np.nanmax(np.abs(F(A + u) - fa - float(d) * u))
            text = (f"window: x within {w:.1e} of a\n"
                    f"largest vertical gap between graph and tangent: {gap:.3e}"
```

```
                    f"  =  {gap / w:.3e} x (half-width of the window)")
        return fig, text

    _slider_view(draw, {'zoom_level': _slider('zoom', 0, 6, 0.25, 0.0)})


def eps_table(f, a, hs=(0.1, 0.01, 0.001, 0.0001, -0.1, -0.01, -0.001, -0.0001), slope=N
    """Errors of the approximation f(a+h) ~ f(a) + slope*h (slope = f'(a) unless given).
    f, a = sp.sympify(f), _exact(a)
    fa = _value(f, a)
    d = sp.diff(f, x).subs(x, a) if slope is None else _exact(slope)
    which = "L(a+h) = f(a) + f'(a)\\,h" if slope is None else "\\ell(a+h) = f(a) + (\\te
    display(Math(rf"f(x) = {sp.latex(f)},\quad a = {sp.latex(a)},\quad f(a) = {sp.latex(
                + (rf"f'(a) = {sp.latex(sp.nsimplify(d))}" if slope is None else rf"\te
                + rf",\qquad {which}"))
    print(f"{'h':>9} {'f(a+h)':>14} {'line at a+h':>14} {'error eps*h':>13} {'eps':>12}
    for h in hs:
        H = _exact(h)
        val = sp.N(f.subs(x, a + H), 40)
        lin = sp.N(fa + d * H, 40)
        err = val - lin
        print(f"{float(H):9g} {float(val):14.10f} {float(lin):14.10f} {float(err):13.4e}
              f"{float(err / H):12.4e} {float(err / H**2):12.6f}")


# ---------------------------------------------------------------- Part 3
def error_bound(f, a, h, tol=None):
    """Linear approximation of f(a+h), its actual error, and the bound M h^2."""
    f, a, h = sp.sympify(f), _exact(a), _exact(h)
    fa, d1, f2 = _value(f, a), sp.diff(f, x).subs(x, a), sp.diff(f, x, 2)
    L = fa + d1 * h
    true = sp.N(f.subs(x, a + h), 30)
    err = true - sp.N(L, 30)
    lo, hi = sorted([float(a), float(a + h)])
    ts = np.linspace(lo, hi, 20001)
    M = float(np.nanmax(np.abs(_np(f2)(ts))))
    bound = M * float(h) ** 2
    display(Math(rf"f(x) = {sp.latex(f)},\quad a = {sp.latex(a)},\quad h = {sp.latex(h)}
                 rf"L(a+h) = f(a) + f'(a)\,h = {_p(fa)} + {_p(d1)}\cdot{_p(h)}"))
    print(f"linear approximation L(a+h) = {float(L):.10f}")
    print(f"true value          f(a+h) = {float(true):.10f}")
    print(f"actual error  f(a+h) - L(a+h) = {float(err):.4e}")
```

```
    display(Math(rf"f''(x) = {sp.latex(f2)}"))
    print(f"M = largest |f''(x)| for x between {lo:.6g} and {hi:.6g}:  {M:.6g}")
    print(f"error bound M h^2 = {bound:.4e}      (actual error)/(bound) = {abs(float(err
    if tol is not None:
        print(f"\nbound < {tol}?  {'yes' if bound < tol else 'NO'}      actual error < {


# ------------------------------------------------------------------ Part 4
def xi_table(f, a, hs=(1, 0.1, 0.01, 0.001, 0.0001, -0.1, -0.001)):
    """For the MVT on the interval between a and a+h: where is xi, relative to the inter
    f, a = sp.sympify(f), _exact(a)
    F, dF = _mp(f), _mp(sp.diff(f, x))
    A = mp.mpf(sp.N(a, 50))
    display(Math(rf"f(x) = {sp.latex(f)},\quad a = {sp.latex(a)}"))
    print(f"{'h':>9} {'xi':>16} {'(xi - a)/h':>14}")
    for h in hs:
        H = mp.mpf(sp.N(_exact(h), 50))
        m = (F(A + H) - F(A)) / H
        lo, hi = (A, A + H) if H > 0 else (A + H, A)
        try:
            xi = mp.findroot(lambda t: dF(t) - m, (lo, hi), solver='anderson')
            print(f"{float(H):9g} {float(xi):16.10f} {float((xi - A) / H):14.8f}")
        except Exception:
            print(f"{float(H):9g}   (no unique xi found)")


def done():
    print('Setup complete.')
done()
```

```
Setup complete.
```

21.1 Finding ξ

Mean Value Theorem. If f is continuous on $[a, b]$ and differentiable on (a, b), there is a $\xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

The theorem says ξ exists, but not where it is. Here we find it.

1.1 On paper. Let $f(x) = x^3$. - On $[0, 3]$: compute the secant slope and find **all** ξ in $(0, 3)$. - On $[-1, 2]$: same question. Careful: which solutions lie in the *open* interval?

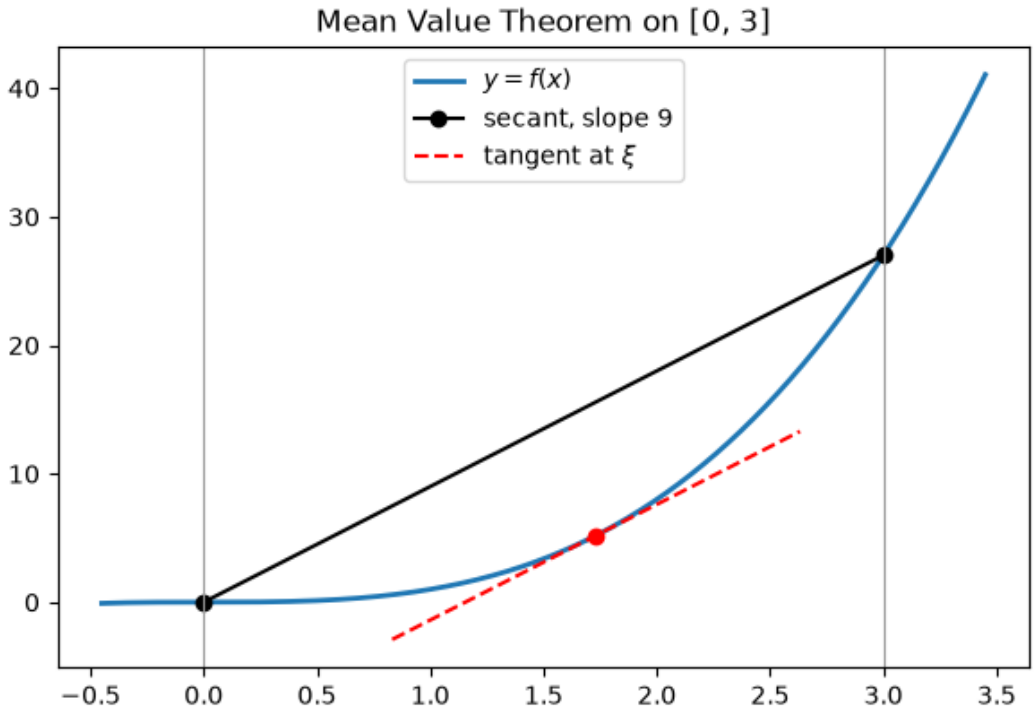
Then run the next two cells.

```
mvt(x**3, 0, 3)
```

$f(x) = x^3, \qquad f'(x) = 3x^2, \qquad [a, b] = [0, 3]$

secant slope $\dfrac{f(b) - f(a)}{b - a} = 9$

```
xi = 1.7320508   (relative position (xi - a)/(b - a) = 0.577350)
```

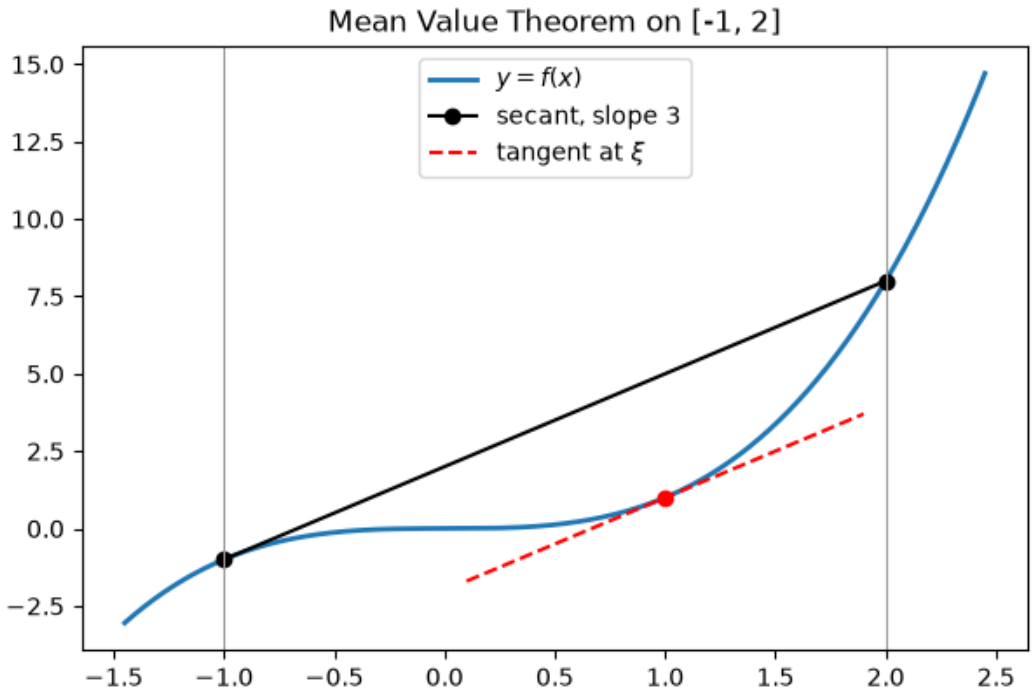



```
mvt(x**3, -1, 2)
```

$f(x) = x^3, \qquad f'(x) = 3x^2, \qquad [a, b] = [-1, 2]$

secant slope $\dfrac{f(b) - f(a)}{b - a} = 3$

```
xi = 1    (relative position (xi - a)/(b - a) = 0.666667)
```



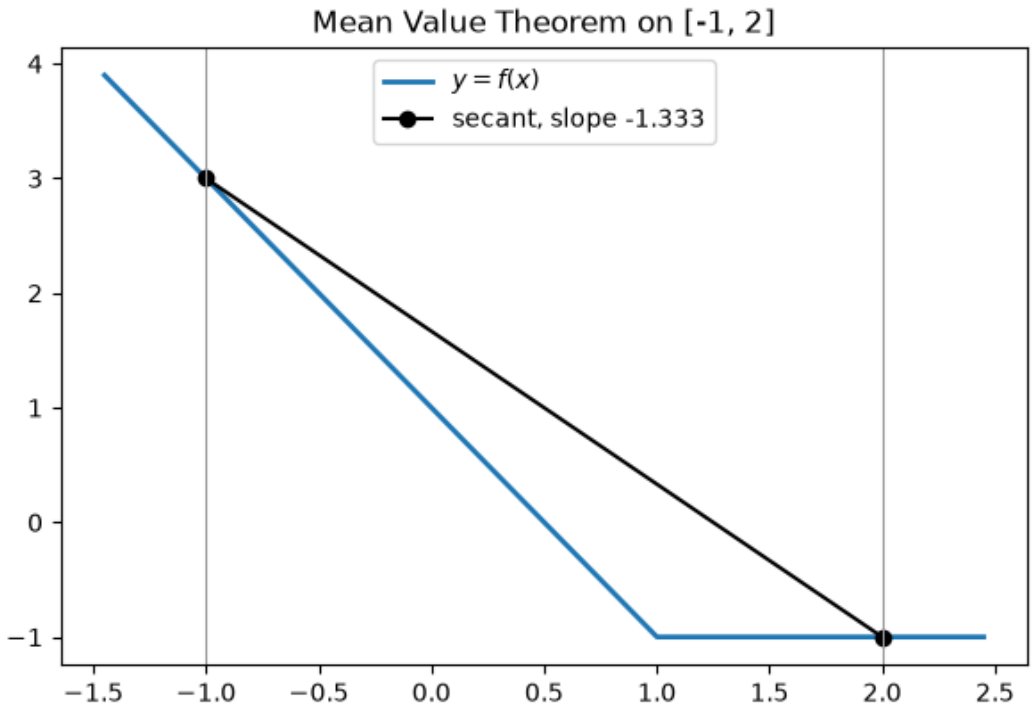
1.2 On paper. Let $f(x) = |x-1| - x$ on $[-1, 2]$. Compute the secant slope. Is there a ξ? Which hypothesis of the MVT fails?

```
mvt(abs(x-1)-x, -1, 2)
```

$$f(x) = -x + |x-1|\,, \qquad f'(x) = \operatorname{sign}(x-1) - 1, \qquad [a,b] = [-1,2]$$

$$\text{secant slope } \frac{f(b)-f(a)}{b-a} = -\frac{4}{3} \approx -1.3333333$$

```
There is NO xi in (a, b) with f'(xi) = secant slope.
```



1.3 Explore. $f(x) = \sqrt[3]{x}$ is **not** differentiable at 0 (vertical tangent). Run the cell below. Does a ξ exist anyway?

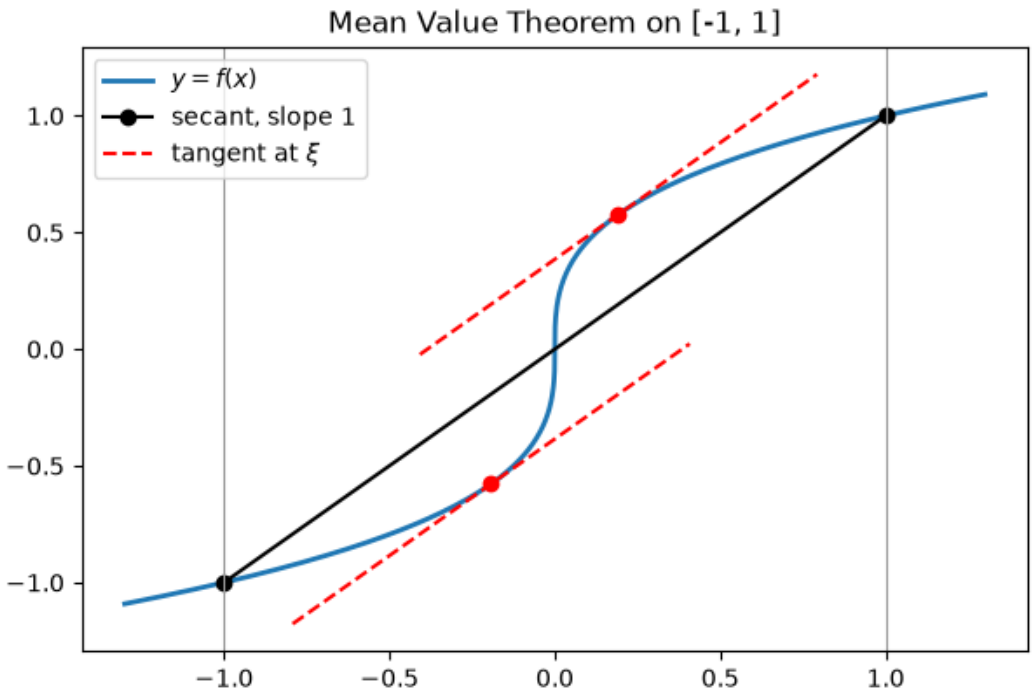
On paper, explain in one sentence why this does not contradict the MVT.

```
mvt(cbrt(x), -1, 1)
```

$$f(x) = \sqrt[3]{x}, \qquad f'(x) = \frac{1}{3\,(\sqrt[3]{x})^2}, \qquad [a,b] = [-1,1]$$

secant slope $\dfrac{f(b)-f(a)}{b-a} = 1$

```
xi = -0.19245009   (relative position (xi - a)/(b - a) = 0.403775)
xi = 0.19245009   (relative position (xi - a)/(b - a) = 0.596225)
```



1.4 Explore. Move the sliders for a and b. The output shows the **relative position** $\dfrac{\xi - a}{b - a}$ of ξ in the interval (0 = left end, 1 = right end).

- For $f(x) = x^2$: where is ξ, for every choice of a and b? Now replace `x**2` by another quadratic, such as `3*x**2 - 5*x + 1`. Same?
- For `x**3` with $a = 0$: record the relative position for three different $b > 0$. What do you notice?

1.5 On paper. Explain both observations by computation: - For $f(x) = px^2+qx+r$, compute the secant slope on $[a, b]$ and solve \$f'( ) = \$ slope. - For $f(x) = x^3$ on $[0, b]$, find ξ and the relative position.

```
mvt_explorer(x**2)
```

```
VBox(children=(FloatSlider(value=-1.0, description='a', layout=Layout(width='400px'), ma
3.0, step=…
```

21.2 The tangent line as an approximation

Let us write the approximation through the tangent line as follows:

$$f(a+h) = f(a) + f'(a)\,h + \varepsilon\,h.$$

Here ε is an **error term that depends on** h.

Then f is differentiable at a exactly when $\varepsilon \to 0$ as $h \to 0$. Geometrically, this says: **zoom in on the graph, and it looks more and more like the tangent line.**

2.1 On paper (predict). For each function, decide: if you zoom in on the graph at $x = a$, will it look like a straight line? If so, which line?

	$f(x)$	a
(a)	x^2	1
(b)	$\lvert x\rvert$	0
(c)	$\sqrt[3]{x}$	0
(d)	$x\sin(1/x)$ (with $f(0)=0$)	0

Then check each one by changing the function and a in the cell below. Move the zoom slider all the way.

```
zoom(x**2, 1)

# other inputs to try:   abs(x), 0        cbrt(x), 0        x*sp.sin(1/x), 0
```

```
VBox(children=(FloatSlider(value=0.0, description='zoom', layout=Layout(width='400px'),
```

2.2 On paper. Let $f(x) = \frac{1}{x}$ and $a = 1$. - Find $L(x) = f(1) + f'(1)(x-1)$. - For $h = 0.1$: compute $f(1.1)$, $L(1.1)$, the error $\varepsilon h = f(1.1) - L(1.1)$, and ε, as exact fractions. - For general h: show that $\varepsilon h = \dfrac{h^2}{1+h}$.

Check with the table (the column `error eps*h` is $f(a+h) - L(a+h)$).

```
eps_table(1/x, 1)
```

$$f(x) = \frac{1}{x}, \quad a = 1, \quad f(a) = 1, \quad f'(a) = -1, \qquad L(a+h) = f(a) + f'(a)\,h$$

```
        h          f(a+h)     line at a+h    error eps*h           eps  error / h^2
      0.1    0.9090909091    0.9000000000     9.0909e-03    9.0909e-
02     0.909091
```

```
     0.01    0.9900990099    0.9900000000    9.9010e-05    9.9010e-
03      0.990099
    0.001    0.9990009990    0.9990000000    9.9900e-07    9.9900e-
04      0.999001
   0.0001    0.9999000100    0.9999000000    9.9990e-09    9.9990e-
05      0.999900
     -0.1    1.1111111111    1.1000000000    1.1111e-02   -1.1111e-
01      1.111111
    -0.01    1.0101010101    1.0100000000    1.0101e-04   -1.0101e-
02      1.010101
   -0.001    1.0010010010    1.0010000000    1.0010e-06   -1.0010e-
03      1.001001
  -0.0001    1.0001000100    1.0001000000    1.0001e-08   -1.0001e-
04      1.000100
```

The error εh goes to 0 much faster than h. The last column compares it with h^2.

2.3 Explore. Run the table for each row below, and on paper fill in the missing columns: the value that `error / h^2` approaches, and $f''(a)$ (the derivative of f', computed by hand).

$f(x)$	a	`error / h^2` $\to$?	$f''(a)$
`1/x`	`1`		
`x**3`	`1`		
`sp.sin(x)`	`sp.pi/6`		
`sp.cos(x)`	`0`		
`x**3 - 2*x + 4`	`1`		

Write down a conjecture: $\lim_{h\to 0} \dfrac{\varepsilon h}{h^2} = \ ?$

```
eps_table(x**3 - 2*x + 4, 1)
```

$f(x) = x^3 - 2x + 4, \quad a = 1, \quad f(a) = 3, \quad f'(a) = 1, \qquad L(a+h) = f(a) + f'(a)\,h$

```
         h          f(a+h)     line at a+h   error eps*h          eps  error / h^2
       0.1    3.1310000000    3.1000000000    3.1000e-02    3.1000e-
01      3.100000
      0.01    3.0103010000    3.0100000000    3.0100e-04    3.0100e-
02      3.010000
     0.001    3.0010030010    3.0010000000    3.0010e-06    3.0010e-
03      3.001000
```

```
   0.0001    3.0001000300    3.0001000000    3.0001e-08   3.0001e-
04     3.000100
    -0.1    2.9290000000    2.9000000000    2.9000e-02  -2.9000e-
01     2.900000
   -0.01    2.9902990000    2.9900000000    2.9900e-04  -2.9900e-
02     2.990000
  -0.001    2.9990029990    2.9990000000    2.9990e-06  -2.9990e-
03     2.999000
 -0.0001    2.9999000300    2.9999000000    2.9999e-08  -2.9999e-
04     2.999900
```

2.4 Explore. Is the tangent special? Use a line through $(a, f(a))$ with a slope slightly *different* from $f'(a)$. Below, $f(x) = x^2$, $a = 1$ and $f'(1) = 2$, but the line uses slope 2.01. What happens to the `eps` column as $h \to 0$? Try other slopes.

On paper: which slope is the only one for which the error is small compared to h?

```
eps_table(x**2, 1, slope=2.001)
```

$$f(x) = x^2, \quad a = 1, \quad f(a) = 1, \quad \text{slope} = \frac{2001}{1000}, \qquad \ell(a+h) = f(a) + (\text{slope})\,h$$

```
        h         f(a+h)    line at a+h   error eps*h          eps  error / h^2
      0.1    1.2100000000    1.2001000000    9.9000e-03   9.9000e-
02     0.990000
     0.01    1.0201000000    1.0200100000    9.0000e-05   9.0000e-
03     0.900000
    0.001    1.0020010000    1.0020010000    0.0000e+00   0.0000e+00     0.000000
   0.0001    1.0002000100    1.0002001000   -9.0000e-08  -9.0000e-
04    -9.000000
     -0.1    0.8100000000    0.7999000000    1.0100e-02  -1.0100e-
01     1.010000
    -0.01    0.9801000000    0.9799900000    1.1000e-04  -1.1000e-
02     1.100000
   -0.001    0.9980010000    0.9979990000    2.0000e-06  -2.0000e-
03     2.000000
  -0.0001    0.9998000100    0.9997999000    1.1000e-07  -1.1000e-
03    11.000000
```

21.3 Bounding the error with the second derivative

We want a **guarantee** for the size of the error εh, without knowing $f(a+h)$. Take $h > 0$.

Step 1. By the MVT on $[a, a+h]$, $\dfrac{f(a+h)-f(a)}{h} = f'(\xi)$ for some $\xi \in (a, a+h)$. Comparing with $\dfrac{f(a+h)-f(a)}{h} = f'(a) + \varepsilon$ gives

$$\varepsilon = f'(\xi) - f'(a).$$

Step 2. Apply the MVT **again**, now to the function f' on $[a, \xi]$. Its derivative is the **second derivative** $f'' = (f')'$. So there is $\eta \in (a, \xi)$ with

$$f'(\xi) - f'(a) = f''(\eta)\,(\xi - a).$$

Step 3. Since $0 < \xi - a < h$, if $|f''| \le M$ on $[a, a+h]$, then $|\varepsilon| \le Mh$. Multiply by h:

> **Theorem (error of the linear approximation).** Let f be twice differentiable on an open interval containing a and $a+h$, and $|f''(x)| \le M$ for all x between a and $a+h$. Then
>
> $$|f(a+h) - L(a+h)| \le Mh^2.$$

(For $h < 0$, use the interval $[a+h, a]$ instead; the result is the same.)

3.1 On paper. For each approximation: find $L(a+h)$, a bound M for $|f''|$ between a and $a+h$, and the error bound Mh^2. - (a) $\dfrac{1}{0.9}$, using $f(x) = \frac{1}{x}$, $a = 1$, $h = -0.1$. (Is $|f''|$ largest at 0.9 or at 1?) - (b) $\sin 31^\circ$, using $f(x) = \sin x$, $a = \frac{\pi}{6}$ $(= 30^\circ)$ and $h = \frac{\pi}{180}$ $(= 1^\circ)$. (A simple choice is $M = 1$.) How many decimal places of $\sin 31^\circ$ does your bound guarantee?

The cells compute M as the exact largest value of $|f''|$; yours may be larger, which is fine.

```
error_bound(1/x, 1, -0.1)
```

$$f(x) = \frac{1}{x}, \quad a = 1, \quad h = -\frac{1}{10}, \qquad L(a+h) = f(a) + f'(a)\,h = 1 + (-1)\cdot\left(-\frac{1}{10}\right)$$

```
linear approximation L(a+h) = 1.1000000000
true value          f(a+h) = 1.1111111111
actual error  f(a+h) - L(a+h) = 1.1111e-02
```

$$f''(x) = \frac{2}{x^3}$$

```
M = largest |f''(x)| for x between 0.9 and 1:  2.74348
error bound M h^2 = 2.7435e-02      (actual error)/(bound) = 0.405
```

```
error_bound(sp.sin(x), sp.pi/6, sp.pi/180)
```

$$f(x) = \sin(x), \quad a = \frac{\pi}{6}, \quad h = \frac{\pi}{180}, \qquad L(a+h) = f(a) + f'(a)\,h = \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{180}$$

```
linear approximation L(a+h) = 0.5151149947
true value          f(a+h) = 0.5150380749
actual error  f(a+h) - L(a+h) = -7.6920e-05
```

$$f''(x) = -\sin(x)$$

```
M = largest |f''(x)| for x between 0.523599 and 0.541052:  0.515038
error bound M h^2 = 1.5689e-04      (actual error)/(bound) = 0.490
```

3.2 On paper (error budget). You want to approximate $\frac{1}{1+h}$ by $L(1+h) = 1-h$, for $h > 0$, with error **guaranteed** below 0.001. Use the theorem to find how large h may be.

Enter your h in the cell and run it. Then increase h to find (to 3 decimals) the largest h for which the **actual** error is still below 0.001.

Your guaranteed h and the actual largest h differ by a factor of about $\sqrt{2}$. Use your conjecture from 2.3 to explain why. (Remember $f''(1) = 2$.)

```
h = 0.05          # <- change this

error_bound(1/x, 1, h, tol=0.001)
```

$$f(x) = \frac{1}{x}, \quad a = 1, \quad h = \frac{1}{20}, \qquad L(a+h) = f(a) + f'(a)\,h = 1 + (-1) \cdot \frac{1}{20}$$

```
linear approximation L(a+h) = 0.9500000000
true value          f(a+h) = 0.9523809524
actual error  f(a+h) - L(a+h) = 2.3810e-03
```

$$f''(x) = \frac{2}{x^3}$$

```
M = largest |f''(x)| for x between 1 and 1.05:  2
error bound M h^2 = 5.0000e-03      (actual error)/(bound) = 0.476

bound < 0.001?  NO      actual error < 0.001?  NO
```

Part VI

The Fundamental Theorem

22 The Fundamental Theorem of Calculus

So far, integration and differentiation have been developed as two separate theories. The integral was defined through limits of Riemann sums (areas), the derivative through limits of difference quotients (slopes). The Fundamental Theorem of Calculus shows that the two operations are inverse to each other. This is the basic fact of calculus, and it turns the computation of integrals, which is hard to do directly from Riemann sums, into the problem of reversing a differentiation.

22.1 The Derivative of the Indefinite Integral

Recall that for a continuous function f on an interval I and a fixed $\alpha \in I$, the **indefinite integral** of f is the function

$$\Phi(x) = \int_\alpha^x f(u)\,du \qquad (x \in I)$$

(Definition 14.1). We know that Φ is continuous (Theorem 14.1). In fact, much more is true.

Theorem 22.1 (Fundamental Theorem of Calculus, Part 1)**.** *Let f be continuous on an interval I and let $\alpha \in I$. Then the indefinite integral*

$$\Phi(x) = \int_\alpha^x f(u)\,du$$

is differentiable at every $x \in I$, and

$$\Phi'(x) = f(x), \qquad \textit{that is,} \qquad \frac{d}{dx}\int_\alpha^x f(u)\,du = f(x).$$

(If x is an endpoint of I, $\Phi'(x)$ is the one-sided derivative. The most important case is $f\colon \mathbb{R} \to \mathbb{R}$, where this does not arise.)

Proof. Let $x \in I$, and let $h \neq 0$ with $x + h \in I$. By Proposition 14.1 and the Mean Value Theorem for Integrals (Theorem 13.1; for $h < 0$ use Remark 13.4, part 3),

$$\Phi(x+h) - \Phi(x) = \int_x^{x+h} f(u)\,du = h \cdot f(\xi)$$

for some ξ between x and $x+h$. Here $f(\xi)$ is the average value of f on the interval between x and $x+h$ (Definition 13.2). Dividing by h,

$$\frac{\Phi(x+h)-\Phi(x)}{h} = f(\xi).$$

The point ξ depends on h, but it always lies between x and $x+h$, so $|\xi - x| \le |h|$. Hence, as $h \to 0$, $\xi \to x$, and since f is continuous, $f(\xi) \to f(x)$.

To make the last step precise, let $\varepsilon > 0$. Since f is continuous at x (Definition 6.2), there is $\delta > 0$ such that $|f(u) - f(x)| < \varepsilon$ for all $u \in I$ with $|u - x| < \delta$. If $0 < |h| < \delta$, then $|\xi - x| \le |h| < \delta$, and therefore

$$\left|\frac{\Phi(x+h)-\Phi(x)}{h} - f(x)\right| = |f(\xi) - f(x)| < \varepsilon.$$

This shows

$$\lim_{h\to 0} \frac{\Phi(x+h)-\Phi(x)}{h} = f(x),$$

so Φ is differentiable at x with $\Phi'(x) = f(x)$ (Definition 15.2). □

Geometrically, $\Phi(x)$ is the (signed) area under the graph of f between α and x, and $\Phi(x+h) - \Phi(x)$ is the area of the thin strip over the interval between x and $x+h$. The Mean Value Theorem for Integrals replaces this strip by a rectangle of the same area, with width h and height $f(\xi)$. The difference quotient is the height of that rectangle. As h shrinks, ξ is squeezed toward x, and by continuity the height $f(\xi)$ is pinned to $f(x)$.

The interactive figure below shows this in two ways. In the left panel, the dashed coral rectangle over the interval between x and $x+h$ has the same area as the blue strip, so its height is the difference quotient $\frac{\Phi(x+h)-\Phi(x)}{h}$; the coral dot marks a point ξ with $f(\xi)$ equal to that height. In the right panel, the same quotient is the slope of the coral secant of the graph of Φ through x and $x+h$. As $h \to 0$, the secant turns onto the dashed teal line through $(x, \Phi(x))$ with slope $f(x)$: by Theorem 22.1, this is the tangent to the graph of Φ.

The default integrand is the cross-section $A(u) = 2\sqrt{u(2-u)}$ of the oil tank in Example 14.2, with $R = L = 1$, so the right panel shows the tank's volume function. The slider for h is logarithmic, from $|h| = 1$ down to $|h| = 0.001$; "Let h → 0" animates it, and the "h > 0 / h < 0" button switches the sign of h, so that both one-sided limits can be checked.

An interactive version of this figure, with sliders for x and h, is included in the HTML version of these notes.

Remark 22.1. The third integrand in the figure shows why continuity is needed. It is the step function

$$f(u) = \begin{cases} 0.6 & u < 1 \\ 1.6 & u \geq 1, \end{cases}$$

which jumps at $u = 1$. Integrating piece by piece from $\alpha = 0$ gives the piecewise linear function $\Phi(x) = 0.6\,x$ for $x \leq 1$ and $\Phi(x) = 0.6 + 1.6\,(x - 1)$ for $x \geq 1$. At $x = 1$,

$$\frac{\Phi(1+h) - \Phi(1)}{h} = \begin{cases} 1.6 & h > 0, \\ 0.6 & h < 0. \end{cases}$$

The two one-sided limits differ, so Φ has a corner at 1 and is **not** differentiable there. Whenever the strip straddles the jump, the figure finds no ξ: the Mean Value Theorem for Integrals fails for discontinuous functions (Remark 13.4, part 2), and with it the proof of Theorem 22.1.

Since differentiable functions are continuous (Theorem 16.1), Theorem 22.1 also gives a second proof of Theorem 14.1.

22.2 Antiderivatives

Theorem 22.1 says that the indefinite integral solves the following problem: given f, find a function whose derivative is f. This problem, reversing differentiation, gets its own name.

Definition 22.1.
Let f be a function on an interval I. A function F is called an **antiderivative** (or **primitive**) of f on I if F is differentiable on I and

$$F'(x) = f(x) \qquad \text{for all } x \in I.$$

In this language, Theorem 22.1 reads:

> *Every indefinite integral of a continuous function f is an antiderivative of f.*

At first sight, finding antiderivatives looks like a problem of a quite different character from computing areas. That it is solved by the indefinite integral is what makes the theorem "fundamental". But it leaves open a question.

Are there any other antiderivatives?

To answer it, we need one consequence of the Mean Value Theorem.

Proposition 22.1.
Let f be continuous on $[a, b]$ and differentiable on (a, b). If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.

Proof. Let $x, y \in [a, b]$ with $x < y$. Then f is continuous on $[x, y]$ and differentiable on (x, y), so by Theorem 18.1 there is $\xi \in (x, y)$ with

$$f(y) - f(x) = f'(\xi)\,(y - x) = 0 \cdot (y - x) = 0.$$

Hence $f(x) = f(y)$. Since x and y were arbitrary, $f(x) = f(a)$ for all $x \in [a, b]$, so f is constant with value $c = f(a)$. □

Lemma 22.1.
The difference between any two antiderivatives F_1, F_2 of f on an interval I is constant:

$$F_1(x) - F_2(x) = c \qquad \text{for all } x \in I.$$

Proof. Let $G(x) := F_1(x) - F_2(x)$. By the sum and constant multiple rules for derivatives (Proposition 16.4, Proposition 16.5), G is differentiable on I with

$$G'(x) = F_1'(x) - F_2'(x) = f(x) - f(x) = 0.$$

Now let $a < b$ in I. On $[a, b]$, G is differentiable and hence continuous (Theorem 16.1), so Proposition 22.1 shows $G(b) = G(a)$. Since a and b were arbitrary, G is constant on I. □

Remark 22.2. It matters that the domain is an **interval**. The function

$$G(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

on $\mathbb{R} \setminus \{0\}$ has $G'(x) = 0$ at every point of its domain, but it is not constant. Consequently $F_1(x) = -\frac{1}{x}$ and $F_2(x) = -\frac{1}{x} + G(x)$ are both antiderivatives of $\frac{1}{x^2}$ on $\mathbb{R} \setminus \{0\}$, but their difference is not constant. The proof of Lemma 22.1 breaks down because there is no interval $[a, b]$ in the domain containing both a negative and a positive point.

The converse of Lemma 22.1 is easy.

Lemma 22.2.
If F is an antiderivative of f and $G(x) = F(x) + c$ for a constant c, then G is also an antiderivative of f.

Proof. By Proposition 16.4, and since every difference quotient of a constant function is 0,

$$G'(x) = \big(F(x) + c\big)' = F'(x) + \frac{d}{dx}\, c = F'(x) + 0 = f(x).$$

□

22.3 The Fundamental Theorem

Combining Theorem 22.1 with the two lemmas, we can describe **all** antiderivatives of a continuous function.

Theorem 22.2 (Fundamental Theorem of Calculus)**.** *Let f be continuous on an interval I. Every antiderivative F of f can be represented in the form*

$$F(x) = c + \Phi(x) = c + \int_{\alpha}^{x} f(u)\, du, \qquad (*)$$

where c is a constant and $\alpha \in I$. Conversely, for any choice of the constant c and of $\alpha \in I$, the expression $(*)$ yields an antiderivative of f.

Proof. Fix $\alpha \in I$. By Theorem 22.1, $\Phi(x) = \int_{\alpha}^{x} f(u)\, du$ is an antiderivative of f. If F is any antiderivative of f, then $F - \Phi$ is constant by Lemma 22.1, say $F - \Phi = c$, and so $F = c + \Phi$.

Conversely, for any $\alpha \in I$ the function $\Phi(x) = \int_{\alpha}^{x} f(u)\, du$ is an antiderivative of f by Theorem 22.1, and so is $c + \Phi$ for every constant c, by Lemma 22.2. □

Remark 22.3. Changing the lower limit α only changes Φ by an additive constant (Proposition 14.1). One might therefore guess that the constant c in $(*)$ is superfluous, that is, that every antiderivative is an indefinite integral $\int_{\alpha}^{x} f(u)\, du$ for a suitable α. Sometimes this is true. The antiderivatives of $f(x) = x^2$ are the functions $\frac{x^3}{3} + c$, and since $\int_{\alpha}^{x} u^2\, du = \frac{x^3}{3} - \frac{\alpha^3}{3}$, the choice $\alpha = \sqrt[3]{-3c}$ produces each of them. But in general it is false. For $f(x) = 0$, every indefinite integral is $\int_{\alpha}^{x} 0\, du = 0$, whatever α is, while every constant function is an antiderivative of f. So the constant c cannot be dropped.

22.4 Evaluating Definite Integrals

The most important application of the Fundamental Theorem is the evaluation of definite integrals

$$\int_a^b f(u)\,du.$$

Suppose we know **one** antiderivative F of f. Let $\Phi(x) = \int_a^x f(u)\,du$ be the indefinite integral with lower limit a. Both F and Φ are antiderivatives of f (the latter by Theorem 22.1), so by Lemma 22.1,

$$F(x) = \Phi(x) + c$$

for some constant c. Since

$$\Phi(a) = \int_a^a f(u)\,du = 0$$

(Definition 13.1), we get $F(a) = 0 + c$, that is, $c = F(a)$. Hence $\Phi(x) = F(x) - F(a)$, and in particular $\int_a^b f(u)\,du = \Phi(b) = F(b) - F(a)$.

Corollary 22.1 (Evaluation of definite integrals)**.** *Let f be continuous on an interval I, let $a, b \in I$, and let F be **any** antiderivative of f on I. Then*

$$\int_a^b f(u)\,du = F(b) - F(a).$$

Any antiderivative works! Compare this with Proposition 14.1, part 1, which gave the same formula for an indefinite integral Φ in place of F. There, we first had to compute Φ, which means computing integrals. Now any function whose derivative is f will do, and such functions can often be found by reading formulas for derivatives backwards. Note also that $a < b$ is not required: the formula holds for all $a, b \in I$.

Notation. The difference $F(b) - F(a)$ is written

$$F(b) - F(a) = F(x)\Big|_a^b = \Big[F(x)\Big]_a^b.$$

For an antiderivative of f one writes

$$\int f(x)\,dx = F(x) + c,$$

without limits of integration, and calls it the **indefinite integral** of f as well.

Remark 22.4. This "indefinite" indefinite integral needs some care. By Theorem 22.2, $\int f(x)\,dx$ stands for *some* antiderivative of f, determined only up to the additive constant c; the formula $\int f(x)\,dx = F(x) + c$ is just a symbolic way of writing $F'(x) = f(x)$. Strictly speaking, it is also slightly inconsistent to use the same letter x for the variable of integration and for the independent variable of F. And it is worth remembering that, a priori, integrating (computing areas by limits of Riemann sums) and reversing differentiation are two different things. It is only the Fundamental Theorem that justifies using the name "indefinite integral" for both.

Example 22.1.
(a) The function $F(x) = x^3 - x^2$ has $F'(x) = 3x^2 - 2x$, so

$$\int_1^2 (3x^2 - 2x)\,dx = \Big[x^3 - x^2\Big]_1^2 = (8 - 4) - (1 - 1) = 4.$$

(b) Since $(-\cos x)' = \sin x$,

$$\int_0^\pi \sin x\,dx = \Big[-\cos x\Big]_0^\pi = -\cos\pi + \cos 0 = 1 + 1 = 2.$$

This agrees with the average value $\frac{2}{\pi}$ of sin on $[0, \pi]$ found in Example 13.5.

In the same way, each formula in Proposition 13.4 now follows from Corollary 22.1 as soon as we know the derivative of the function on the right-hand side: no Riemann sums are needed anymore.

22.5 Summary

- **FTC, Part 1** (Theorem 22.1): if f is continuous, the indefinite integral $\Phi(x) = \int_\alpha^x f(u)\,du$ is differentiable with $\Phi' = f$. The proof combines the Mean Value Theorem for Integrals, $\frac{\Phi(x+h)-\Phi(x)}{h} = f(\xi)$ with ξ between x and $x + h$, with the continuity of f. Without continuity the conclusion can fail (Remark 22.1).
- An **antiderivative** (or primitive) of f is a function F with $F' = f$ (Definition 22.1). Every indefinite integral of a continuous function is one.
- If $f' = 0$ on an interval, then f is constant (Proposition 22.1, from the Mean Value Theorem). Hence two antiderivatives of the same function on an interval differ by a constant (Lemma 22.1), and adding a constant to an antiderivative gives another one (Lemma 22.2). On domains that are not intervals, this fails (Remark 22.2).

- **Fundamental Theorem of Calculus** (Theorem 22.2): the antiderivatives of a function f that is continuous on an interval are exactly the functions $c + \int_\alpha^x f(u)\,du$. The constant c cannot be dropped in general (Remark 22.3).
- **Evaluation of definite integrals** (Corollary 22.1): for any antiderivative F of f, $\int_a^b f(u)\,du = F(b) - F(a) = \big[F(x)\big]_a^b$. The notation $\int f(x)\,dx = F(x) + c$ stands for an arbitrary antiderivative (Remark 22.4).